

Intel's Core-based client processors

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Vers. 8.0

Februar 2026

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Intel's Core-based client processors

- 1. Introduction and overview of Intel's Core-based client processors
- 2. Naming of Intel's Core-based client processors
- 3. Key enhancements in the microarchitecture of Intel's Core-based client processors over time
- 4. Case examples for recent client processors
- 5. References

Intel's Core-based client processors

- This Chapter focuses on **client processors** that power **desktops and laptops**.
- We begin with an **introduction** and a **generational overview** of Intel's Core-based client processors.
- Next, we discuss Intel **naming conventions** for these processors.
- The main Section explores the **key architectural features over time**, including their CPU architecture, core count, core width, and memory subsystem.
- Finally, we present **two case studies** of Intel's latest processor, including its **Lunar Lake desktop** and **Arrow Lake laptop** processors, highlighting their microarchitecture and relevance for the company.

A remark on the terminology

In this Lecture Note, we use the terms “processor family”, “processor line”, or “processor series” as synonyms.

1. Introduction and overview of Intel's Core-based client processors

1. Introduction and overview of Intel's Core-based client processors (1)

1. Introduction and overview of Intel's Core-based client processors

What are client processors?

- Processors powering desktops (DTs) and laptops are named **client processors**.
- In contrast, Intel and AMD refer to their laptop processors as **mobile processors**; nevertheless, in this lecture, we will use the original term "**laptop processors**".
- Using this terminology, **client and mobile processors can be contrasted** as follows:
 - First, **client processors**, which include both **DT and laptop processors**, are typically built on the **x86 ISA**, while **mobile processors**, such as those found in **tablets and smartphones**, primarily use the **ARM ISA**.
 - Secondly, **client devices** - namely DTs and laptops - **generally do not provide touchscreen displays**, unlike **mobile devices**, which include tablets and smartphones, that typically **do**.

Intel and AMD likely chose the term "mobile" for its marketing appeal.

Remark

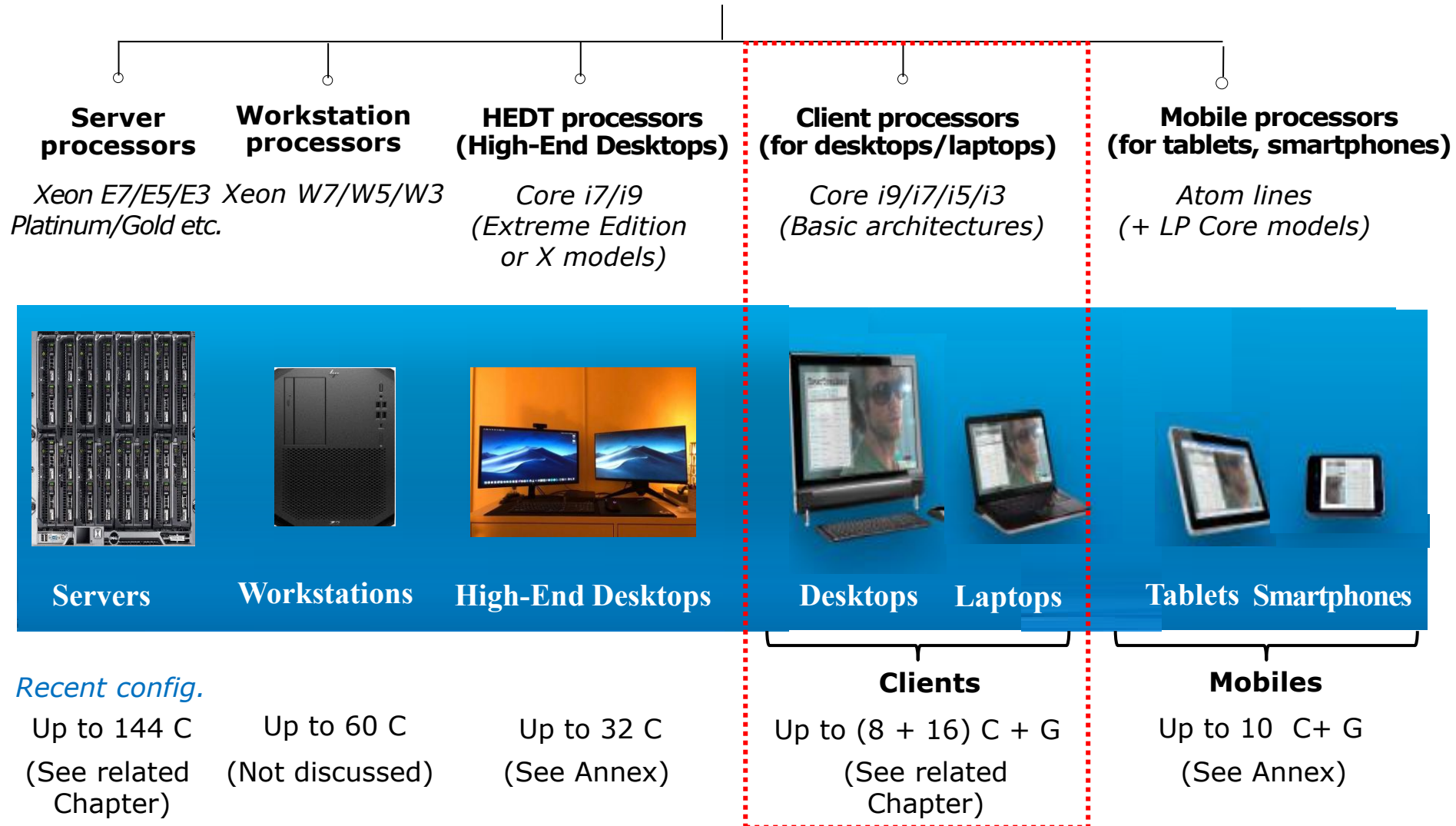
The above statement about the use of the x86 and ARM ISA has recently become **blurred** as an **increasing number of ARM ISA-based clients are emerging in the market offering lower power consumption**, as discussed in a separate Chapter.

- In the following Figure, we provide an overview of how Intel's processors can be classified according to their market aim.

1. Introduction and overview of Intel's Core-based client processors (2)

The client processor category in Intel's Core family

Main processor categories in Intel's Core family

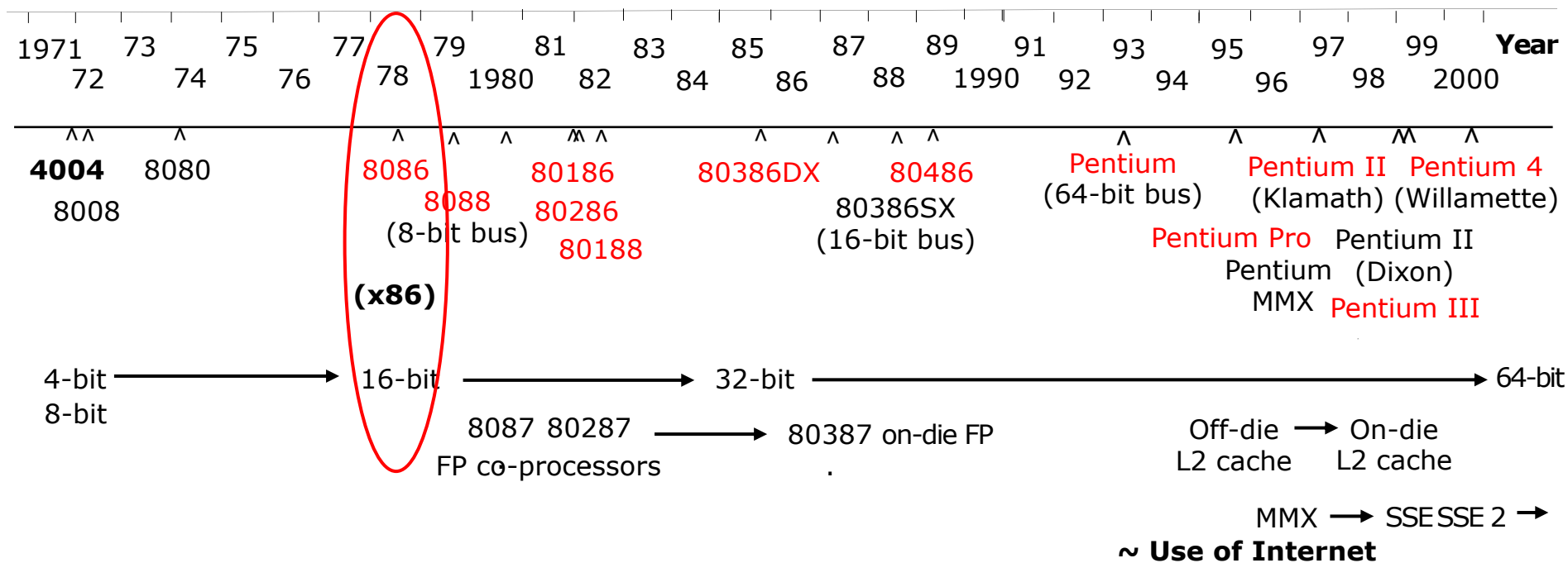


1. Introduction and overview of Intel's Core-based client processors (3)

Intel's early mainstream processors and DRAM memories -1

Intel's journey to the Core family -1

The Figure below illustrates Intel's processor lines that preceded the Core family.



See the next slide for a brief discussion of the above processors.

1. Introduction and overview of Intel's Core-based client processors (4)

Intel's journey to the Core family -2

- Intel introduced the **x86 ISA** with its **16-bit 8086** processor in **1978**.
- This was soon followed by the release of the 8088 along with two other 16-bit processors - the 80186 and 80286.
- **IBM selected the 8088 for its original IBM PC**, launched in 1981.

The 8088 was essentially an 8086 with an 8-bit wide IO bus, making the IBM PC more cost-effective.

Later, IBM released the **PC AT**, an upgraded version of the original IBM PC, featuring a hard drive and adopting the more advanced **8086**.

- **As the IBM PC gained dominance in the global market, Intel's x86 processor family became the most successful** and widely adopted architecture in personal computing.
- Intel subsequently introduced a long line of **32-bit processors**, including the **80386, 80486, Pentium, Pentium II, Pentium III, and Pentium 4**.
- These increasingly advanced processors allowed Intel to maintain a dominant position in the market, consistently capturing around **90 % market share**.
- In 2004, Intel entered the **64-bit** era with its **Prescott F Pentium 4 model**, marking **a significant evolution** in its processor architecture.
- Then, **in 2006**, Intel launched its first **64-bit dual-core** processor line, **the Core family**, specifically designed to deliver **maximum performance per Watt**.
- Since that milestone, Intel has introduced **new generations of Core-based client processor lines almost every year**, continually improving performance and power efficiency, as will be discussed subsequently.

1. Introduction and overview of Intel's Core-based client processors (5)

Overview of Intel's Core-based client processor lines and their key features -1

Gen.	Processor	Technology	Date of intro.	Segment	Max core count	Conneding memory	Mem. speed (up to)	Mem. channels	Socket	MCH/ PCH
1.	Core 2	65 nm	6/2006	L D	2C	Via the NB	DDR2-800	2	LGA 775	945-975
	Core 2 Quad	65 nm	6/2007	D	2x2C		DDR3-1067		LGA 775	3 Series
	Penryn	45 nm	6/2008	D	2x2C		DDR3-1067		LGA 775	4 Series
	2. G. Nehalem (Lynnfield)	45 nm	9/2009	D	4C	Via the processor	DDR3-1333		LGA 1156	5 Series
	Westmere (L: Arrandale) D: Clarkdale/	32 nm	1/2010	L D	2C+G				L: BGA 1288 D: LGA 1156	5 Series
2.	Sandy Bridge	32 nm	1/2011	L D	2C/4C+G		DDR3-1333		L: PGA 988 D: LGA 1155	6 Series
3.	Ivy Bridge	22 nm	4/2012	L D	4C+G		DDR3-1600			7 Series
4.	Haswell	22 nm	6/2013	L D	4C+G		DDR3-1600		L: BGA 1168 D: LGA 1150	D: 8 Ser.
4.	Haswell Refr.	22 nm	5/2014	D	4C+G		DDR3-1666			D: 9 Ser.
5.	Broadwell	14 nm	9/2014	L D	4C+G		DDR3-1866			D: 9 Ser.
6.	Skylake	14 nm	10/2015	L D	4C+G		DDR4-2133		L: BGA 1356 D: LGA 1151	D: 100 Series
7.	Kaby Lake	14 nm	8/2016	L D	4C+G		DDR4-2400			D: 200 Series
8.	Coffee Lake	14 nm	10/2017	L D	6C+G		DDR4-2666		L: BGA 1440 D: LGA 1151	D: 300 Series
8.	Amber Lake	14 nm	8/2018	L	2C+G		LPDDR3-2133		BGA 1515	--
8.	Whiskey Lake	14 nm	8/2018	L	4C+G		DDR4-2400		BGA 1528	--
9.	Coffee Lake R.	14 nm	10/2018	D	8C+G		DDR4-2666		LGA 1151	300 Series
10.	Comet Lake	14 nm	8/2019	L D	10C+G		DDR4-2666		L: BGA 1528 D: LGA 1200	D: 400 Series
11.	Rocket Lake	14 nm	3/2021	D	8C+G		DDR4-3200		LGA 1200	

1. Introduction and overview of Intel's Core-based client processors (6)

Overview of Intel's Core-based client processor lines and their key features -2

	Processor	Technology	Date of intro.	Market Segment	Max. core count	Mem. speed (up to)	Mem. channels	Socket	PCH
8.	Cannon Lake	10 nm	5/2018	L:U	2C	DDR4-2400	2x64- bit	BGA 1440	--
10.	Ice Lake	10 nm	7/2019	L: G	4C+G	DDR4-3200 LPDDR4- 3733		BGA 1526	--
11.	Tiger Lake	10 nm	09/2020 01/2021 05/2021	L: G L: H L: H	4C+G 4C+G 8C+G	LPDDR4x-4267 DDR4-3200		BGA 1449 BGA 1449 BGA 1787	--
11.	Tiger Lake	10 nm	01/2021	D	8C+G	DDR4-3200		LGA 1700	500 Series
12.	Alder Lake	Intel 7	11/2021	D	8+8C+G	DDR5-4800		D: LGA 1700	600 Series
12.	Alder Lake	Intel 7	02/2022	L:H,P L: U	6+8C+G 2+8+G	DDR5-4800 LPDDR5-5200 LPDDR4x-4267		BGA 1744	--
12.	Alder Lake	Intel 7	05/2022	L: HX	8+8+G	DDR5-4800		BGA: 1781	600 Series
13	Raptor Lake	Intel 7	10/2022	D	8+16C+G	DDR5-5600		LGA 1700	700 Series
13.	Raptor Lake	Intel 7	01/2023	L:HX	8+16C+G	DDR5-5600		BGA 1964	700 Series
13.	Raptor Lake	Intel 7	01/2023	L:H,P L:U	6+8C+G 2+8C8C	DDR5-5200 LPDDR5x-6400 LPDDR4x-4267		BGA 1744	--
14	Raptor Lake Refresh	Intel 7	10/2023	D	8+16C+G	DDR5-5600		LGA 1700	700 Series
14	Raptor Lake Refresh	Intel 7	01/2024	L:HX	8+16C+G	DDR5-5600 DDR4-3200		BGA 1964	700 Series
14.	Raptor Lake Refresh	Intel 7	01/2024	L:U	2+8C+G	DDR5-5200 LPDDR5x-6400 LPDDR4x-4267		BGA-1744	--

L: Laptop D: Desktop

1. Introduction and overview of Intel's Core-based client processors (7)

Overview of Intel's Core Ultra client processor lines and their key features

	Processor	Technology	Date of intro.	Segment	Max. core count	Memory (up to)	TDP	Socket	PCH
14/CUS 1	Meteor Lake (Series 1)	CPU: Intel 4 GPU: N5 SOC: N6	12/2023	L:H L:U	6+8+2C+G 2+8+2C+G+NPU	DDR5-5600 LPDDR5X-7467	H: 45/28 W U: 15 W	BGA 2049	-integrated
14/CUS 1	Meteor Lake (Series 1)	CPU: Intel 4 GPU: N5 SOC: N6	12/2023	L:U- 9W	2+8+2C+G+NPU	LPDDR5X-6400	9 W	BGA 2551	integrated
CUS 2	Lunar Lake (Series 2) (2 dies)	Compute: N3B Platform.: N6	09/2024	L:V	4+4C+G+ NPU (no hyperthreading)	LPDDR5x-8533 (in-package integrated)	17/30W	BGA 2833	Integrated
CUS 2	Arrow Lake (Series 2)	CPU: N3B GPU: N5 SOC/IO: N6	10/2024 01/2025	D:S	8+16C+G+ NPU (no hyperthreading)	DDR5-5600	125 W	LGA 1851	800 Series
CUS 2	Arrow Lake (Series 2)	CPU: N3B GPU: N5 SOC/IO: N6	01/2025	L:HX	8+16C+G+ NPU (no hyperthreading)	DDR5-6400	55 W	BGA 2114	800 Series
CUS 2	Arrow Lake (Series 2)	CPU: N3B GPU: N5 SOC/IO: N6	01/2025	L:H	4+8+2C+G+ NPU (no hyperthreading)	DDR5-6400 LPDDR5x-8400	28 W	BGA 2049	Integrated
CUS 2	Arrow Lake (Series 2)	CPU: Intel 3 GPU: N5 SOC/IO: N6	07/2025	L:UA	2+8+2C+G+ NPU (P: hyperthreading)	DDR5-6400 LPDDR5x-8400	15 W	BGA 2049	Integrated
CUS 3	Panther Lake (Series 3)	CPU: Intel 1.8A GPU: N3E SOC/IO: N6	01/2026	L:H	4+8+4+G	DDR5-7200 LPDDR5x-9600		BGA 2540	Integrated

The tiles with the specified IC technologies N3B, N3E N5, and N6 are manufactured by TSMC.

CUS: Core Ultra Series

2. Naming of Intel's Core-based client processors

2. Naming of Intel's Core-base client processors (1)

2. Naming of Intel's Core-base client processors

- Beginning with its 2nd gen. Sandy Bridge family (in 2011), Intel introduced a **standard branding** for its client, HEDT, and server processors, as follows:
 - Client processors are branded as **Core i3/i5/i7/i9**,
 - HEDT processors are branded as **Core i7/i9** and
 - Server processors are branded as **Xeon E3/E5/E7**, where E refers to Enterprise.
- From these categories, we will now focus solely on **Intel's naming conventions for client processors**.

However, Intel has adopted **two distinct approaches to naming** its processors:

- The **first** naming convention was used **through the 13th generation** of Intel's Core processors.
- Beginning **with the 14th generation Meteor Lake client lineup**, Intel introduced a new naming convention for its **Core Ultra processor lines**.
- Both naming schemes will be explained in the following sections.

2. Naming of Intel's Core-based client processors (2)

a) Intel's original naming scheme for Core-based client processors (1-14 generations)

- From the Sandy Bridge line (2011) through to the 14th gen. Raptor Lake Refresh line (2023), Intel named its client processors by three name segments.
 - The first segment identifies the **brand name**, which is the "Intel Core" designation.
 - the next part is the **brand modifier**, typically i3/i5/i7/i9, followed by a **hyphen**.
 - The final segment consists of **three parts**, which are:
 - The **generation identifier**, it may consist of 1 or 2 digits (2-14) while
 - the **model number (SKU (Stock Keeping Unit))** it is 3 digits long and
 - the **product line suffix**, which consists of one or two letters, and will be discussed in detail later.
- The Figure below presents a **naming example**; it refers to the 7th gen. i7-7500U (Kaby Lake DT) processor with the suffix U.

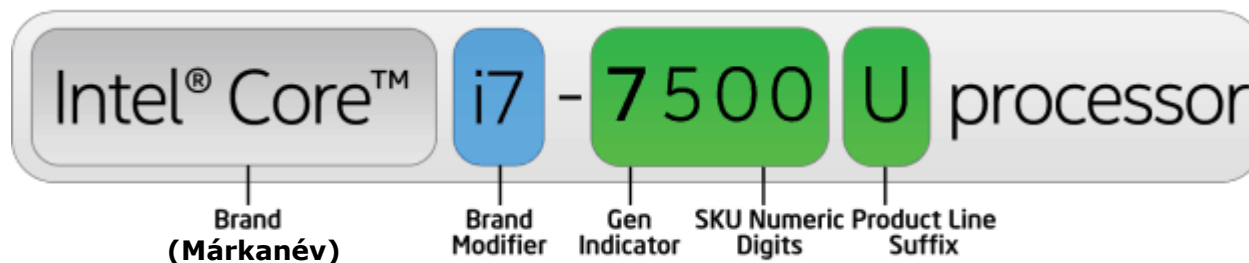


Figure: Intel's naming scheme used from the 2. up the 14. generations of the Core family (2011-2023) [43]

- The following Figures illustrate the **brand modifiers**, **generation identifiers** and **model numbers** of Intel's client processors across the 2nd to 14th generation Core processors.

2. Naming of Intel's Core-based client processors (3)

Brand modifiers, generation identifiers, and model numbers of Intel's Core-based client processors up to the 14 nm lines

1. gen.		2. gen.		3. gen.	4. gen.	5. gen.	
Core 2 New microarch. 65 nm	Penryn New process 45 nm	Nehalem New microarch. 45 nm	Westmere New process 32 nm	Sandy Bridge New microarch. 32 nm	Ivy Bridge New process 22 nm	Haswell New microarchi. 22 nm	Broadwell New process 14 nm
<i>TICK</i> (2006)		<i>TOCK</i> (2007)		<i>TOCK</i> (2008)	<i>TICK</i> (2010)	<i>TOCK</i> (2011)	<i>TICK</i> (2012)

4/5/6xxx

6/7/8/9xxx

i5/i7-xxx

i3/i5/i7-xxx

i3-i7-2xxx

i3-i7-3xxx

i3-i7-4xxx

i3-i7-5xxx

6. gen.	7. gen.	8. gen. ¹	9. gen.	10. gen.
Skylake New microarch. 14 nm <i>TOCK</i> (2015) i3/i5/i7 --6xxx	Kaby Lake New microarch. 14 nm <i>TOCK</i> (2016) -7xxx	Kaby Lake R/G Coffee Lake Amber Lake-Y Whiskey Lake-U Cannon Lake (10 nm) 14 nm <i>TOCK</i> (2017/18) -8xxx (-10xxx for AL-Y R)	Coffee Lake R New microarch. 14 nm <i>TOCK</i> (2018) i3-i9 -9xxx	Comet Lake New microarch. 14 nm <i>TOCK</i> (2019) i3-i9 -10xxx

¹The 8th generation encompasses the following processor lines:

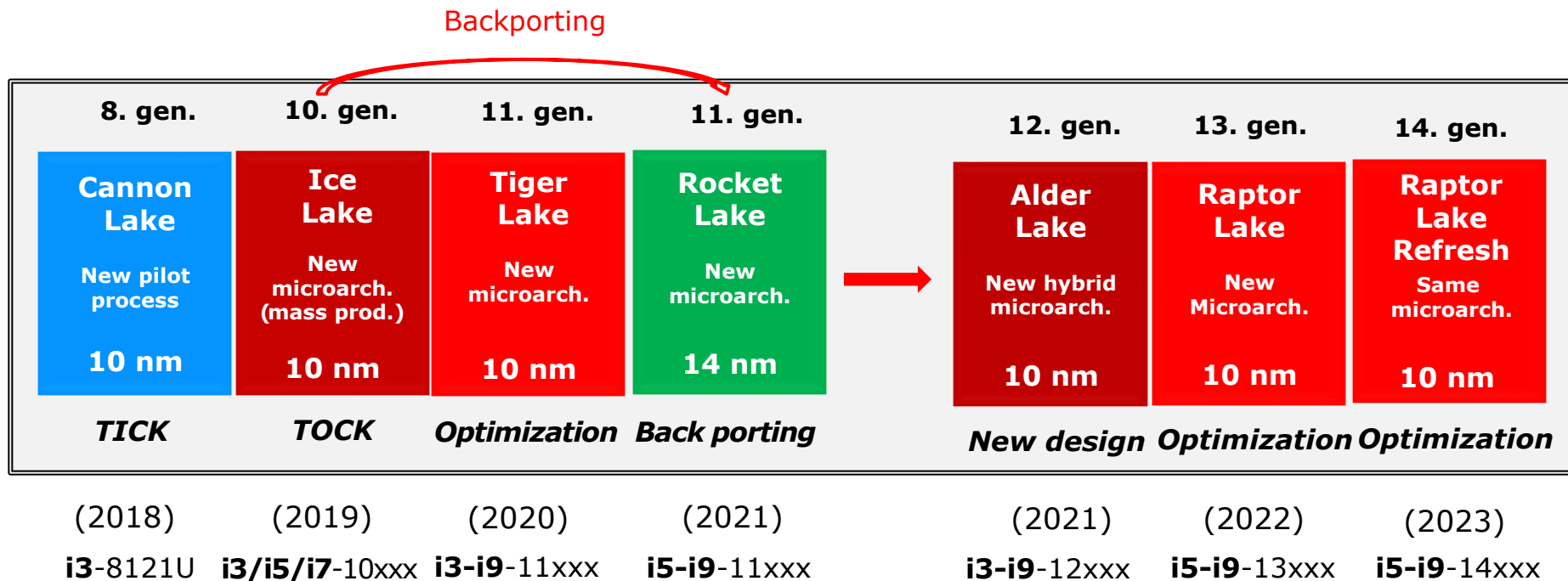
- Kaby Lake Refresh
- Kaby Lake G with AMD Vega graphics
- Coffee Lake
- Amber Lake Y
- Whiskey Lake U
- (all 14 nm) and
- Cannon Lake (10 nm)

lines [29].

R: Refresh

2. Naming of Intel's Core-based client processors (4)

Brand modifiers and model numbers of Intel's Core-based 10-nm client processors



Remarks on discrepancies in the naming of Intel's processor generations

- Note that Intel named **both** its 14 nm Comet Lake (2019) and the 10 nm Ice Lake (2019) processors as part of the **10th generation**.
- Similarly, the company assigned the **11th generation label** to **both** its 10 nm Tiger Lake (2020) and the 14 nm Rocket Lake (2021) lines.
- In addition, Intel has confusingly labeled **both** the **Raptor Lake Refresh** and **Meteor Lake** lines as **14th-generation** (the latter not indicated in the Figure above).

2. Naming of Intel's Core-based client processors (5)

The product line suffix of Core-based client processors up to the 10 nm lines

- It is the final part of the third segment, it indicates both the **target market sector** and the associated **TDP** (Thermal Design Power) of the processor, as outlined below.

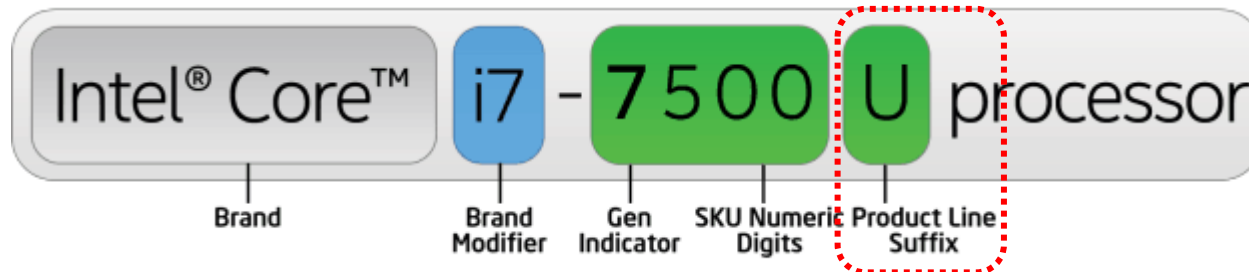


Figure: Naming scheme of client processors in Intel's Core family [43]
(used from the 2nd gen. Sandy Bridge line until the 14th Core generation (2011-2023)

- Most frequently used client suffixes and related TDP values:**

Mobile suffixes (SoC: System on Chip)

Y: ultra-low-power (11.5/13 W), when regular naming (i3/i5/i7-nxxxY) is used or extreme-low-power (4.5/5 W, when irregular naming **m3/m5/m7-nYxx**) is used

U: low power (typically 15/28 W)

H: high-performance laptop (typically 35/45 W)

Desktop suffixes

T: power-optimized models (ultra-low power with 35/45 W TDP)

S: performance-optimized models (low power with 65 W TDP)

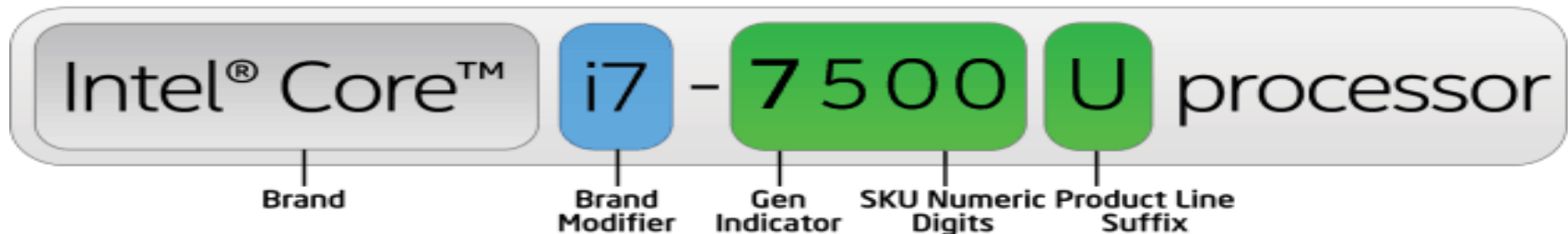
K: unlocked (adjustable clock multiplier (60-95 W))

2. Naming of Intel's Core-base client processors (6)

b) Intel's new naming scheme for the Core Ultra series (introduced in 2023) [44]

- With the introduction of the 14th gen. Meteor Lake Refresh line, Intel revised its client processor naming scheme, replacing the format it had used for over 10 years.

For a comparison, we show both naming schemes, first the original format.



The new naming scheme is, as shown below:



- It consists further on of three parts:
 - 1) the brand name
 - 2) the brand level (called brand modifier previously) and
 - 3) the processor number

however, they have modified interpretations, which will be described in the following.

2. Naming of Intel's Core-base client processors (7)

b1) Interpretation of the new brand names

There are **two differences** compared to the original naming scheme:

- Intel **interprets** now the **original "Intel Core" brand** as the **mainstream-oriented brand**,
- in addition, the company introduces a **new – premium brand**, and designates it as the **"Intel Core Ultra" brand**.

2. Naming of Intel's Core-base client processors (8)

b2) Interpretation of the brand levels

- The **brand levels** (called previously brand modifier) refer further on to the **performance level** of the processor concerned.

However, in the new naming system they **no longer include the letter "i"**- i.e. instead of i3, i5 etc. Intel uses now **only 3, 5.** etc.

- **Mainstream**-oriented "**Core processors**" may have the brand levels **3/5/7**, whereas
- **Premium** "**Core Ultra processors**" may have the brand levels **5/7/9**.

2. Naming of Intel's Core-base client processors (9)

b3) Interpretation of the processor numbers [45]

The 3rd naming segments, called **Processor numbers**, continue to have **three parts**, as shown to the right.

These are: the **Series identifier**, the **model number** (SKU: Stock Keeping Unit) and the **Suffix**, as detailed next.

Series identifier: It disrupts the initially introduced generation numbering, instead, it introduces a new **Series sequencing**, beginning with **Meteor Lake** as **Series 1**.

SKU#: **Model number**; it only includes **two digits**.

Suffix: It relates to the **market segment** and the **base TDP** value, as before.

Most frequently used client suffixes and related TDP values:

Mobile suffixes

Y: ultra-low-power (9 W)

U: low power (typically 15 W)

V: low-power (typically 17/30 W)

H: high-performance laptop (typically 28/45 W)

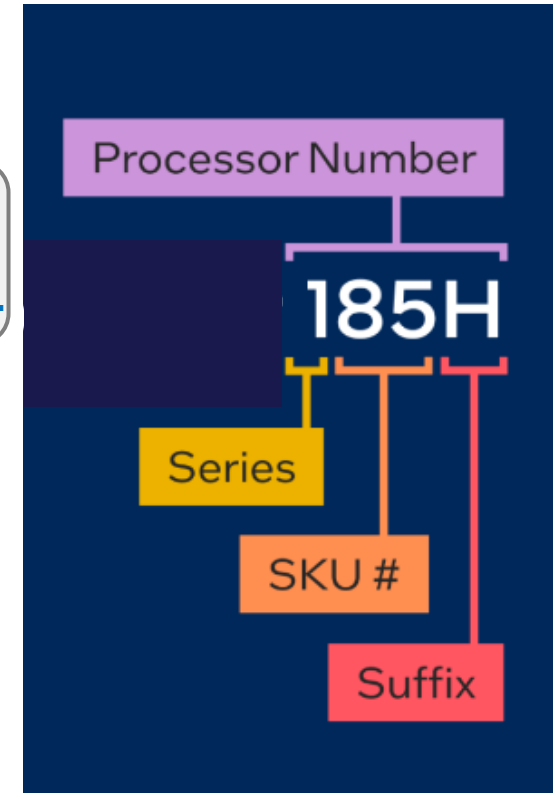
HX: extreme performance (typically 65 W)

Desktop suffixes

T: power-optimized models (35 W)

K: unlocked (adjustable clock multiplier (125 W)

KF: unlocked without a GPU (adjustable clock multiplier (125 W)



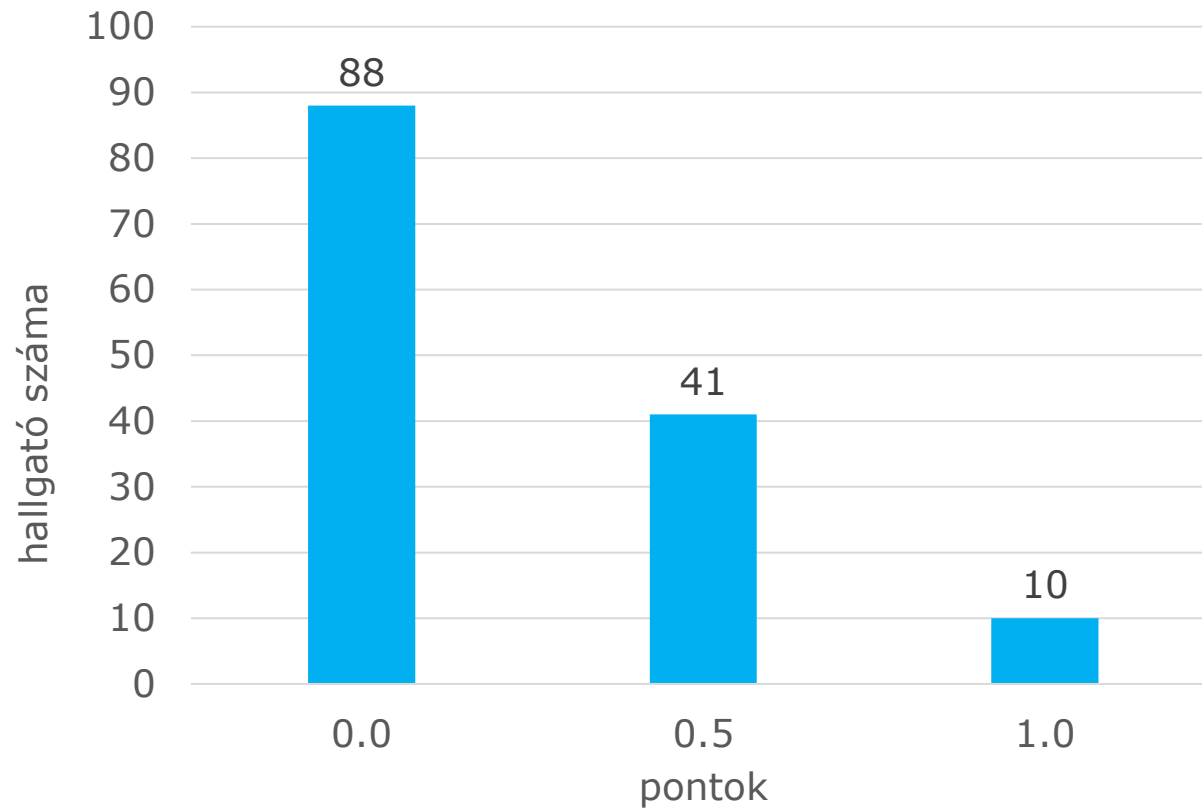
2. Naming of Intel's Core-base client processors (10)

Remark

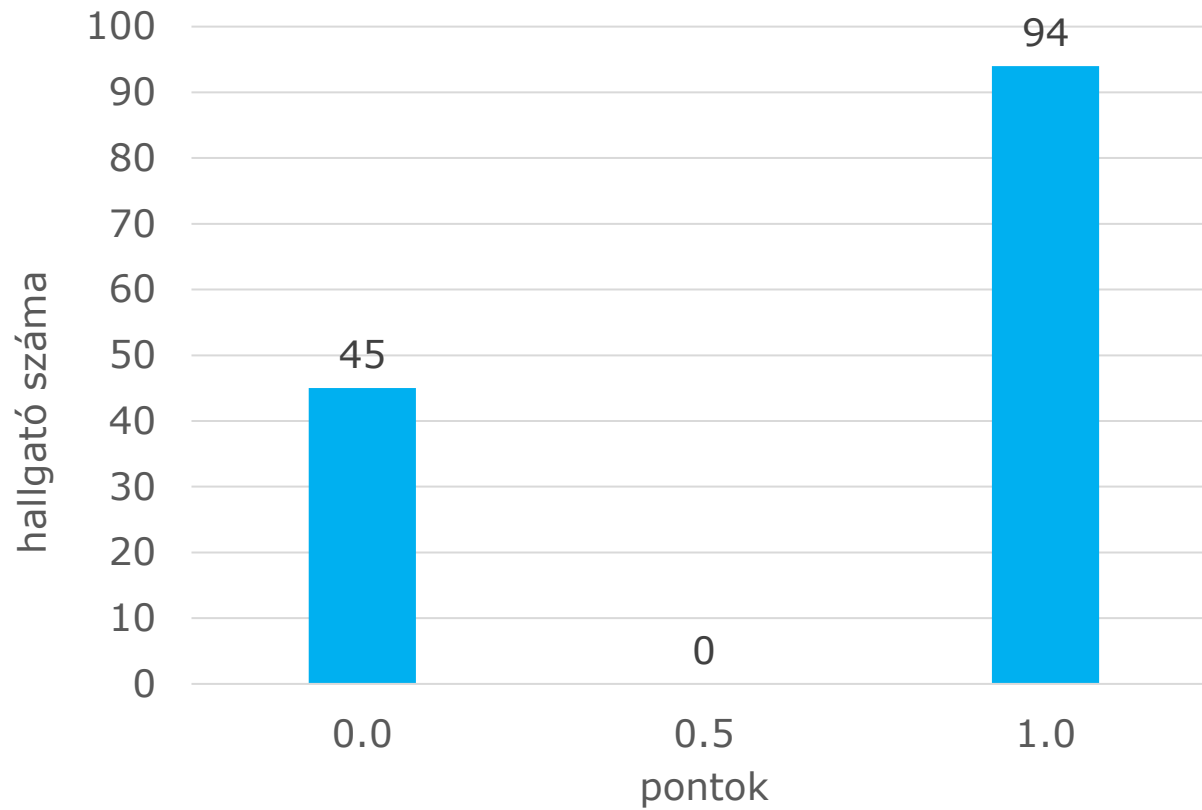
It is worth noting that the new TDP values are generally similar to those of previous ones, however, the HX and K/KF suffixes correspond to higher TDP ratings than before, reaching up to 15/28 W and 125 W respectively.

Milyen előnyökkel járnak a szimmetrikus többmagos processzorok?

Egyszerűsítik a magok ütemezését [1/2 pont] és disszipáció kezelését [1/2 pont]

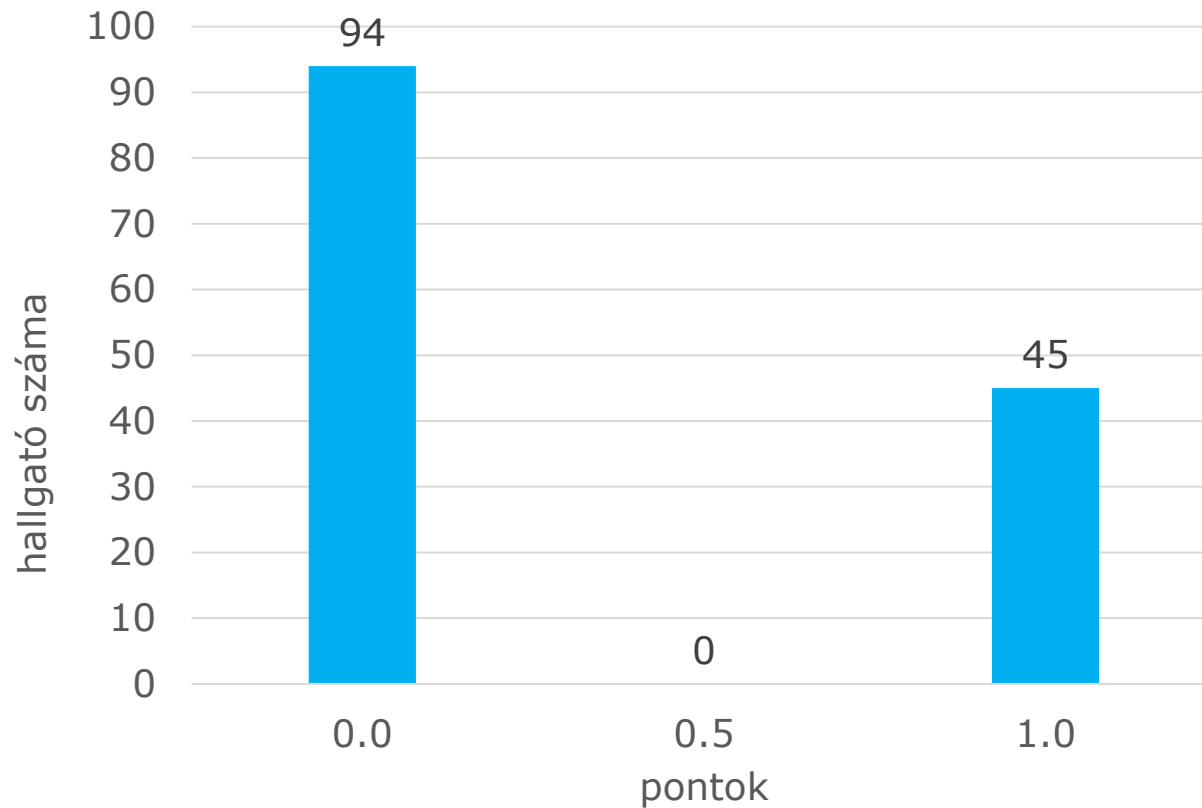


Milyen céllal vezette be Intel 12. generációs Alder Leake családjában a big.little architektúrákat?
Fogyasztás csökkentése céljából.

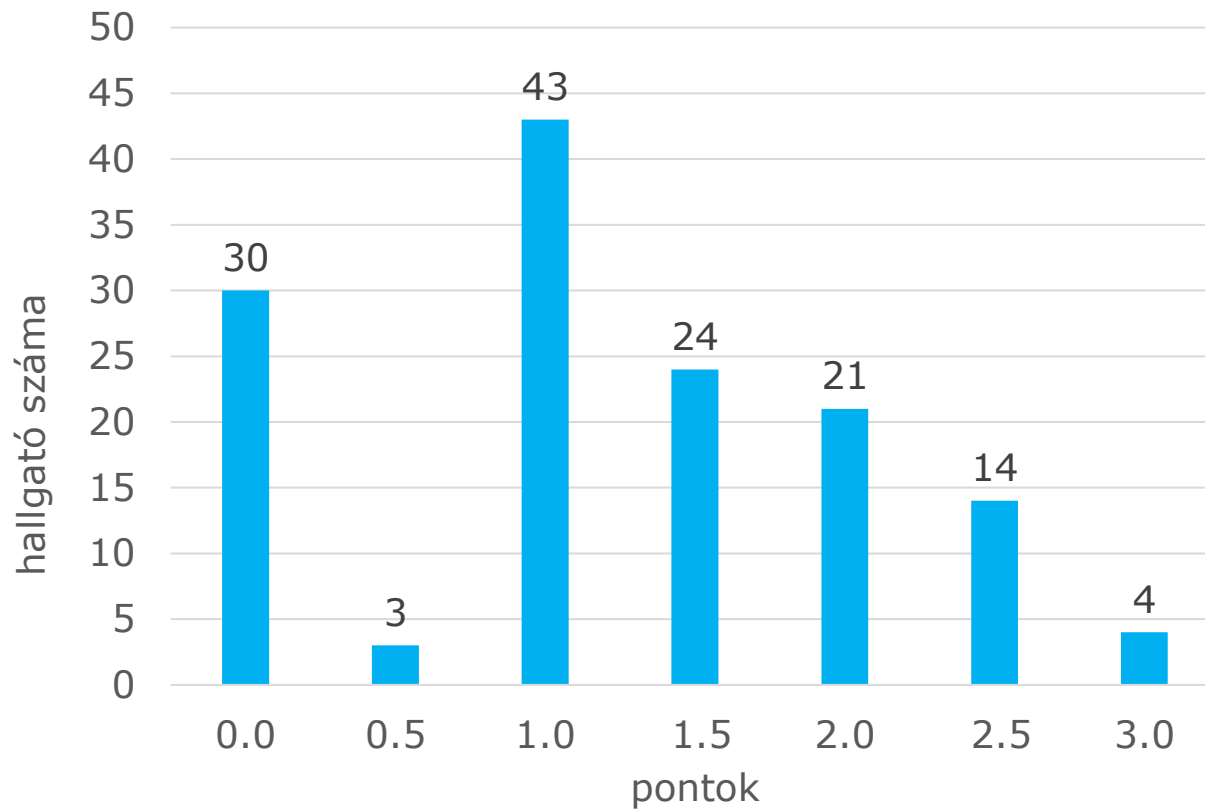


14. generációs Core családját követően, Intel milyen céllal vezette be kliens processzorai heterogén többlapkás kialakítását?

Gyártási költségek csökkentése érdekében



Három kérdés – összesített eredménye



2. Naming of Intel's Core-base client processors (11)

Mainstream
processors

Intel® Core™ 7 processor 150U

Brand

Brand Level

Processor Number

Series

SKU #

Suffix

Premium
processors

Intel® Core™ Ultra 9 processor 185H

Brand

Brand Level

Processor Number

Series

SKU #

Suffix

Raptor Lake Refresh
U series 01/2124

2. Naming of Intel's Core-base client processors (12)

Using the new naming scheme of client processors beginning with the 14th Core generation (called Meteor Lake)

Multi-die heterogeneous big. little designs

(14 th gen.)				
Core Ultra Series 1	Core Ultra Series 2	Core Ultra Series 2	Core Series 2	Core Ultra Series 3
Meteor Lake New process and μ arch. Intel 4 TSMC N5/N6	Lunar Lake New process and μ arch. TSMC N3/N6	Arrow Lake New μ arch. TSMC N3/N5/N6	Meteor Lake R. New μ arch. Intel 4 TSMC N5/N6	Panther Lake New process new μ arch. Intel 18A/3 TSMC N3/N6
4 tiles (12/2023) 100U/H	2 tiles (09/2024) 200V	4 tiles (10/2024) 200K (01/2025) 200H/HX/T	4 tiles (01/2025) 200U	3 tiles (01/2026) 300H
(Laptops)	(Low-Power Laptops)	(Laptops/DTs)	(Laptops)	(Laptops)

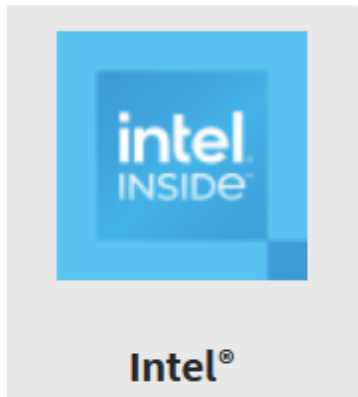
Remark

There is an **inconsistency** in Intel's naming scheme, as follows:

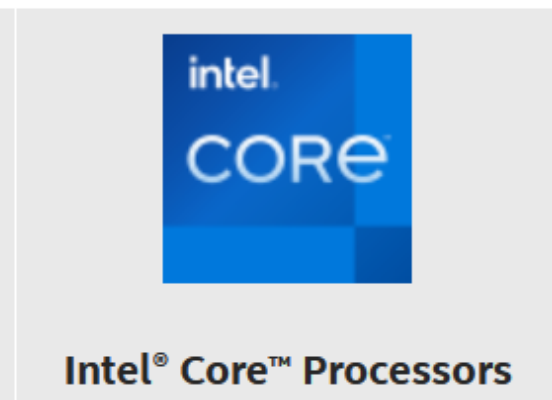
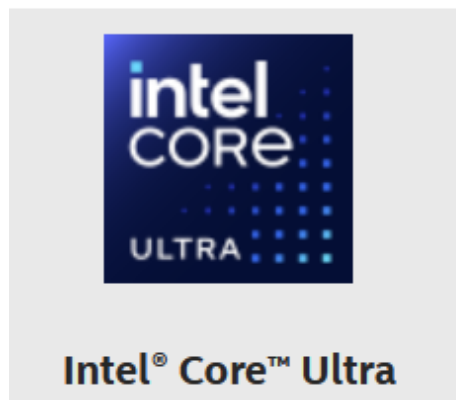
In 01/2014, Intel introduced the **Raptor Lake Refresh U-series DT processors** and **retroactively applied its new naming scheme to it**, labelling it as **Core Ultra Series 1**.

2. Naming of Intel's Core-base client processors (13)

Previous and recent badges of Intel's Core processors [45], [44]



The previous badge



Renewed badges (since the 14th gen.)

3. Key enhancements in the microarchitecture of Intel's Core-based client processors over time

- 3.1 Overview of the key enhancements of the microarchitecture
- 3.2 Evolution of the CPU core configuration and introduction of multi-die designs
- 3.3 How are the cores of big.little architectures scheduled?
- 3.4 How are Intel's recent multicore processors packaged?
- 3.5 How has the width of the CPU cores increased?
- 3.6 How has the core count in Intel's client CPUs increased?
- 3.7 How has the memory subsystem been enhanced?
- 3.8 Basics of 3D graphics processing

3.1 Overview of the key enhancements in the microarchitecture (1)

3. Key enhancements in the microarchitecture of Intel's Core-based client processors

3.1 Overview of the key microarchitectural enhancements

Processor **microarchitectures** can be enhanced in **two different aspects**, as shown below:

Key microarchitectural enhancements in Intel's Core-based client processors

Key enhancements made
to increase processor performance

Key enhancements made
to improve power management
(Not discussed)

- For **processor performance**, **CPU cores** play a crucial role, particularly regarding single-core or single-thread performance.
- For this reason, it is appropriate to **discuss improvements in CPU core designs separately** from those made in other processor components, as indicated below.

Key enhancements made to increase processor performance

Key microarchitectural enhancements
made to the **CPU cores**

Further microarchitectural enhancements
made to the **processor**

- The following Sections outline these improvements made to boosting processor performance.

3.1 Overview of the key enhancements in the microarchitecture (2)

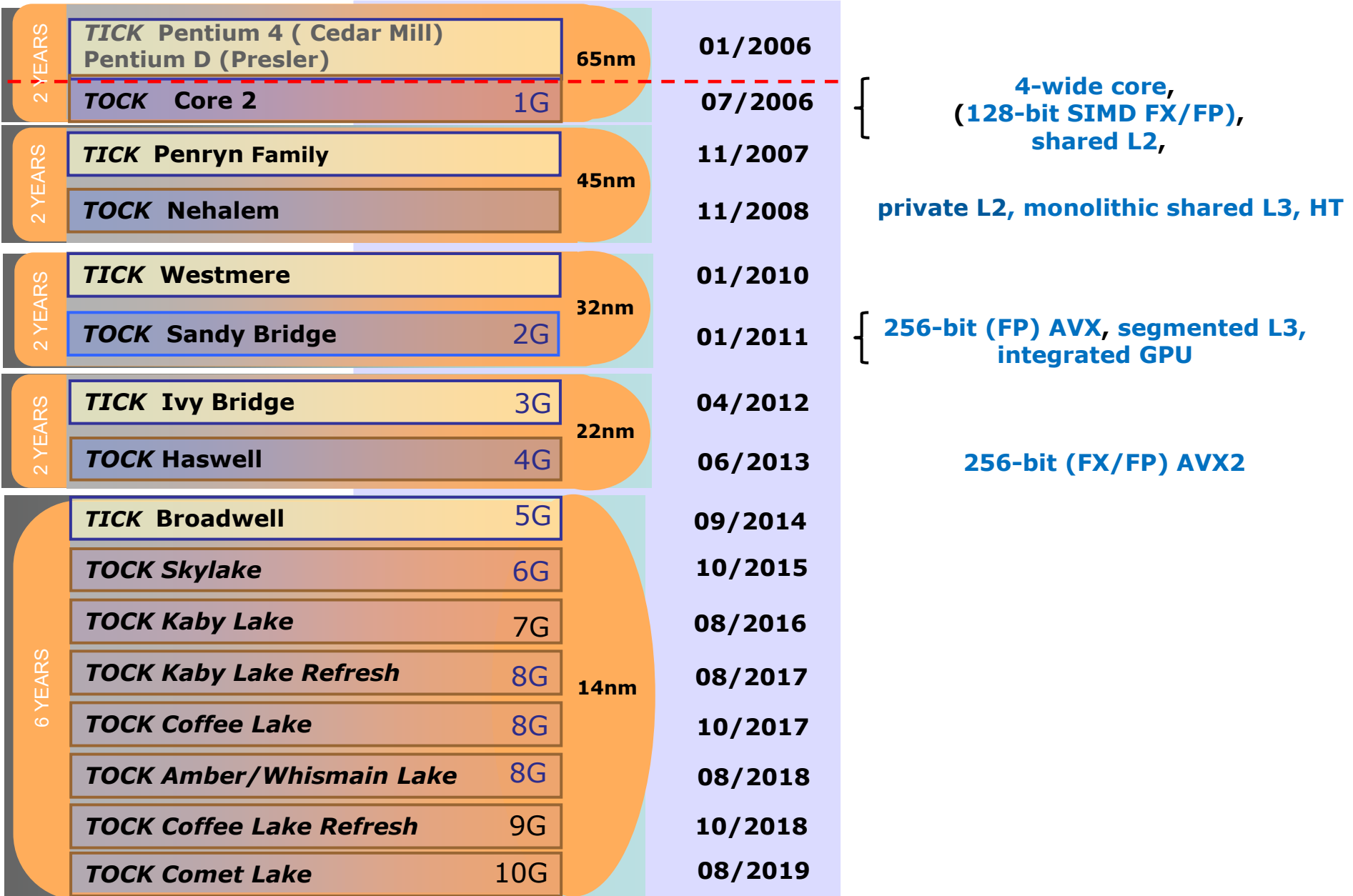
What key microarchitectural enhancements have been made to CPU cores?

- *Increasing the CPU core width (from 4 to 9)*
- *Increasing the back-end width of CPU cores (from 5 to 17)*
- *Shared L3 cache with private per-core L2 caches*
- *Segmented L3 cache*
- *Hyperthreading*
- *Widening the SIMD execution width (from 128-bit, to 256-bit and finally to 512-bit).*

The next two Figures show [when Intel introduced the above-mentioned enhancements](#).

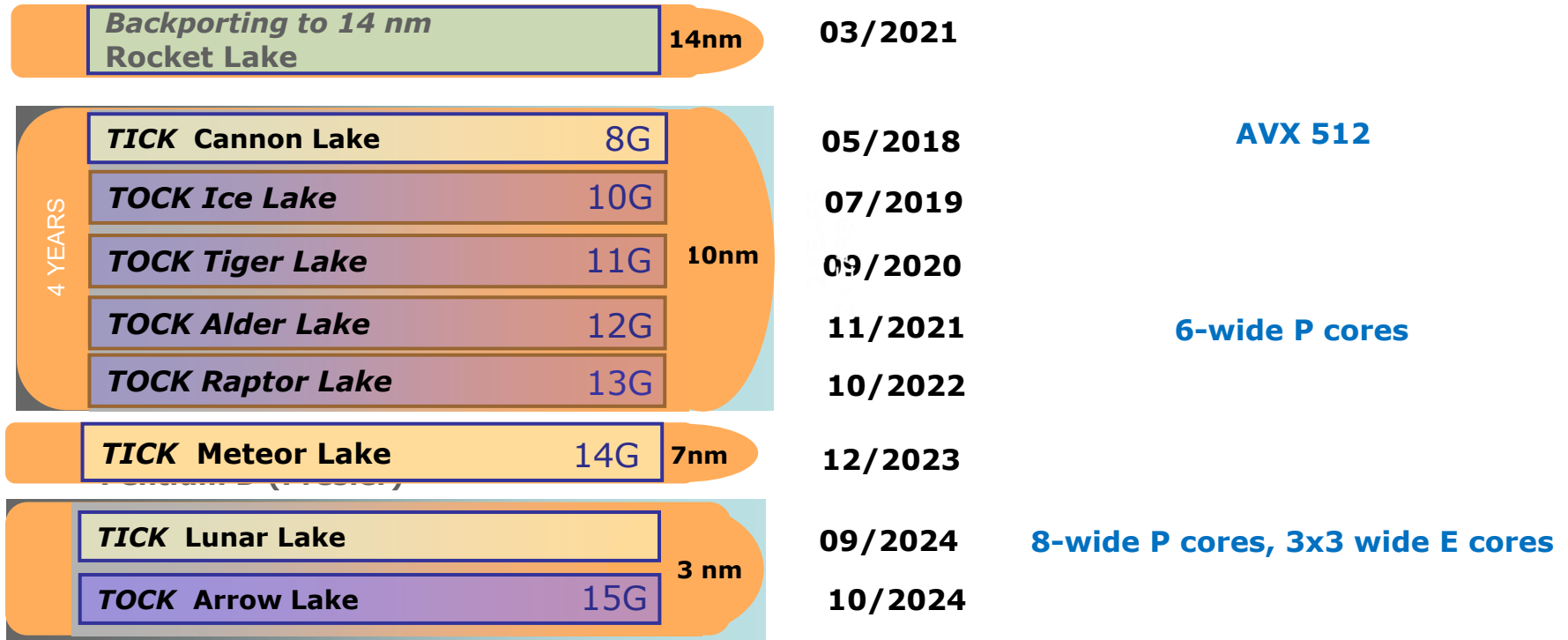
3.1 Overview of the key enhancements in the microarchitecture (3)

Key microarchitectural enhancements of the CPU cores -1



3.1 Overview of the key enhancements in the microarchitecture (4)

Key microarchitectural enhancements of the CPU cores -2



These enhancements focus on boosting performance.

3.1 Overview of the key enhancements in the microarchitecture (5)

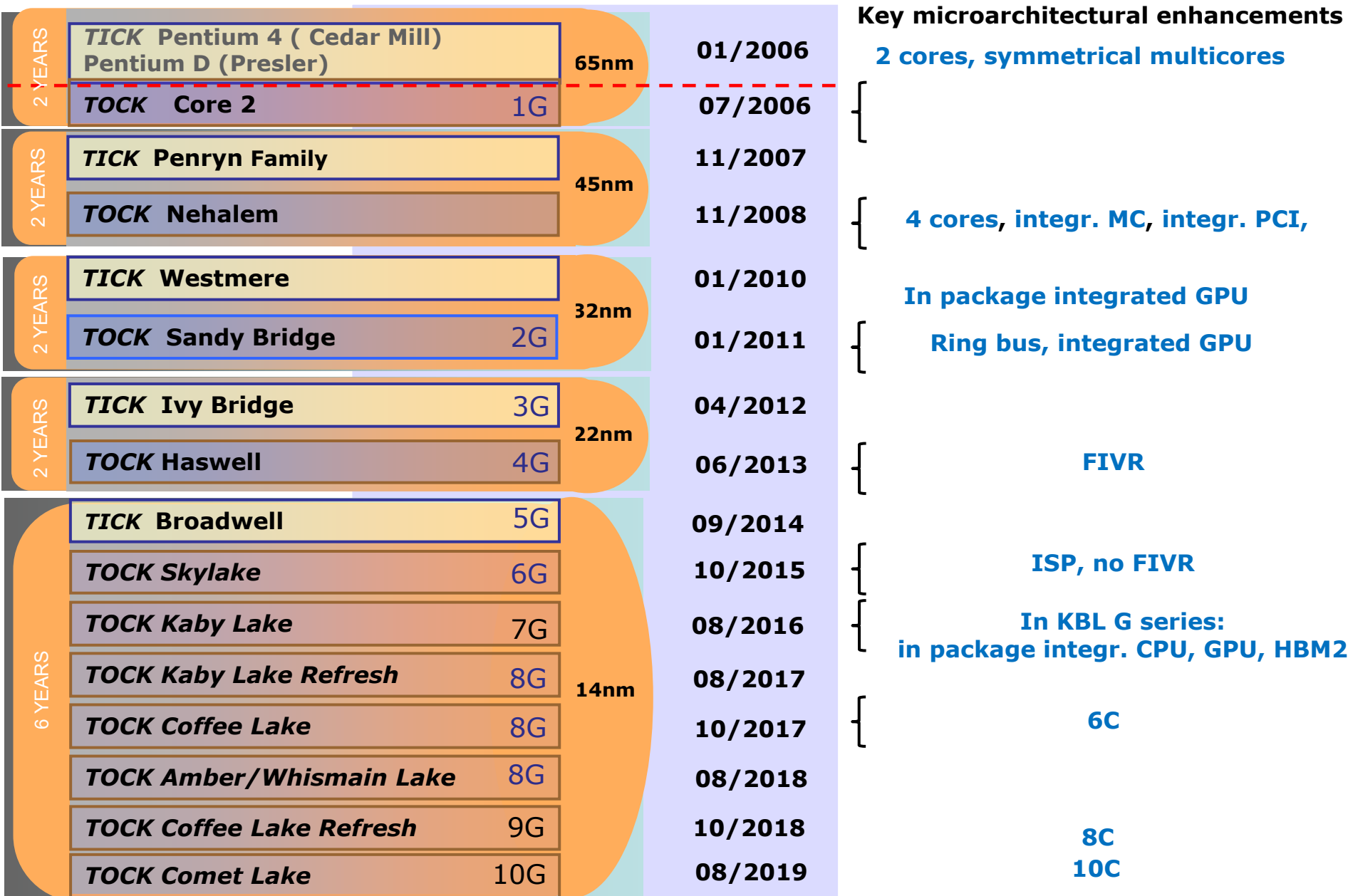
What further microarchitectural enhancements have been made to the processors?

- *Evolution of the CPU core configuration*
- *Increasing the core count (from 2 up to 24 in laptops/DTs)*
- *Introducing the heterogeneous multi-die approach*
- *Ring bus interconnect*
- *Integrated memory controller (NUMA architecture)*
- *Integrated PCI controller*
- *Integrated ISP (Signal Processor)*
- *Integrated GPU*
- *Integrated FIVR (Fully Integrated Voltage Regulator)*
- *Integrated NPU (Neural Processing Units)*
- *Enhancing the memory subsystem*
- *Enhancing the PCIe subsystem*

The next two Figures show [when Intel introduced the above-mentioned enhancements](#).

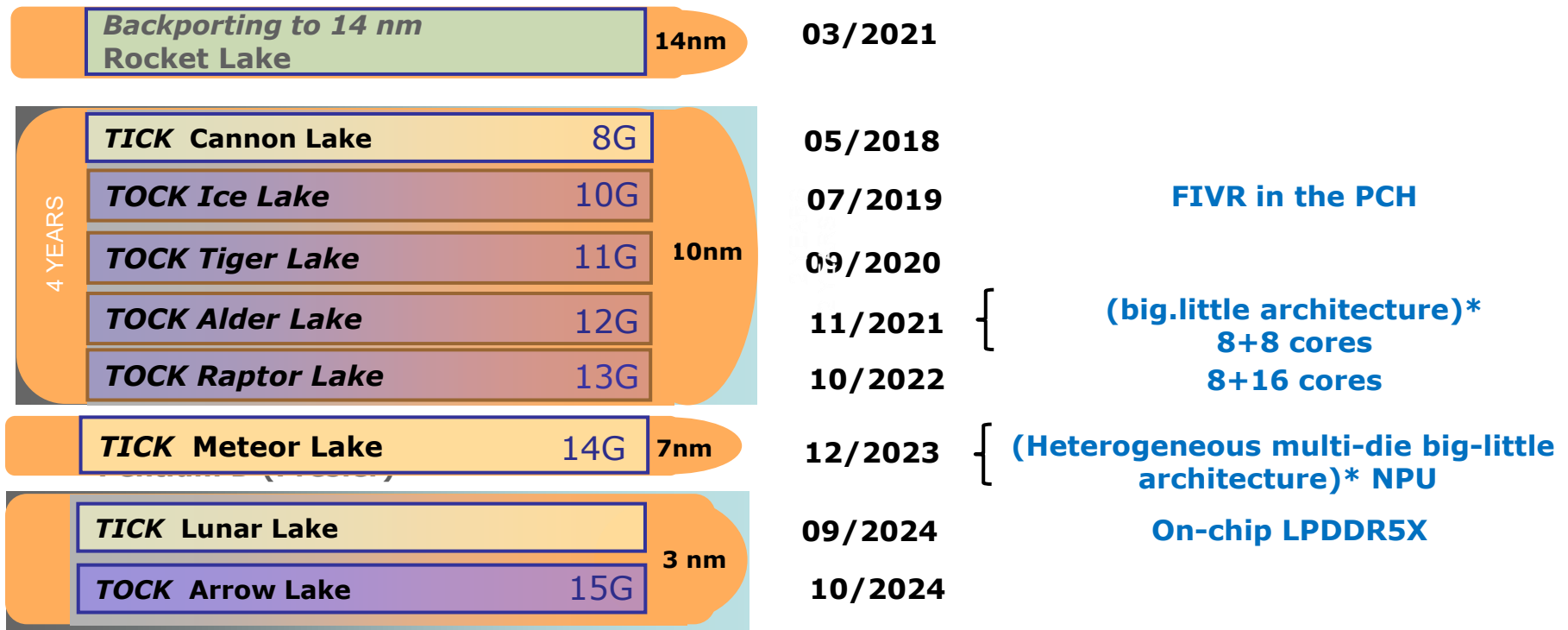
3.1 Overview of the key enhancements in the microarchitecture (6)

Introducing further key microarchitectural enhancements to the processors -1



3.1 Overview of the key enhancements in the microarchitecture (7)

Introducing further key microarchitectural enhancements to the processors -2



*These enhancements focus on other aspects than boosting performance.

Particularly, big.little architectures aim to reduce power consumption, while multi-die (heterogeneous) implementations seek to lower manufacturing costs.

3.1 Overview of the key enhancements in the microarchitecture (8)

Evolution of selected microarchitectural features in Intel's Core-based client processors

- It would be tempting to trace the dynamic evolution of the microarchitecture in the Core family across all the enhancements mentioned, but space constraints make that impractical.
- Instead, we'll focus on a few **key enhancements**, which highlight the major trends in CPU core design.

The selected design aspects discussed in the following Sections are:

3.2 Evolution of the CPU core configuration and introduction of multi-die designs

3.3 How are the cores of big.little architectures scheduled?

3.4 How are Intel's recent multicore processors packaged?

3.5 How has the CPU core width increased?

3.6 How has the core count in Intel's client CPUs increased?

3.7 How has the memory subsystem been enhanced? and

3.8 Basics of 3D graphics processing.

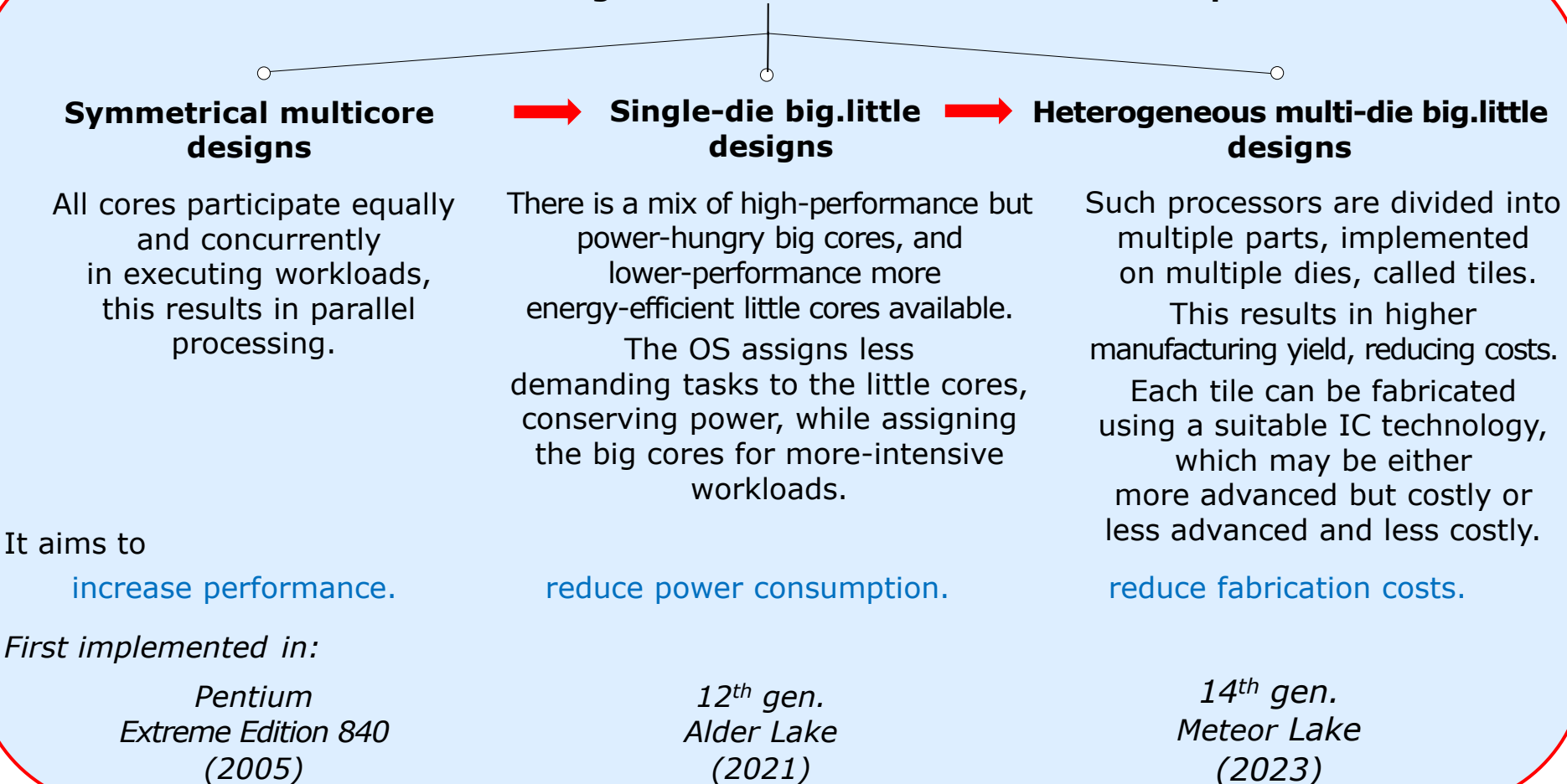
3.2 Evolution of the CPU core configuration and introduction of multi-die designs

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (1)

3.2 Evolution of the CPU core configuration and introduction of multi-die designs -1

The CPU core configurations and the multi-die approaches have evolved in Intel's Core family through **three consecutive steps**, as shown below.

Evolution of CPU core configurations in Intel's Core-based client processors



3.2 Evolution of the CPU core configuration and introduction of multi-die designs (2)

Evolution of the CPU core configuration and introduction of multi-die designs -2

The above-mentioned evolution steps will be discussed next.

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (3)

a) The 1st evolution step: symmetrical multicore designs (2006-2021)

- From the very beginning up to the 11th generation Core processors, which were released in 2021, Intel's CPUs were designed as symmetrical multicores. In these designs, all cores participate equally and concurrently in executing workloads.
- Obviously, the primary goal of symmetrical multicores is to increase performance through parallel processing.

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (4)

Intel's first symmetrical multicore design within the Core family: the Core 2 (released 2006)

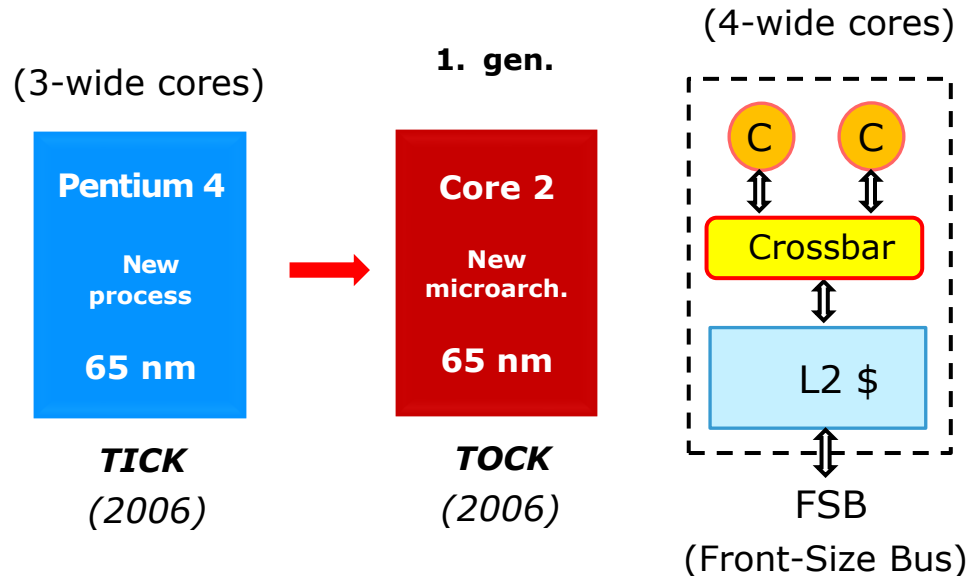
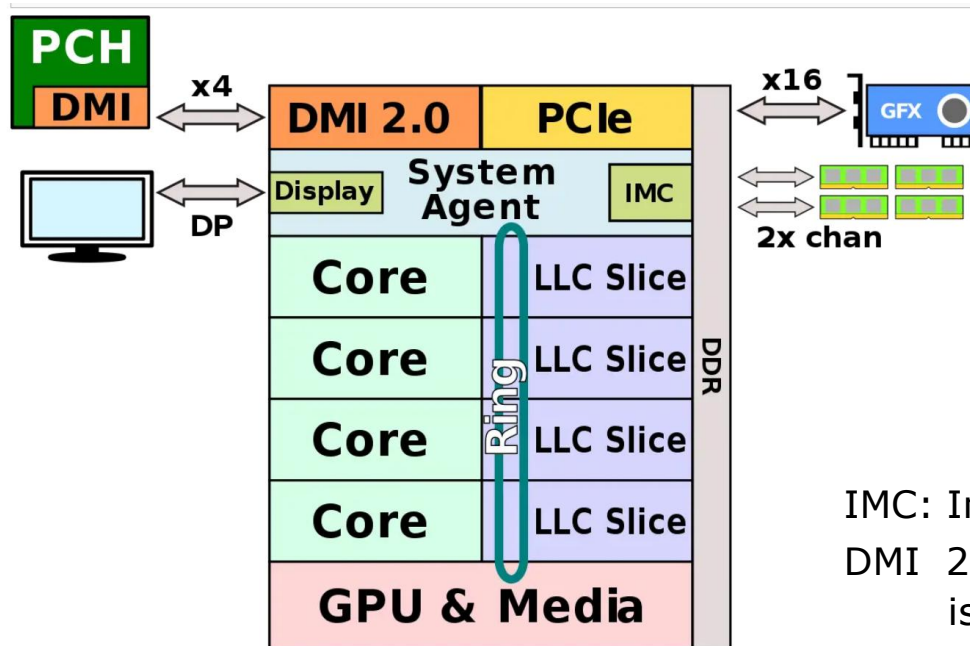


Figure: simplified block diagram of Intel's Core 2 processor

- The **Core 2** was a **4-wide** superscalar design, including **two cores** which were **connected** via a **crossbar switch** to the **1 MB shared L2 cache**.
- The L2 cache was linked to the **system bus**, known as the **Front-Side Bus (FSB)** through a bus unit (which is not shown in the Figure above).
- The FSB provided the connection to the **chipset**, which comprised the North Bridge (NB) and the South Bridge (SB) (although these components are not indicated).

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (5)

Example of Intel's advanced symmetrical multicore designs – the microarchitecture of the 2nd gen. Sandy Bridge processor [46]



IMC: Integrated Memory Controller
DMI 2.0 (Direct Media Interface)
is essentially a PCIe 2.0 link.
MU: Media Unit

- 2. gen. Core processors became **ring bus-based** and included a **segmented L3 cache**.
- These designs introduced two key components: a **GPU + MU** and a **System Agent (SA)**.
- The **SA** is responsible for providing **memory accesses** and handling **PCIe-based connections** to
 - the external graphics (GFX) through x16 PCIe 3.0 links,
 - the Platform Control Hub (PCH) via an x4 DMI link and
 - the display using a Display Port (DP) link.

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (6)

The last symmetrical multicore designs – Intel's 10 nm Tiger Lake processors

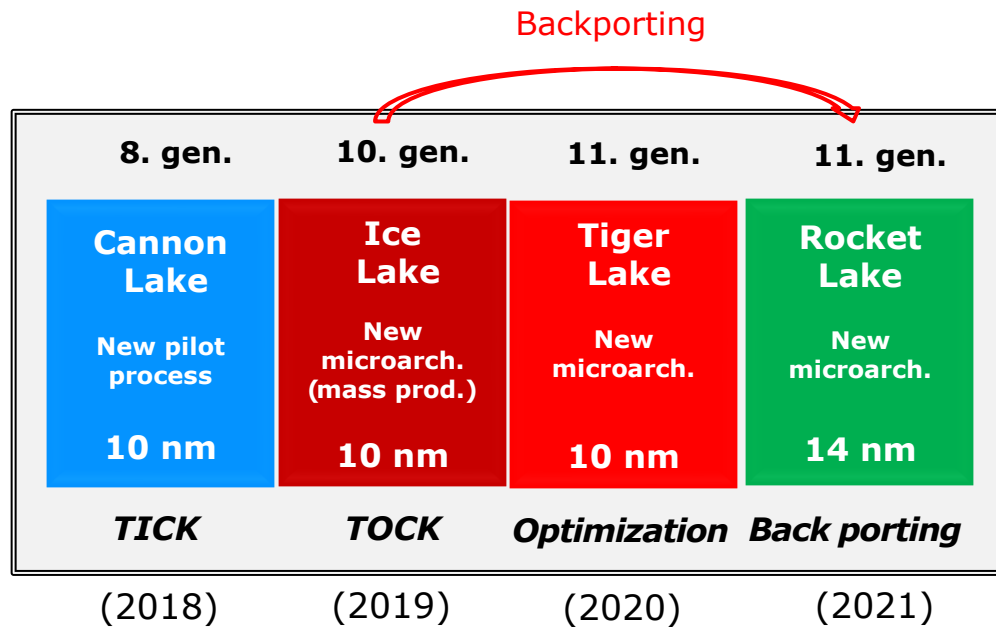
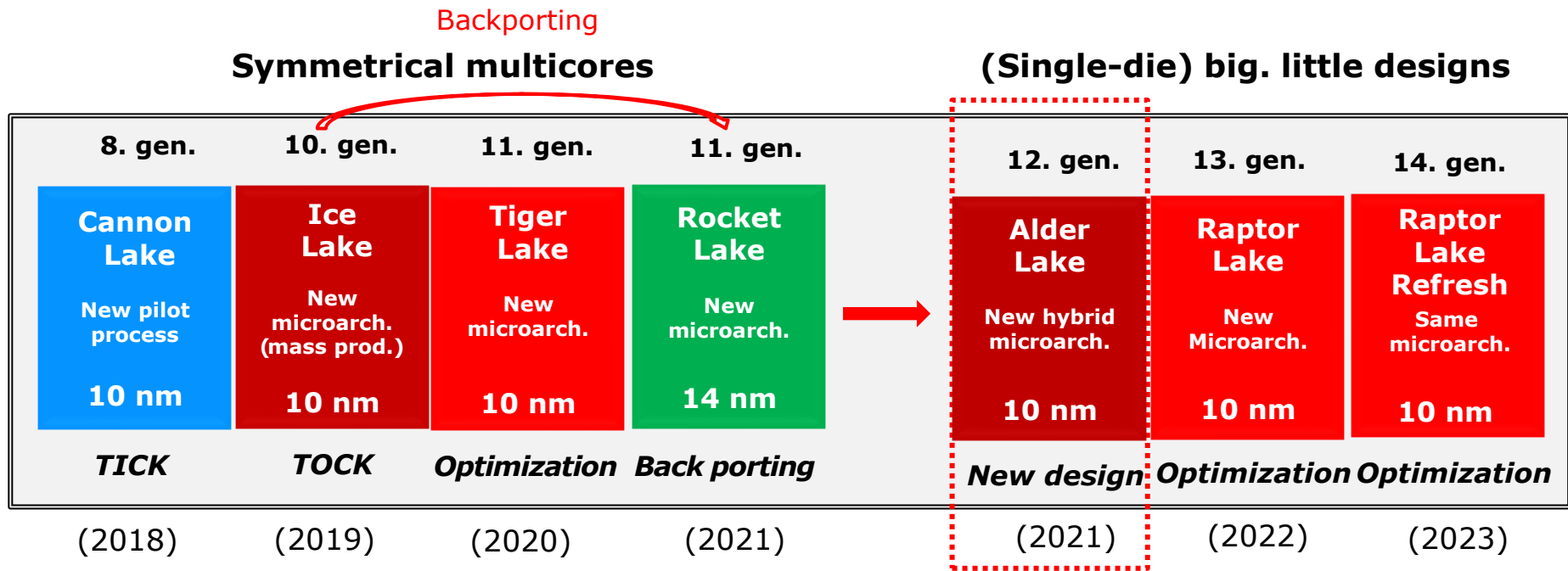


Figure: Intel's last symmetrical multicore designs implemented using the 10 nm process

- Following Ice Lake, Intel launched the performance-optimized **11th gen. Tiger Lake** line.
- Next, Intel released the **Rocket Lake** design, which is a **backported implementation** of the 10th gen. 10 nm Ice Lake line and also got the **11. gen.** label.
- **Backporting** refers to the **implementation** of a given microarchitecture **on an older, less advanced technology**, typically to reduce manufacturing costs.
- Note that confusingly, Intel labeled **both the 10 nm Ice Lake and the 14 nm Rocket Lake processors are 11th-generation products.**

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (7)

b) The 2nd evolution step: introducing big.little processor designs -1



- With the introduction of the 12th gen. Alder Lake lineup, Intel transitioned to big.little (also known as **hybrid**) architectures in 2021 to reduce power consumption.
- Intel's **big.little designs** incorporate **high-performance (P) cores** and **high-efficiency (E) cores** (as detailed next).

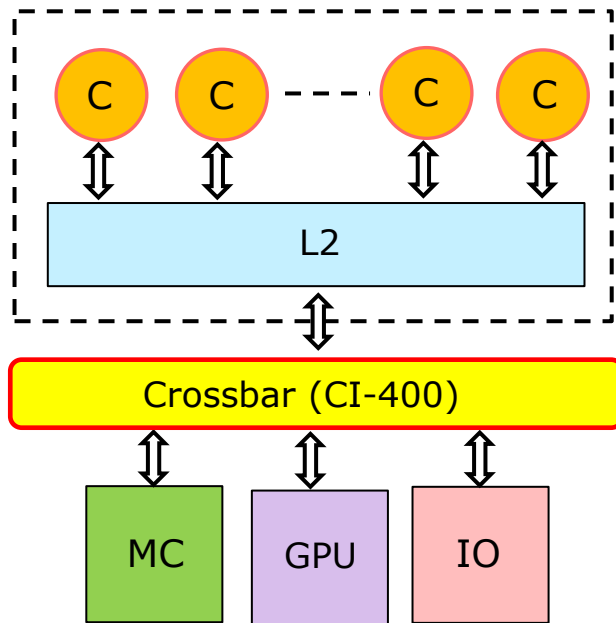
3.2 Evolution of the CPU core configuration and introduction of multi-die designs (8)

Principle of big.little architectures at the time of their introduction by ARM in 2012 (assuming the use of L2 caches and crossbar interconnects)

Use of homogeneous cores or big and little cores in multicore CPUs

Symmetric multicores

They include homogeneous cores..



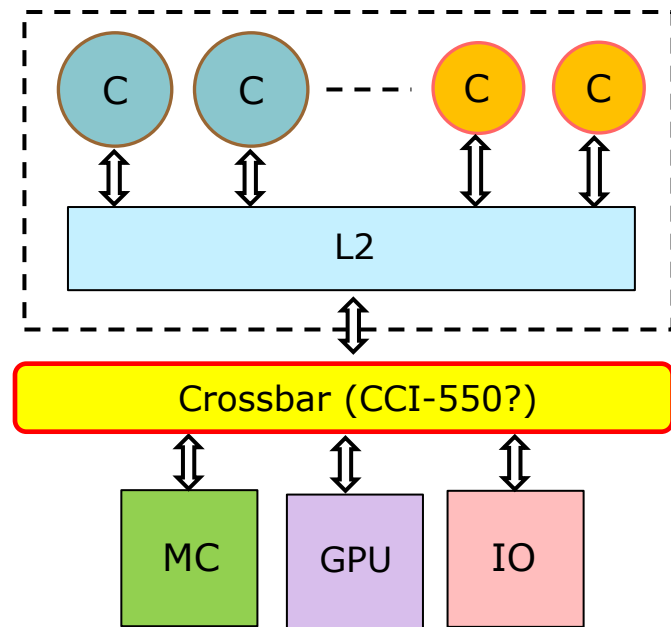
Symmetrical multicore

big.little multicores

They include hybrid cores:

higher performance/higher power consuming **big** cores,
and lower performance/lower power consuming **little** cores.

The OS allocates less demanding tasks to less power-hungry little cores, thereby **reducing the overall power consumption.**



big.little multicore

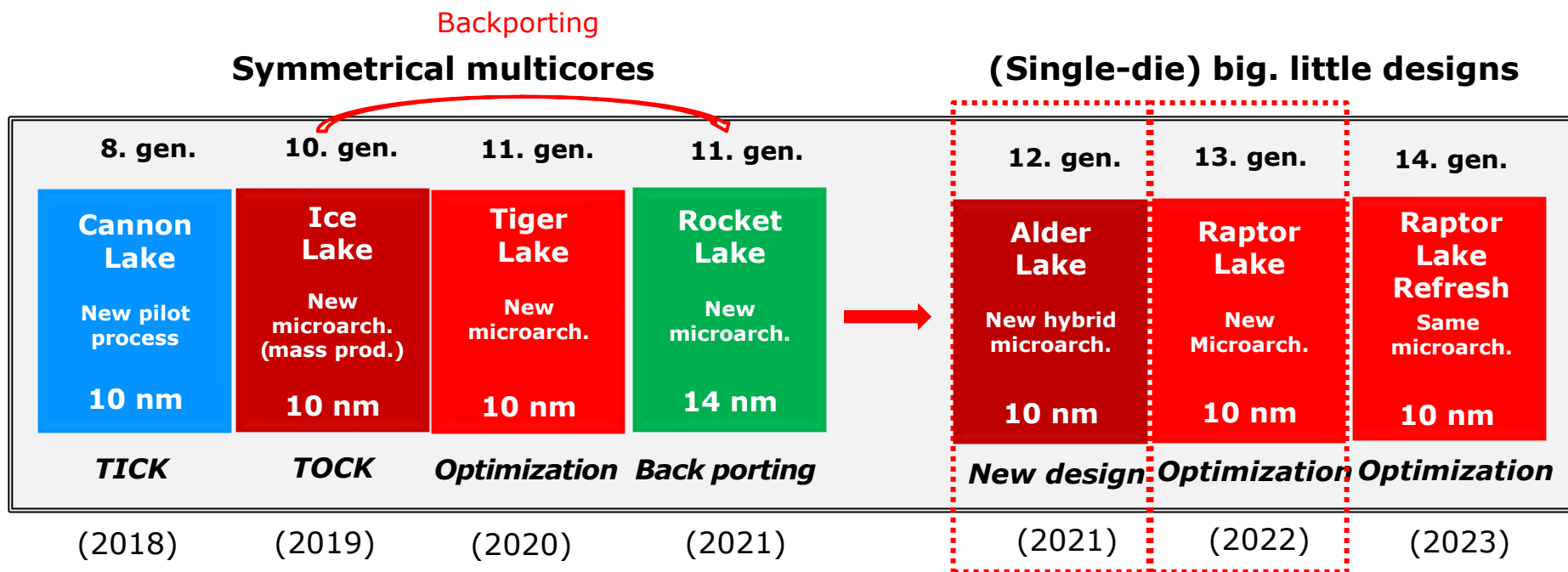
3.2 Evolution of the CPU core configuration and introduction of multi-die designs (9)

Comment on the terms “hybrid” and “heterogeneous”

- In the previous Figure, big.little architectures were classified as **hybrid designs**. However, they are sometimes referred to as **heterogeneous designs** as well.
- Both terms are **closely related** but used in **slightly different contexts**.
- The term **“hybrid”** is more commonly applied to **CPU designs**, emphasizing that the CPU **consists of different types of cores**, which have different characteristics, e.g., there exist both high-performance and high-efficiency cores in the CPU.
- Relating **processor designs**, the term **“heterogeneous”** typically denotes a microarchitecture that incorporates **different types of execution units**, such as CPUs, GPUs, NPU, etc.
- In **multi-die designs**, the term **“heterogeneous”** refers to **implementations using different types of dies** rather than identical ones.

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (10)

The 2nd evolution step: introducing big.little processor designs -2

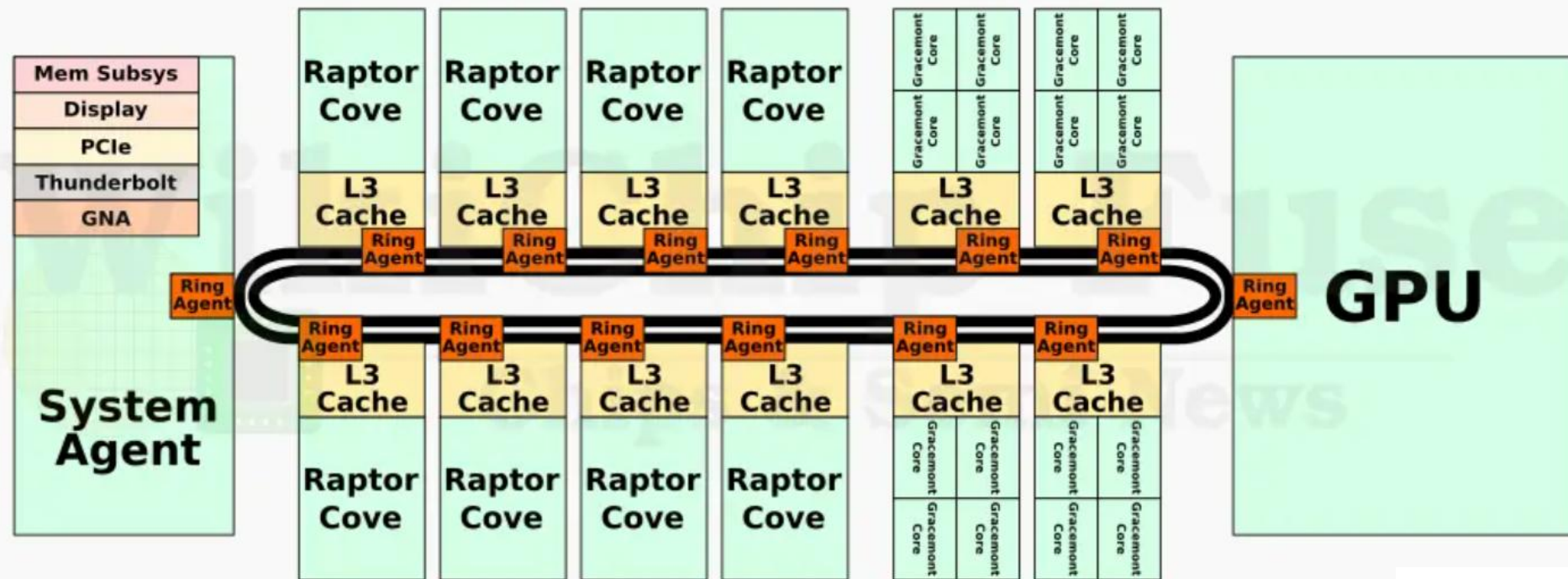


- 12th gen. **Alder Lake** processors, in particular, include up to 8 P (Performance) cores and up to 8 E (Efficiency) cores.
- In contrast, the upcoming 13. gen. **Raptor Lake** line enhances this design by providing up to 8 P cores and up to 16 E cores.

We note that Intel's **server processors** did not transition from symmetrical multicores to **big.little designs**, since server workloads often demand consistent performance across the cores, e.g., database servers.

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (11)

Example: Layout of the 13th gen. Raptor Lake big.little processor featuring 8 big and 16 little cores (2022) [48]

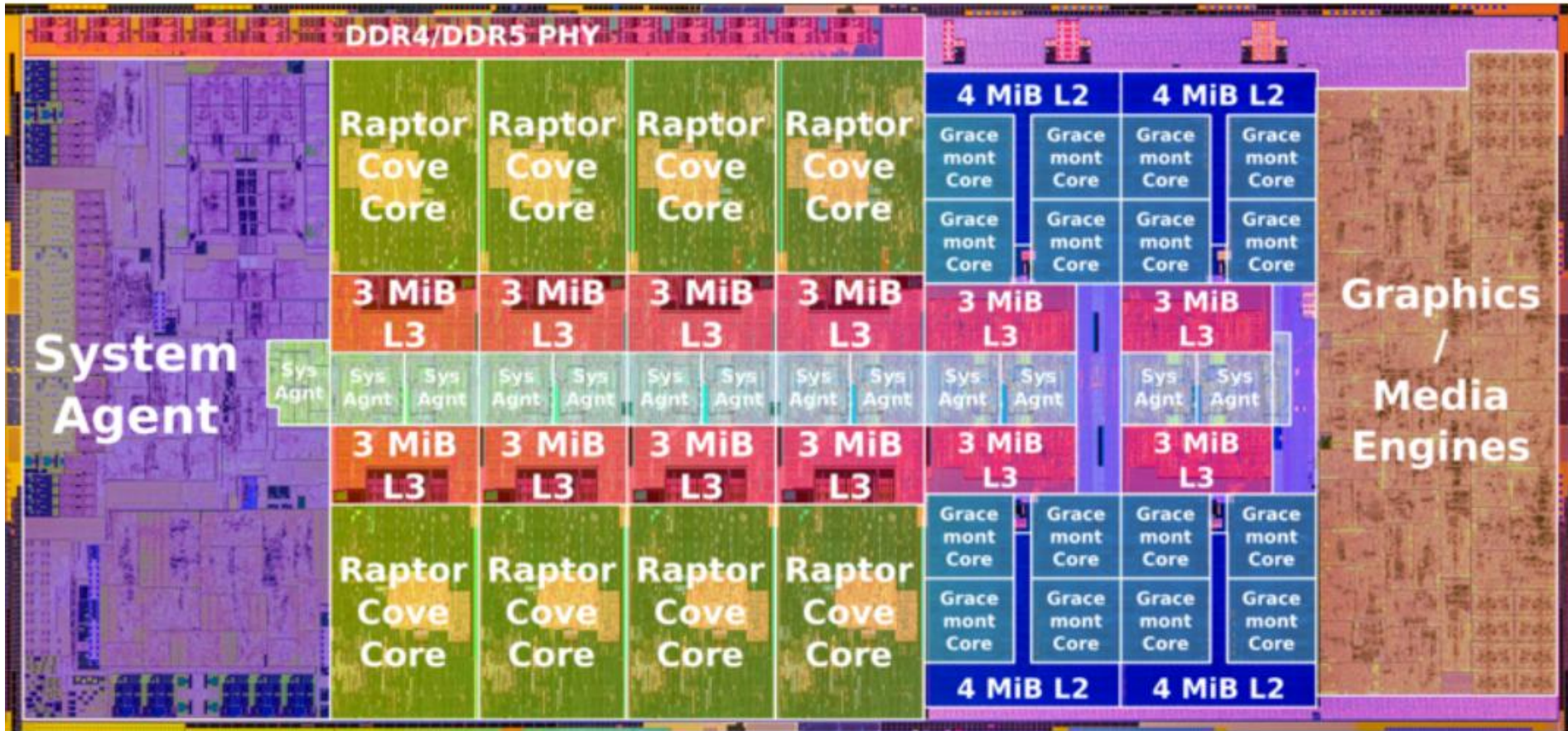


Note that the **little cores** are grouped into **core blocks**.

- There are **four core blocks in total**, each containing **4-cores**.
- Each **core block includes a shared L2 cache** (not shown here, but illustrated in the next Figure), and an **L3 cache segment**.

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (12)

Floor plan of the 13th gen. Raptor Lake big.little processor featuring 8 big and 16 little cores (2022) [49]



The die size is $\sim 215 \text{ mm}^2$

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (13)

Remarks on Intel's first big.little processor implementation

- It's important to clarify that Intel's Alder Lake line is not its first hybrid (big.little) design. Prior to Alder Lake, Intel introduced the Lakefield processor in 2020, featuring one big and four little cores.

Additionally, the Lakefield processor employed a dual-die implementation, rather than a single-die design (22 nm Base die/10 nm Compute die).

- However, this processor was not widely adopted, and did not receive a formal identifier.

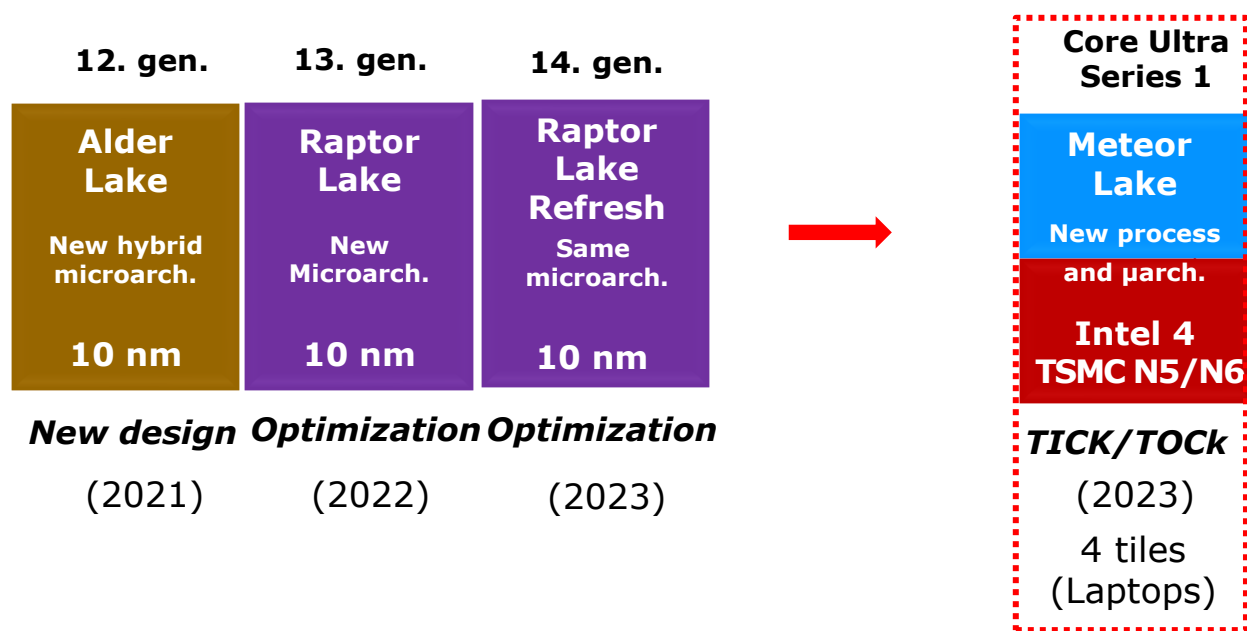
As such, Lakefield is best regarded as a pilot project – a preliminary step toward Intel's hybrid (big.little) designs.

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (14)

c) The 3rd evolution step: introducing heterogeneous multi-die big.little designs -1

- In 2023, Intel began to implement **big.little client processors on multiple dies**, known as **heterogeneous multi-die designs** alongside their 14th gen. Meteor Lake lineup, as indicated below.

(Single-die) big. little designs Heterogeneous multi-die big. little designs



- In **heterogeneous multi-die designs**, the processor is divided into **multiple non-identical components**, known as **tiles**.

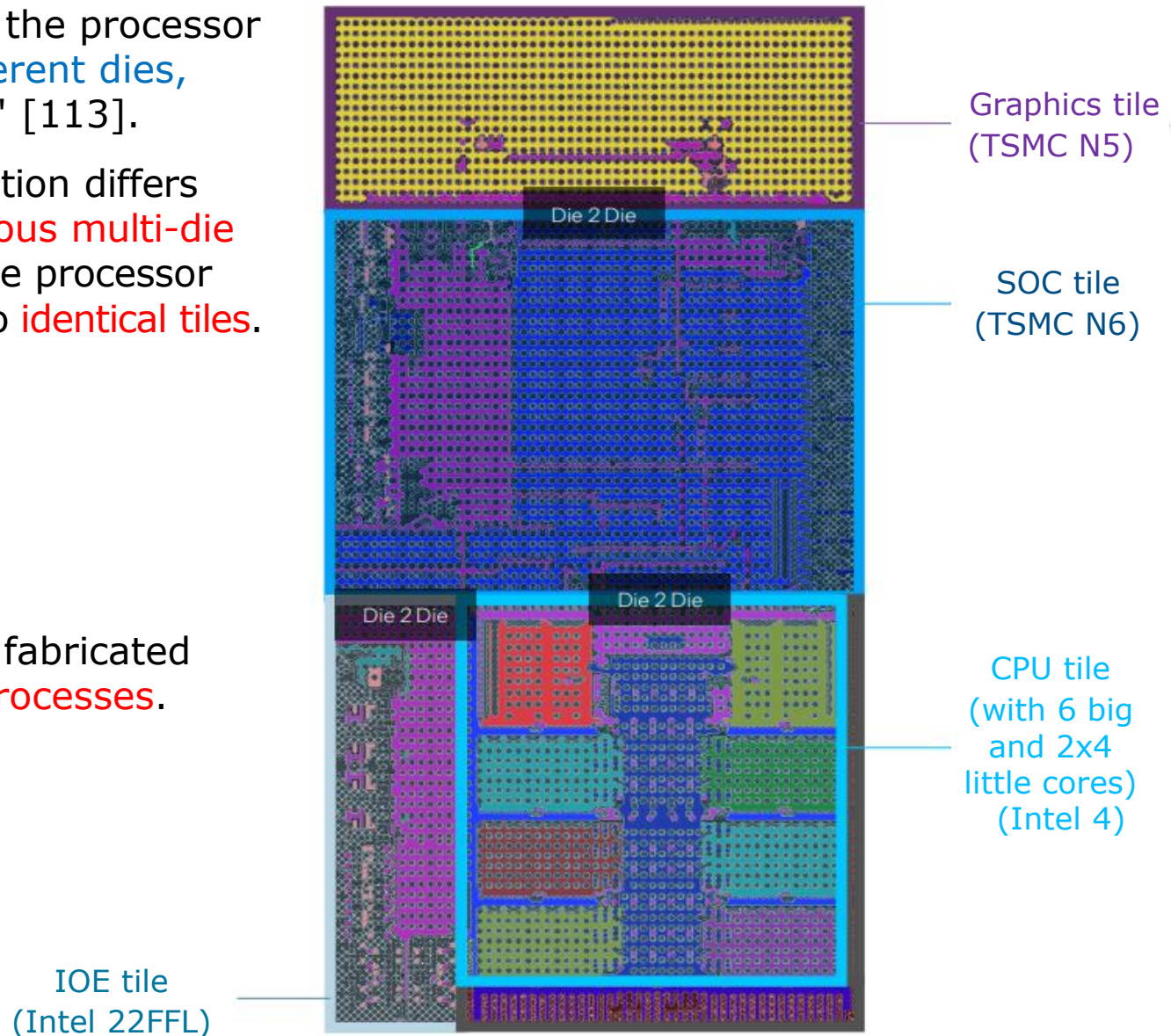
These tiles are then **interconnected** using high-speed interfaces.

- Refer to the Meteor Lake design in the following Figure.

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (15)

The tiles make up the 14th. gen. heterogeneous multi-die big.little Meteor Lake design

- As the Figure shows, the processor consists of **four different dies**, referred to as "**tiles**" [113].
- This type of segmentation differs from the **homogeneous multi-die approach**, where the processor die is segmented into **identical tiles**.
- The four tiles are:
 - CPU tile
 - Graphics tile
 - SOC tile
 - IOE tile
- Note that each tile is fabricated using **different IO processes**.



3.2 Evolution of the CPU core configuration and introduction of multi-die designs (16)

Benefits of heterogeneous multi-die designs compared to single-die (monolithic) designs

- Heterogeneous multi-die designs offer two key benefits.
 - a) the advantage of using multi-die designs over single-die designs, as well as
 - b) the additional benefit of using heterogeneous (multi-die) designs over single-die designs.
- Subsequently, we will discuss both advantages of heterogeneous multi-die designs separately.

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (17)

a) The benefit of multi-die designs over single-die designs -1

- Enlarging dies increases the likelihood of defects, which in turn reduces the yield.

Typically, doubling the die size will cut the yield in half.

A lower yield, on the other hand, will lead to higher selling prices.

- The multi-die approach, which involves dividing a die into multiple smaller dies, can help improve the overall yield, as illustrated in the following Figure. Improving the overall yield, clearly leads to a reduction in fabrication costs.

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (18)

Example of increasing the yield by dividing dies into multiple smaller ones [143]

- In the Figure below, each large die is divided into 4 smaller dies, and there are 7 error spots.

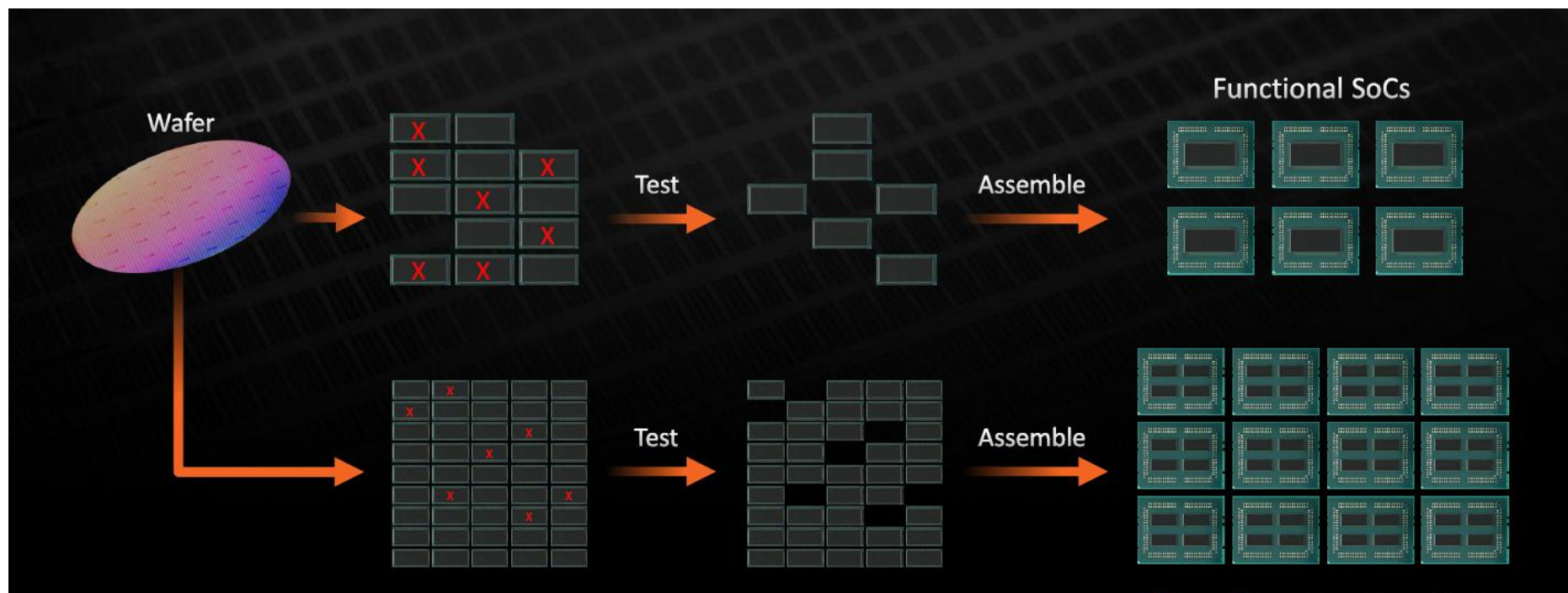


Figure: increasing the yield by dividing dies into multiple smaller ones [143]

- Without segmentation, 7 large dies become defective.
 - However, after segmentation, only 7 smaller dies became faulty.
- Thus, twice as many large dies remain error-free, **effectively doubling the yield**.
- This benefit **applies to both homogeneous and heterogeneous multi-die segmentations**.

Drawbacks of the multi-die approach

When using the multi-die approach, **two key requirements** arise:

- a) **The dies need to be linked together** by a proper **cache coherent interconnect fabric**.

The necessary interconnections **increase transfer latencies**, which degrade performance and **incur additional power consumption** due to the required communication.

- b) **Multiple dies must be packed together** using a suitable **packaging technology**.

This point is discussed in Section 3.5 (How are Intel's recent multicore processors packaged?)

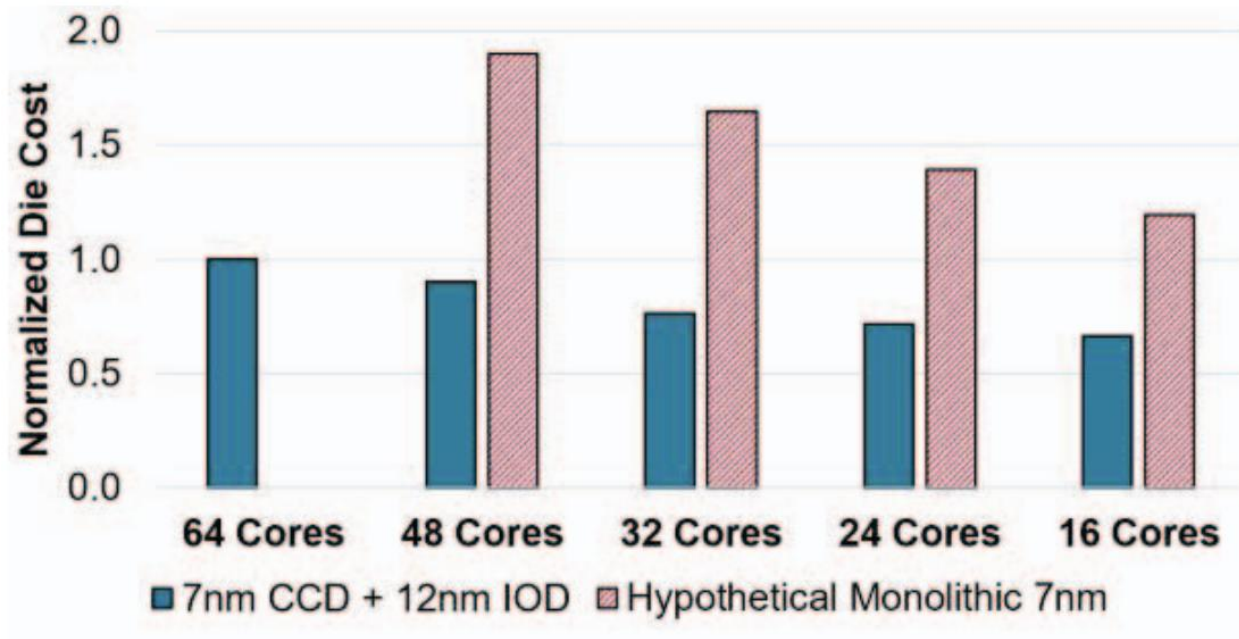
3.2 Evolution of the CPU core configuration and introduction of multi-die designs (20)

b) The additional benefit of using heterogeneous multi-die designs over single-die designs

- In a **heterogeneous multi-die design**, a processor is built up of **multiple dies**, each **with specific capabilities**, such as computing or graphics.
 - Then, **different dies** can be fabricated using **different IC processes**.
 - This is crucial because not every die needs the most advanced and costly technology. For instance, CPU tiles require a cutting-edge technology, since they determine the processor's performance.
In contrast, GPU dies, and first of all, IO dies can be manufactured using less advanced, more cost-effective processes without notably affecting overall system performance.
 - Accordingly, **by manufacturing specific die types using appropriate IC technologies** — such as fabricating CPU cores by cutting-edge processes, while fabricating I/O dies with less advanced technologies — **the overall manufacturing cost can be reduced compared to single die processors**.
- Note that using less advanced but less costly processes for less demanding dies to reduce manufacturing costs follows the same principle as using less performant but less power-hungry cores for less-demanding workloads to reduce power consumption.

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (21)

Example showing how much Zen 2 server die costs can be lowered by using heterogeneous design compared to a hypothetical monolithic layout [222]



- In the example provided, server dies are composed of CCD compute dies and an IO die, which are manufactured using 7 nm and 12 nm processes, respectively.
- As shown in the Figure, the **heterogeneous design significantly reduces the die cost compared to a hypothetical monolithic design.**

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (22)

Remark

We note that using **heterogeneous multi-die segmentation** for processors is clearly **more cost-effective than employing homogeneous multi-die processors**, which consist of multiple dies with the same functionality, as seen in certain high-core-count Intel processors.

This is because **in homogeneous multi-die processors, all dies must be manufactured using the same demanding and costly technology.**

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (23)

Block diagram of the 14th gen. heterogeneous multi-die big.little Meteor Lake processor

This architecture resembles that of the 13th gen. Raptor Lake processor, discussed previously, but it has been implemented across four tiles, as shown below.

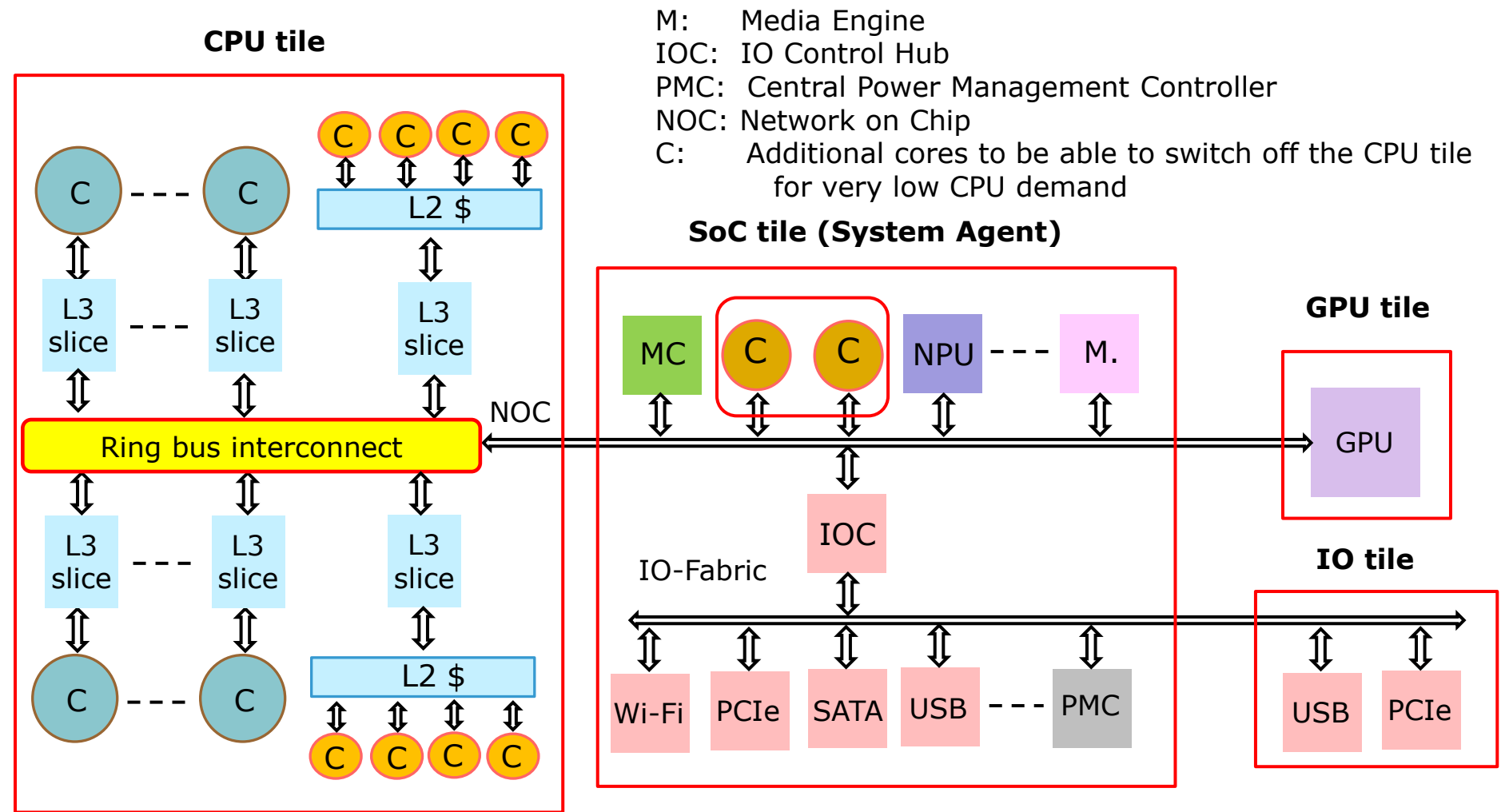


Figure: Block diagram of the 14th gen. heterogeneous multi-die big.little Meteor Lake processor

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (24)

Meteor Lake Overview: In-depth with Intel Architects and Engineers - Talking Tech, Intel Technology [188]

Meteor Lake System on Chip (SoC)

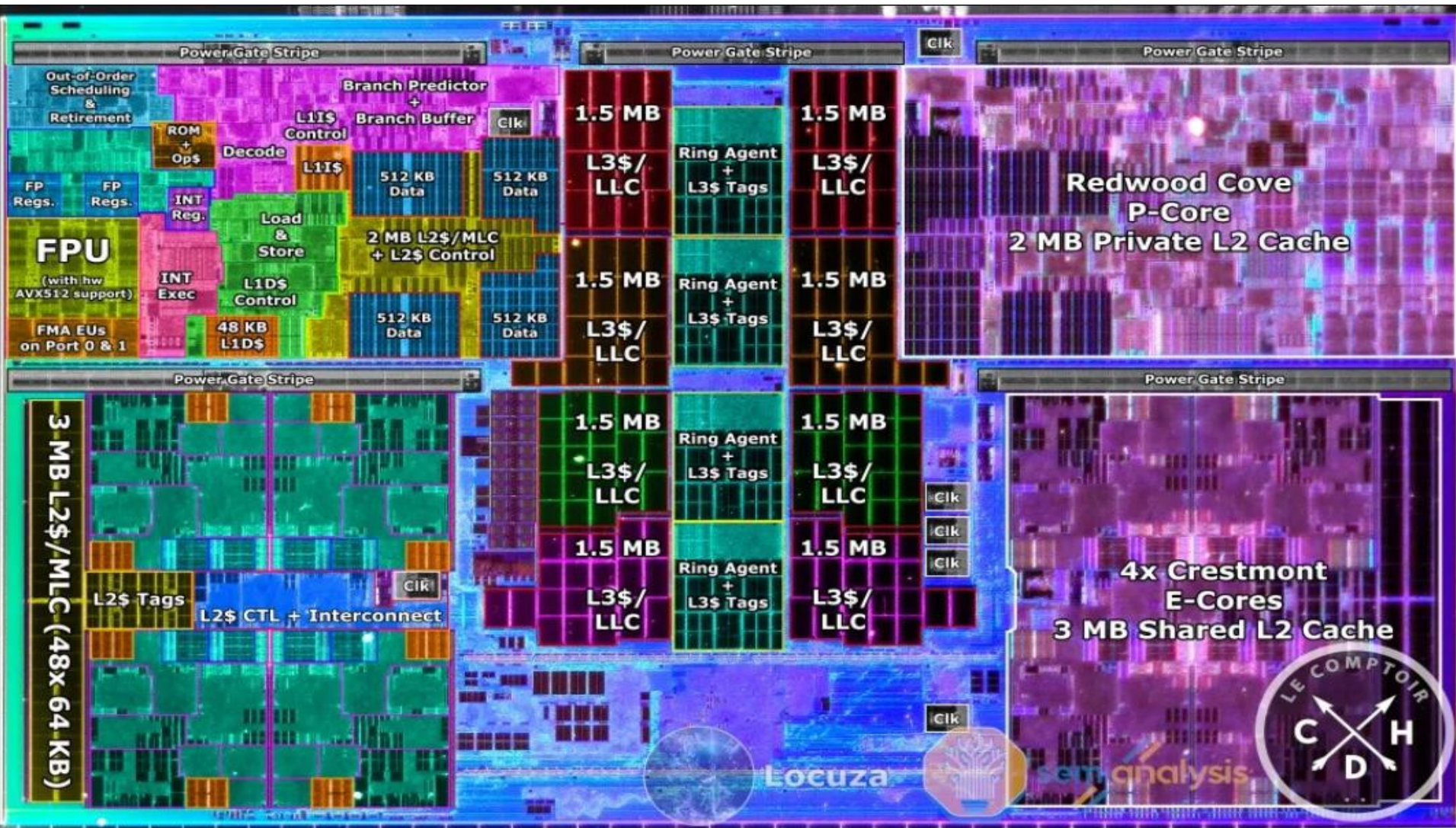


(inspirational music)



3.2 Evolution of the CPU core configuration and introduction of multi-die designs (25)

Example: Floor plan of the CPU tile of the heterogeneous multi-die 14th gen. Meteor Lake processor with 2 P (Performance) and 8 E (Efficiency) cores [50]



3.2 Evolution of the CPU core configuration and introduction of multi-die designs (26)

The evolution of Intel’s heterogeneous multi-die big.little designs, referred to as Intel’s Core Ultra processor series

Intel’s heterogeneous multi-die big. little designs

Core Ultra Series 1	Core Ultra Series 2	Core Ultra Series 2	Core Series 2	Core Ultra Series 3
Meteor Lake	Lunar Lake	Arrow Lake	Meteor Lake R.	Panther Lake
New process and µarch.	New process and µarch.	New µarch.	New µarch.	New process new µarch.
Intel 4 TSMC N5/N6	TSMC N3/N6	TSMC N3/N5/N6	Intel 4 TSMC N5/N6	Intel 18A/3 TSMC N3E/N6
TICK/Tock	TICK/Tock	~ Optimization	Optimization	TICK/Tock
(2023)	(2024)	(2024)	(2025)	(2026)
4 tiles	2 tiles	4 tiles	4 tiles	3 tiles
(Laptops)	(LP Laptops)	(Laptops/DTs)	(Laptops)	(Laptops)

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (27)

Remarks on the manufacturing processes used in Intel's heterogeneous multi-die big.little designs, known as the Lunar Lake and Arrow Lake processors

- Originally, Intel intended to fabricate its Lunar Lake's and Arrow Lake's processors using its Intel 20A and 18A nodes.

However, the company encountered unexpected challenges in developing these advanced processes.

To address its manufacturing setbacks, Intel cancelled developing the 20A process and redirected its effort toward the 18A node.

- At the same time, Intel asked TSMC to produce its Lunar Lake and Arrow Lake processors in order to avoid delays in market entries.

Specifically, the compute dies of both lines were fabricated using TSMCs most advanced available process, the N3 process.

- A significant point to note is that in both transitions (Meteor Lake to Lunar Lake, and Meteor Lake to Arrow Lake) Intel unexpectedly changed both the IC process and the microarchitecture at the same time.

By doing so, Intel adopted a single-phase development model in these cases, to accelerate the market entry of these processors.

This approach marks a deviation from its original two-phase model that was introduced with the Core family.

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (28)

Remarks on the manufacturing processes used in Intel's heterogeneous multi-die big.little designs, known as the Panther Lake processor

- Again, Intel made simultaneous changes to both the IC process and the microarchitecture.

This means that the company adopted again a single-phase development model, aimed at accelerating the market entry of its Panther Lake processors.

- This decision was likely made because Intel was not yet ready with suitable processes for these components.
- Note also that Intel let manufacture the GPU and SOC tiles by TSMC's (using its N3E and N6 processes, respectively).

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (29)

Remark to Intel's design policy: running two competing design centers -1

1. gen.		2. gen.		3. gen.	4. gen.	5. gen.	
Core 2 New microarch. 65 nm	Penryn New process 45 nm	Nehalem New microarch. 45 nm	Westmere New process 32 nm	Sandy Bridge New microarch. 32 nm	Ivy Bridge New process 22 nm	Haswell New microarchi. 22 nm	Broadwell New process 14 nm
TOCK	TICK	TOCK	TICK	TOCK	TICK	TOCK	TICK
(2006) Haifa	(2007) Oregon	(2008) Oregon	(2010) Oregon	(2011) Haifa	(2012) Haifa	(2013) Oregon	(2014) Oregon

6. gen.	7. gen.	8. gen. ¹	9. gen.	10. gen.
Skylake New microarch. 14 nm	Kaby Lake New microarch. 14 nm	Kaby Lake R/G Coffee Lake Amber Lake-Y Whiskey Lake-U (Cannon Lake 10 nm) 14 nm	Coffee Lake R New microarch. 14 nm	Comet Lake New microarch. 14 nm
TOCK	TOCK	TOCK	TOCK	TOCK
(2015) Haifa	(2016) Haifa	(2017/18) Haifa	(2018) Haifa	(2019) Haifa

¹The 8th generation encompasses the following processor lines:

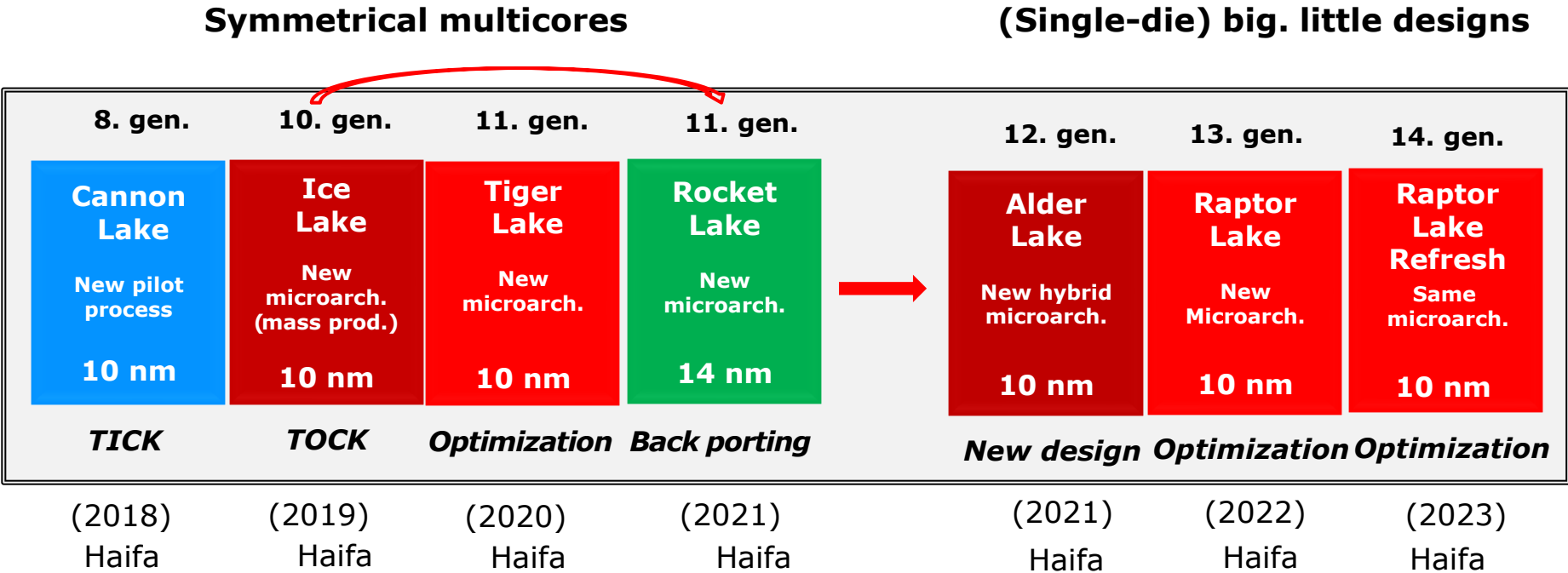
- Kaby Lake Refresh
- Kaby Lake G with AMD Vega graphics
- Coffee Lake
- Amber Lake Y
- Whiskey Lake U
- (all 14 nm) and
- Cannon Lake (10 nm)

lines [29].

R: Refresh

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (30)

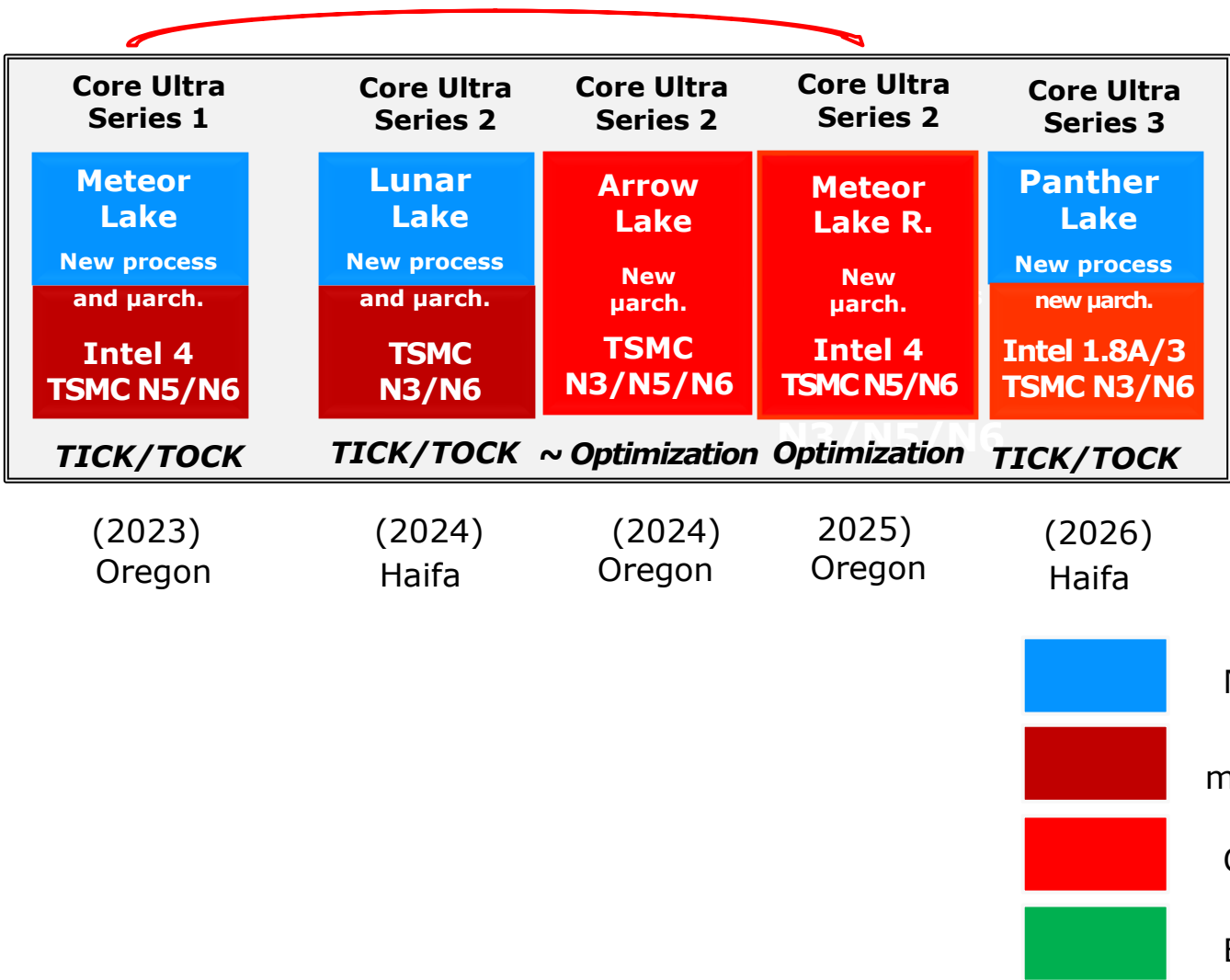
Remark to Intel's design policy: running two competing design centers -2



3.2 Evolution of the CPU core configuration and introduction of multi-die designs (31)

Remark to Intel's design policy: running two competing design centers -3

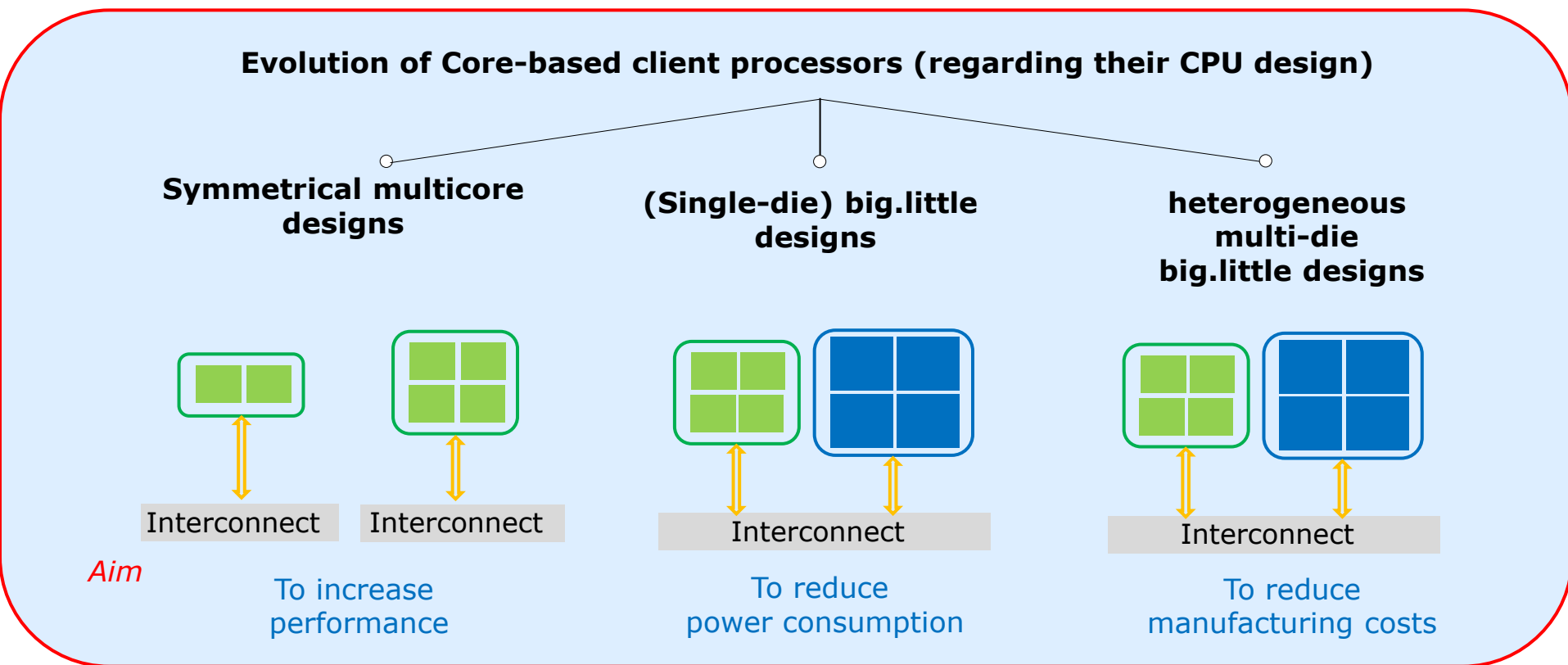
Multi-die heterogeneous big. little designs



3.2 Evolution of the CPU core configuration and introduction of multi-die designs (32)

Summing up the evolution of Intel's Core-based client processors

Considering their **core configuration**, Intel's Core-based client processors have undergone **three main evolution steps**, as illustrated in the Figure below.



*From the 1st gen. Core 2 line (2006)
up to the 11th gen Raptor Lake (2021)*

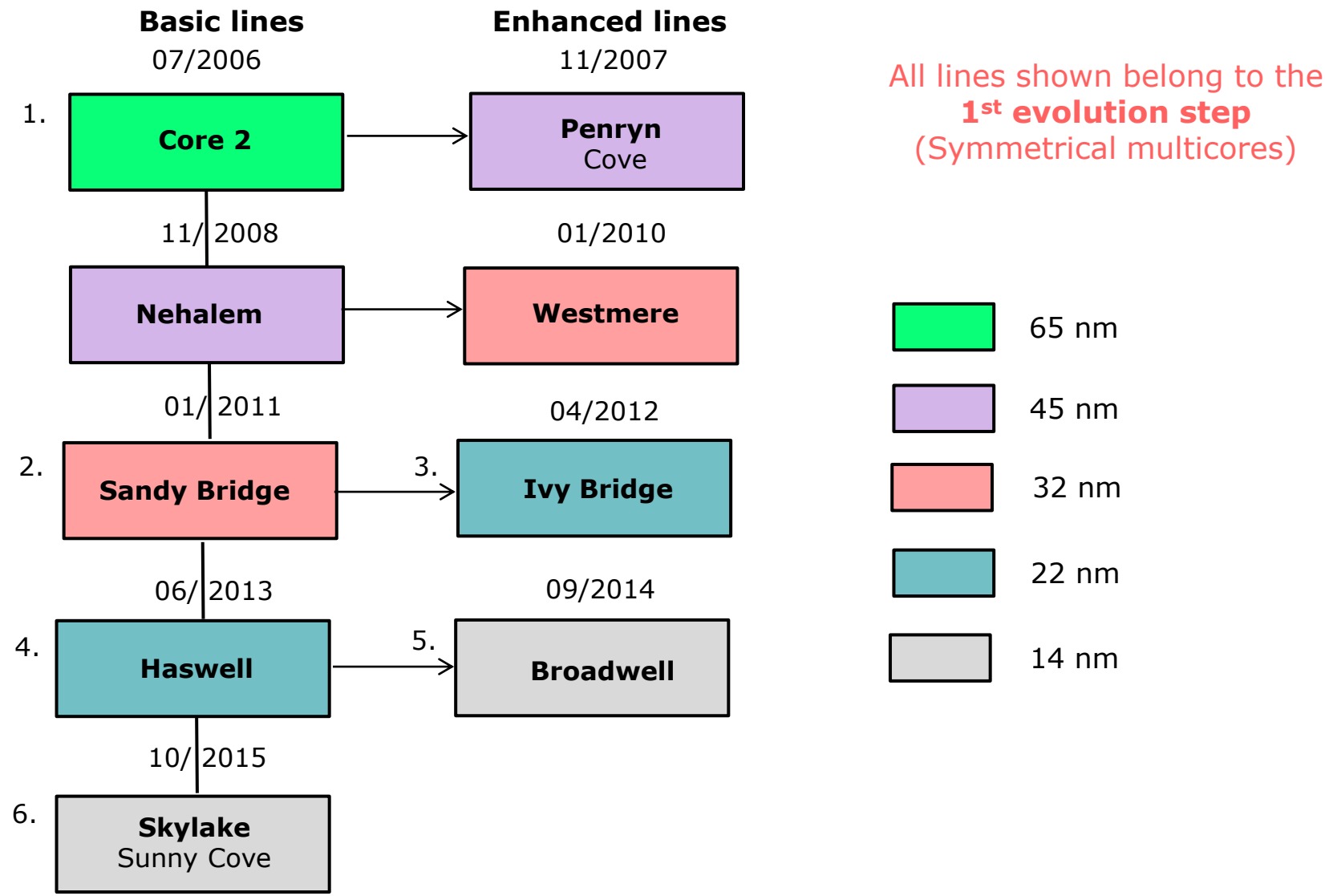
*12th gen. Alder Lake (2021)
13th gen. Raptor Lake (2022)*

*From the 14th gen.
Meteor Lake (2023) on*

Note that the **primary objective** of each core configuration is **highlighted** in the Figure.

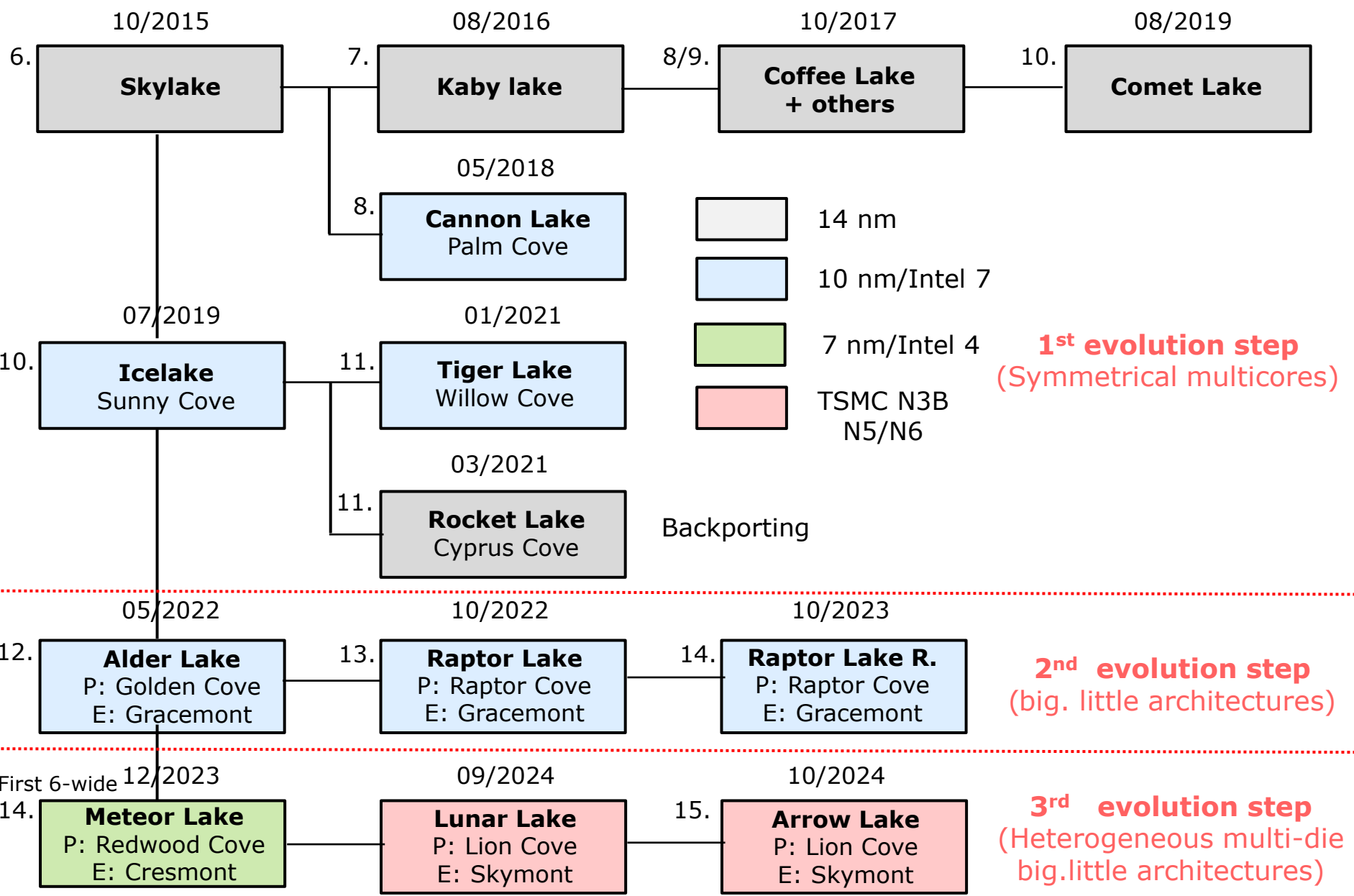
3.2 Evolution of the CPU core configuration and introduction of multi-die designs (33)

Distinguishing Intel's basic and subsequent client lines in the Core family -1



3.2 Evolution of the CPU core configuration and introduction of multi-die designs (34)

Distinguishing Intel's basic and subsequent client lines in the Core family -2



3.2 Evolution of the CPU core configuration and introduction of multi-die designs (35)

Basic microarchitectures, their shrinks and optimizations

Basic architectures	Generation	Basic architectures, their shrinks and optimizations		
Core 2	1.	2006 2007	65 nm 45 nm	Core 2 Penryn
Nehalem		2008 2010	45 nm 32 nm	Nehalem Westmere
Sandy Bridge	2. 3.	2011 2012	32 nm 22 nm	Sandy Bridge Ivy Bridge
Haswell	4. 5.	2013 2014	22 nm 14 nm	Haswell Broadwell
Skylake	6. 7. 8. 8. 9. 10.	2015 2016 2017 2018 2019 2020	14 nm 14 nm 14 nm 10 nm 14 nm 14 nm	Skylake Kaby Lake Kaby Lake Refresh + more Cannon Lake Coffee Lake Refresh Comet Lake
Ice Lake	10. 11.	2019 2020	10 nm 10 nm	Ice Lake Tiger Lake
Alder Lake	12. 13. 14.	2021 2022 2023	10 nm 10 nm 10 nm	Alder Lake Raptor Lake Raptor Lake Refresh
Meteor Lake	14./Core Ultra Series 1	2023	7 nm	Meteor Lake (CPU die)
	Core Ultra Series 2	2024	3 nm	Lunar Lake (CPU die)
	Core Ultra Series 2	2024	3 nm	Arrow Lake (CPU die)
	Core Ultra Series 3	2026	1.8A	Panther Lake

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (36)

Comparing Intel's and AMD's client processor design paths, considering their CPU configurations and multi-die design approaches -1

Intel

**Symmetrical
multicore
designs**

*Pentium
Extreme Edition 840
(2005)*



**Single-die
big.little
designs**

*12th gen
Alder Lake
(2021)*



**Heterogeneous
multi-die
big.little designs**
(for client processors)

*14th gen
Meteor Lake
(2023)*

AMD

**Symmetrical
multicore
designs**

*Athlon 64 X2
(2005)*



**Modular
homogeneous
multi-die
design**

*Zen/Zen+
(2016/2018)*



**Modular
heterogeneous
multi-die
designs**

*Zen 2
(2019)*



**Heterogeneous
multi-die
big.little
designs**
(for client processors)

*Zen 4/Zen 4c
(2023)*

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (37)

Comparing Intel's and AMD's client processor design paths, considering their CPU configurations and multi-die design approaches -2

- The preceding Figure shows that starting from symmetrical multicore designs in 2005, **both Intel and AMD** have gradually transitioned to **heterogeneous multi-die big.little** designs, although they have followed **different paths of the evolution**.
- **AMD** first adopted **homogeneous multi-die designs** with its Zen family in 2017, primarily to reduce fabrication costs.
- The company also adopted a **modular** design approach, to utilize the same set of basic dies across various market segments, including laptops, desktops, HEDTs, or servers.

This approach helps for reducing both development and manufacturing costs.

- Next, in 2019, AMD transitioned from homogeneous, multi-die designs to **heterogeneous multi-die designs** to further decrease manufacturing costs by employing less advanced processes for less demanding dies, such as the IO die.
- Finally, in 2023, AMD adopted the **big.little design paradigm**, to significantly lower power consumption in certain processor models.
- In contrast, **Intel** first introduced **big.little designs** with its 12th gen. Alder Lake line in 2021, primarily to reduce power consumption.
- This was followed by a **heterogeneous multi-die design approach** in its Meteor Lake line in 2023, to lower manufacturing costs.

3.3 How are the cores of big.little architectures scheduled?

3.3 How are the cores of big.little architectures scheduled? (1)

3.3 How are the cores of big.little architectures scheduled? -1

- Concerning this point, please recall that Intel introduced earlier **hardware support for the OS** using DVFS (Dynamic Voltage and Frequency Scheduling) to reduce dissipation in the Core family of processors, in 2006.

With DVFS, Intel determined the **actual utilization of threads** (also known as logical processors) by utilizing **two Model Specific Registers (MSRs) per thread**, which are essentially 64-bit counters.

- While Intel introduced **big.little architectures** in its 12th gen. Alder Lake processor family, in 2021, it continued to provide **hardware support for the OS**, this time **for thread scheduling**.

The introduced hardware guidance for scheduling hybrid architectures is known as **Tread Director (TD)**.

3.3 How are the cores of big.little architectures scheduled? (2)

How are the cores of big.little architectures scheduled? -2

- The necessity for **hardware-supported thread scheduling** by means of the **Thread Director**, arose from the fact that big.little architectures utilize **different core types**, which **require a distinct method** of thread scheduling compared to symmetrical multicores.
- The **Thread Director (TD)** supports **simultaneous execution across different core types** while optimizing various aspects of thread scheduling, such as performance and power consumption.
- The TD works **in cooperation with the Windows OS**, initially, with Windows 11 release 21H2, which was unveiled in 10/2021.
- Unfortunately, the TD is only “**sparsely**” **documented**.
- Therefore, below we can only provide a strongly simplified description of it, while referring to its implementation in the **Panther Lake** line, launched in **01/2026**, known as **Thread Director v2**.
- The implementation details of the TD will not be discussed in the Lecture.

3.3 How are the cores of big.little architectures scheduled? (3)

The Thread Director (TD) in the Panther Lake processor -1 [223], [224]

- The key component of the TD is the **Thread Director Table**.
It keeps the actual **performance and power efficiency capabilities** of the **logical processors**.
- A **logical processor** is a **CPU core** that the OS can use to run a thread.
 - A CPU core **without Hyper-Threading** equals **one** logical processor.
 - A CPU core **with Hyper-Threading** provides **two** logical processors.
- **Performance capabilities** are 8-bit values (0 ... 255), specifying the **relative performance level of a logical processor**.

Higher values indicate higher performance; the lowest performance level of 0 indicates a recommendation to the OS to not schedule any software threads on it for performance reasons.

- **Energy Efficiency capabilities** are 8-bit values (0 ... 255), specifying the **relative energy efficiency level of a logical processor**.

Higher values indicate higher energy efficiency; the lowest energy efficiency capability of "0" indicates a recommendation to the OS to not schedule any software threads on it for efficiency reasons.

An Energy Efficiency capability of 255 indicates which logical processors have the highest relative energy efficiency capability.

In addition, the value "255" is an explicit recommendation for the OS to consolidate work on those logical processors for energy-efficiency reasons.

3.3 How are the cores of big.little architectures scheduled? (4)

The Thread Director (TD) in the Panther Lake processor -2 [224]

The **performance and power efficiency capabilities** are kept in **distinct fields** of the Thread Director Table, which is structured as shown below.

3.3 How are the cores of big.little architectures scheduled? (5)

Assumed structure of the Thread Director Table [224] -1

The logical processors are ordered in **classes** (CL).

The **Panther Lake** processor has **three classes**, as follows:

CL_0 = LP-E cores

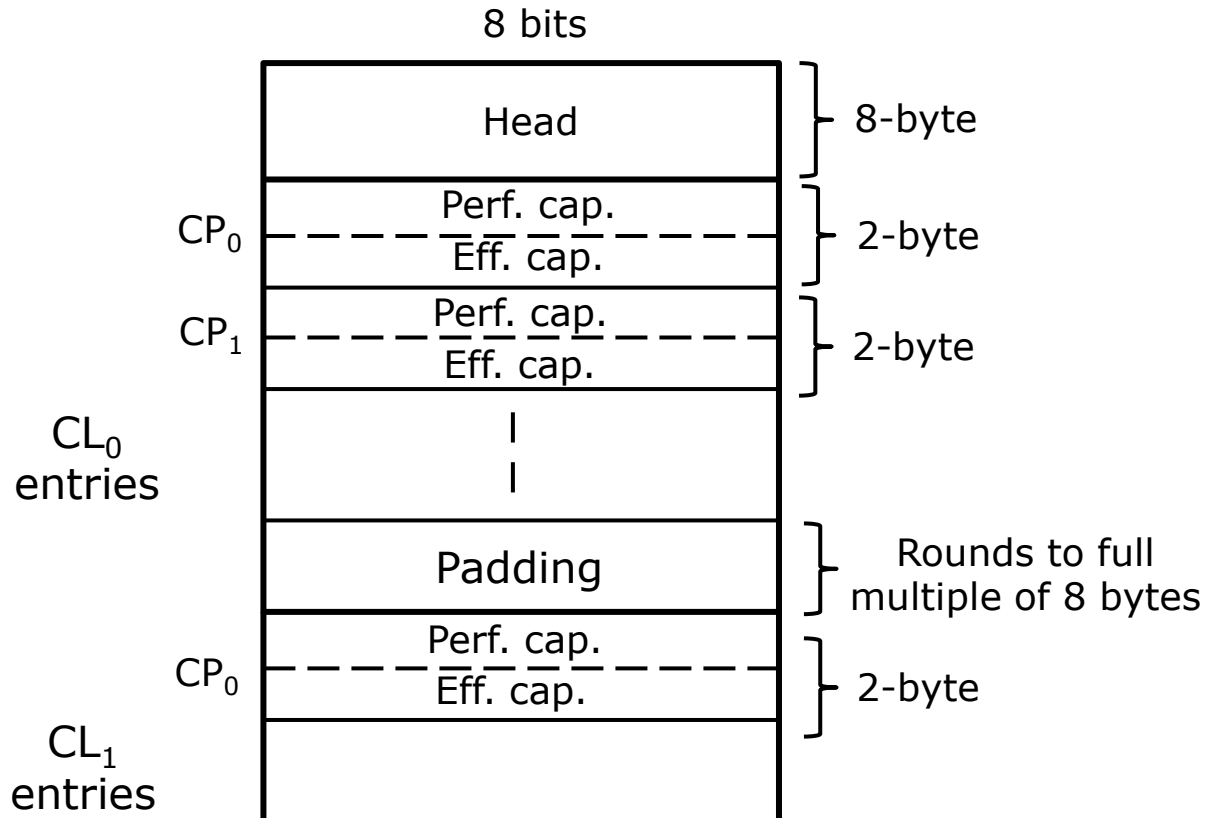
CL_1 = E cores

CL_2 = P cores.

The Thread Director Table lists the actual capability values ordered according to the logical processor classes CL_i as seen in the following Figure.

3.3 How are the cores of big.little architectures scheduled? (6)

Assumed structure of the Thread Director Table [224] -1



In the Table the capability values are ordered according to the logical processor classes CL_i , where the classes may be interpreted as

CL_0 = LP-E cores
 CL_1 = E cores
 CL_2 = P cores.

3.3 How are the cores of big.little architectures scheduled? (7)

Assumed structure of the Thread Director Table [224] -3

- Within each logical core class $(CL)_i$, the capabilities of the logical processors are listed as CP_0, CP_1 etc.

How are the actual capability values determined?

- **Hardware scans internal telemetry data**, such as performance counters and on-die sensors in loops, and determines the capability values of the logical processors on a μs scale.
- Capabilities may change every tens of milliseconds. (223)

For instance, the operating conditions of a thread may be affected by several factors, such as a change in the supply voltage due to frequency/voltage scaling, a reduction in clock frequency and core voltage due to increased temperature or a higher load on the thread shared on the same core etc.

In such cases, the capabilities will be reduced, and the corresponding values must be updated in the TD Table.

Changes of the capability values generate **interrupts**.

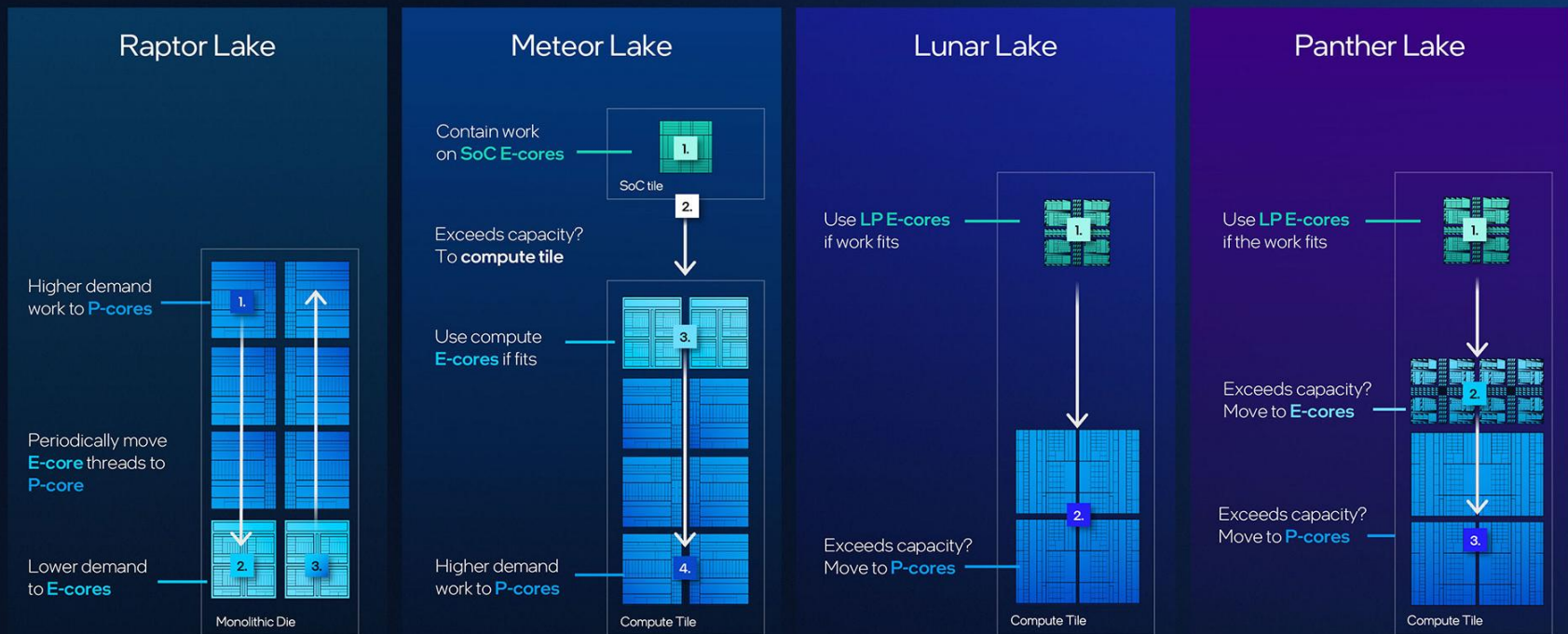
The Thread Director (TD) in the Panther Lake processor -3 [224]

- The OS processes these interrupts and adjusts thread scheduling accordingly while using the latest capability data in the Thread Director Table, and further data not published.
- Currently, there is no publicly available documentation revealing how the TD updates the capability data based on the telemetry data read.
- Also, there is a lack of reliable documentation on how the OS reschedules threads, making it impossible to discuss this point.

3.3 How are the cores of big.little architectures scheduled? (9)

Evolution of scheduling workloads to heterogeneous cores in Intel's processors from the Raptor Lake line to the Panther line [225]

Thread Director Core Scheduling Evolution



3.4 How are Intel's recent multi-die processors packaged?

3.4 How are Intel's recent multi-die processors packaged?

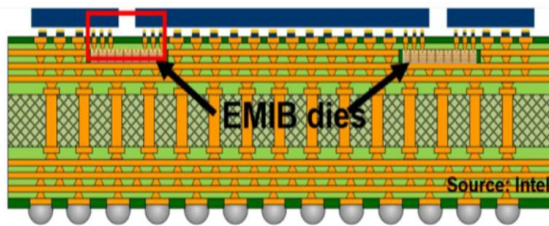
- Intel introduced its **EMIB** and **Foveros** technologies in **2017 and 2018**, respectively.
- However, **around 2023-2025** Intel partially **reinterpreted** and **renamed** these packaging technologies, as outlined below.

3.4 How are Intel's recent multicore processors packaged? (2)

Intel's recent packaging technologies and their use

Intel's recent packaging technologies and their use

EMIB 2.5D



Used in

Stratix 10 FPGA (2017)

(Field-Programmable Logic Array)

Server processors:

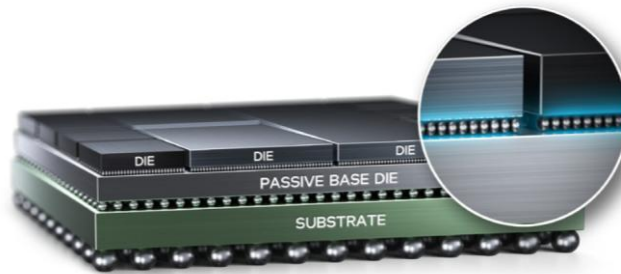
4th gen. Sapphire Rapids (2023)
(4-tile CPU)

5th gen. Emerald Rapids (2024)

6th gen. Sierra Forest (2024)

6th gen. Granite Rapids (2024)

Foveros 2.5D



Lakefield (2019)

Meteor Lake (2023)

Arrow Lake (2024)

Lunar Lake (2024)

Panther Lake (2026)

Foveros Direct 3D



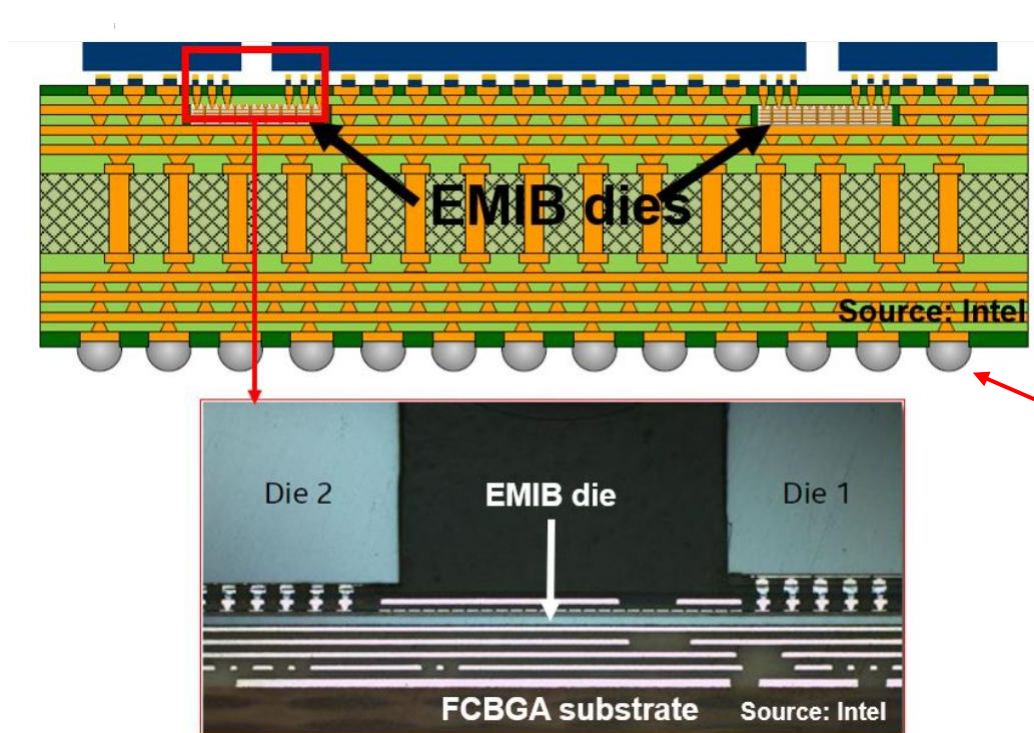
Server processor:

6+ gen. Clearwater Forest
(2026)

The technologies mentioned above will be discussed subsequently.

3.4 How are Intel's recent multicore processors packaged? (3)

The EMIB 2.5D (Embedded Multi-die Interconnect Bridge) packaging technology []



Its main components are:

- Soldered microbumps with a pitch of 45-55 μm .
- Organic package substrate
- Multiple Cu routing layers
- Dielectric layers isolating embedded Si bridges
- BGA ball grid

It was **first** implemented in the FPGA chip (Stratix 10) **in 2017**, referred to as Intel's EMIB technology.

Soldered microbumps: forraszgöbök Pitch: osztás-távolság

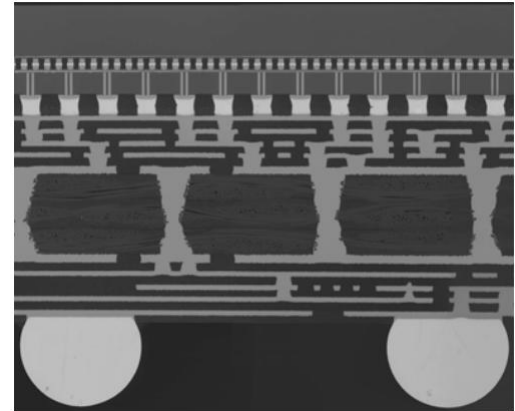
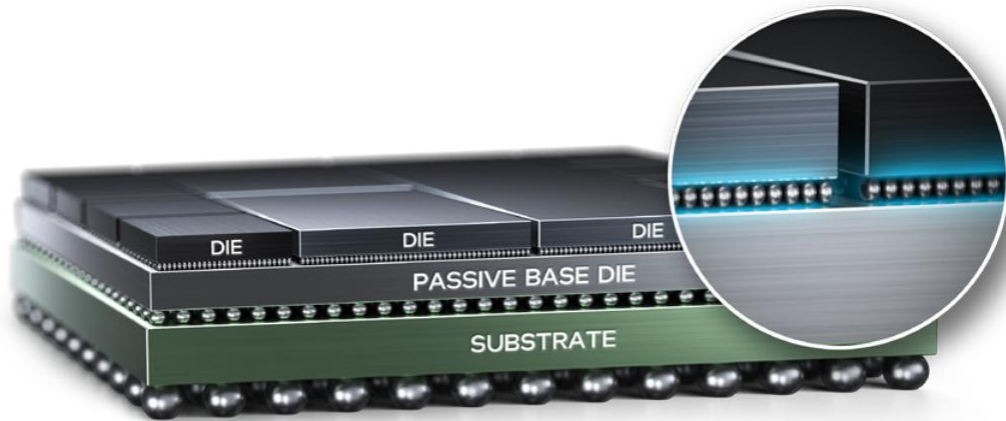
Organic package substrate: Polimer-alapú tokozó anyag

Cu routing layers: réz vezetősávok BGA ball grid array: BGA forraszgolyó-rács

FPGA: Field-Programmable Gate Array

3.4 How are Intel's recent multicore processors packaged? (4)

The Foveros 2.5D packaging technology [199]



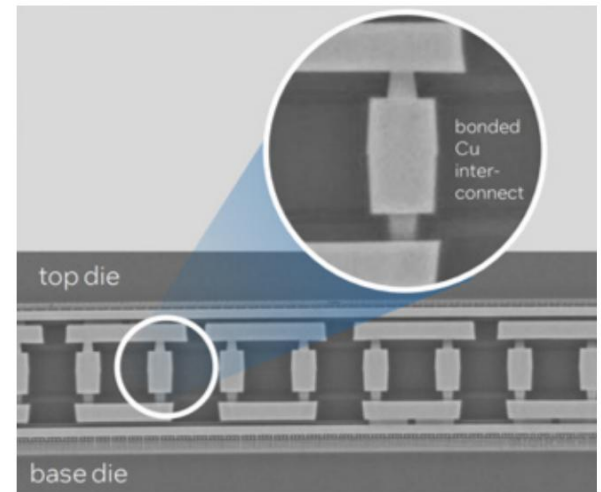
Its main components are:

- Soldered fine-pitch (20-50 μm) micro-bumps
- Si-interposer (base die) with
 - local Cu routing layers
 - TSVs (Through Silicon Vias) for power, ground (réz átvezetők)
 - C4 solder bumps with a much coarser pitch (100-200 μm) than microbumps.
- Organic package substrate with
 - Coarser Cu routing layers
 - power/ground planes
 - embedded decoupling capacitors (beágyazott leválasztó kapacitások)
- BGA Ball grid

It was **first** implemented in the **Lakefield processor in 2019**, referred to as Intel's Foveros technology.

3.4 How are Intel's recent multicore processors packaged? (5)

The Foveros Direct 3D packaging technology [220]



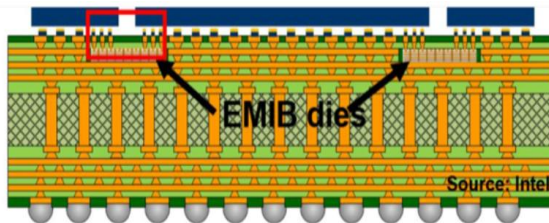
- It **eliminates traditional solder microbumps** by utilizing direct **Cu-to-Cu hybrid bonding** between the chiplets and the base die with a pitch of less than 10 μm .
To achieve direct Cu-to-Cu hybrid bonding, Cu pads (réz érintkezők) are pressed together under heat and pressure, allowing Cu atoms to diffuse across the interface and create a hybrid metallic interconnect.
- This method **removes the resistance and parasitic effects of solder bumps**, while enabling a **much tighter pitch**.
- The roles of the base die and the package substrate remain largely the same as in the Foveros 2.5D technology.
- The Foveros Direct 3D packaging will **first** be implemented in Intel's 6+ gen. Clearwater server processor series, which is set to be launched in **H1 2026**.

3.4 How are Intel's recent multicore processors packaged? (6)

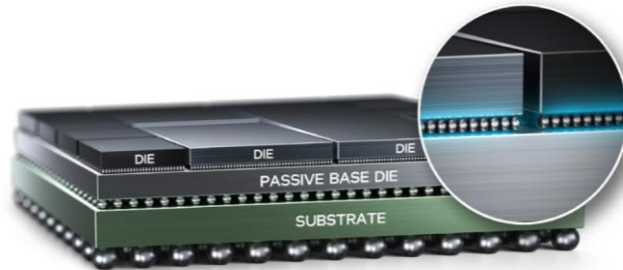
Key features of Intel's recent packaging technologies [221]

Intel's recent packaging technologies and their use

EMIB 2.5D



Foveros 2.5D



Foveros Direct 3D



	EMI 2.5D	Foveros 2.5D	Foveros Direct 3D
Bump pitch	55-45 μm	25 μm	<10 μm
Bump density	770-330/mm ²	772-1600/mm ²	10000/mm ²
C2C power	0.25 pJ/bit	0.15 pJ/bit	0.05 pJ/bit

As indicated above, newer packaging technologies provide more dense and more energy-efficient interconnections between slides.

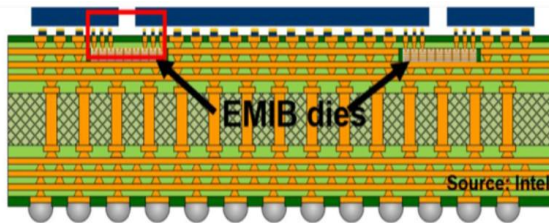
C2C: Chip to Chip

3.4 How are Intel's recent multicore processors packaged? (7)

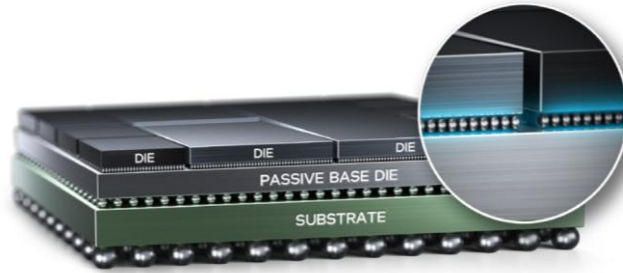
Summing up Intel's recent packaging technologies

Intel's recent packaging technologies and their use

EMIB 2.5D



Foveros 2.5D



Foveros Direct 3D



These technologies are utilized to interconnect and package **side-by-side placed dies**.

EMIB (Embedded Multi-die Interconnect Bridge): In this technology, the dies are **soldered** to the substrate while they are interconnected using EMIB bridges, which are embedded in an organic package substrate.

The substrate provides additional interconnects, power, and ground, and includes BGA balls.

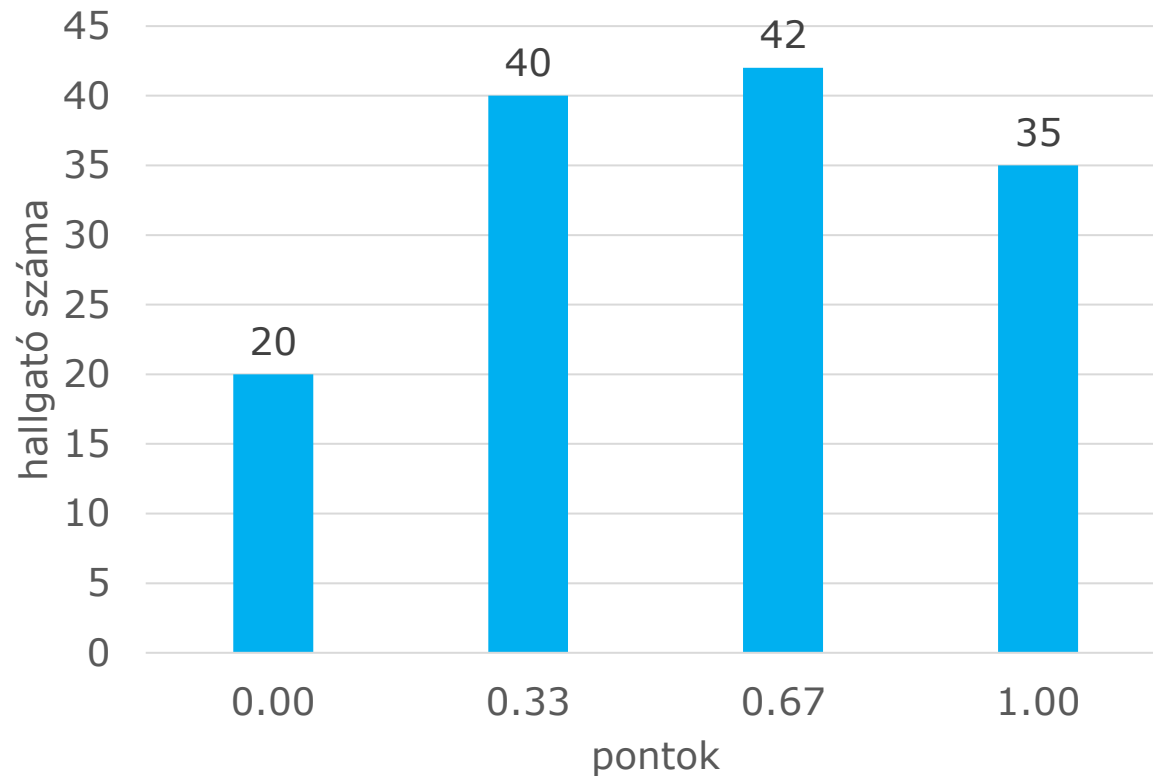
Foveros 2.5D: This method involves **soldering** the dies to a passive base die (Si interposer), which facilitates direct interconnections between the dies and provides power and ground connections.

The base die has the same functions as before but provides a finer interconnection structure.

Foveros Direct 3D: This approach replaces soldering the dies to the base die by **direct Cu-to-Cu interconnects**, which is more dense and more power efficient.

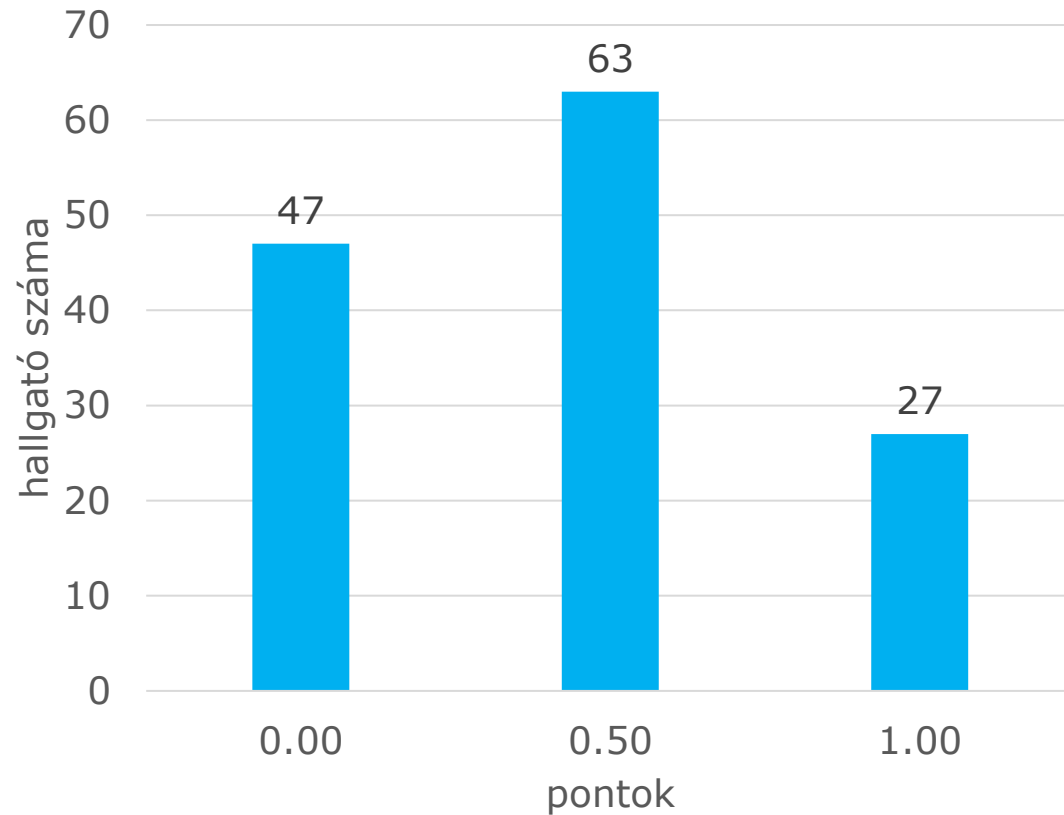
Mely 3 lépésben fejlődtek Intel kliens processzorai a CPU magok konfigurációja tekintetében?

v: Szimmetrikus többmagos, egylapkás (big. little), heterogén (többlapkás) big.little



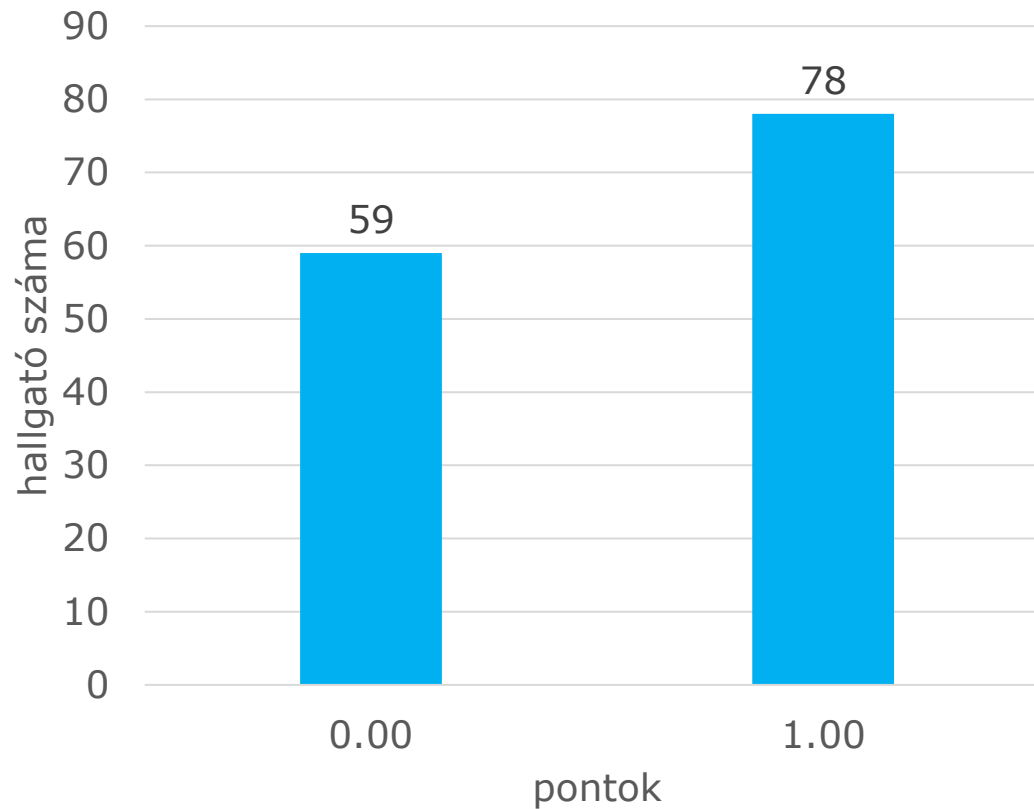
Mi tette szükségessé Intel számára a Thread Director kifejlesztését?

V: big.little (hibrid) architektúrák megjelenése / eltérő magtípusok használata



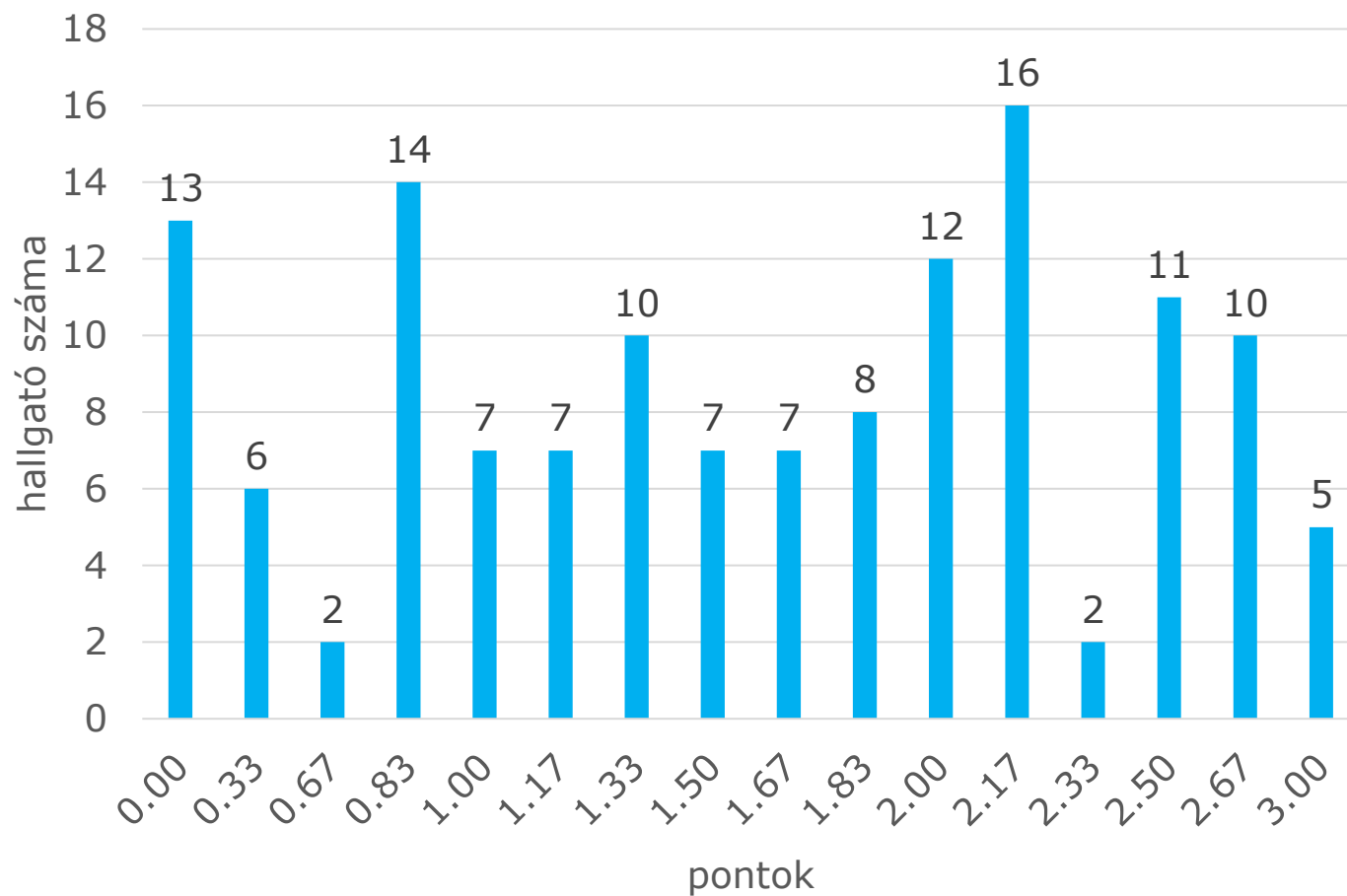
Alapvetően milyen előnnyel jár a processzorok többlapkás megvalósítása?

V: Gyártási költségek csökkentése



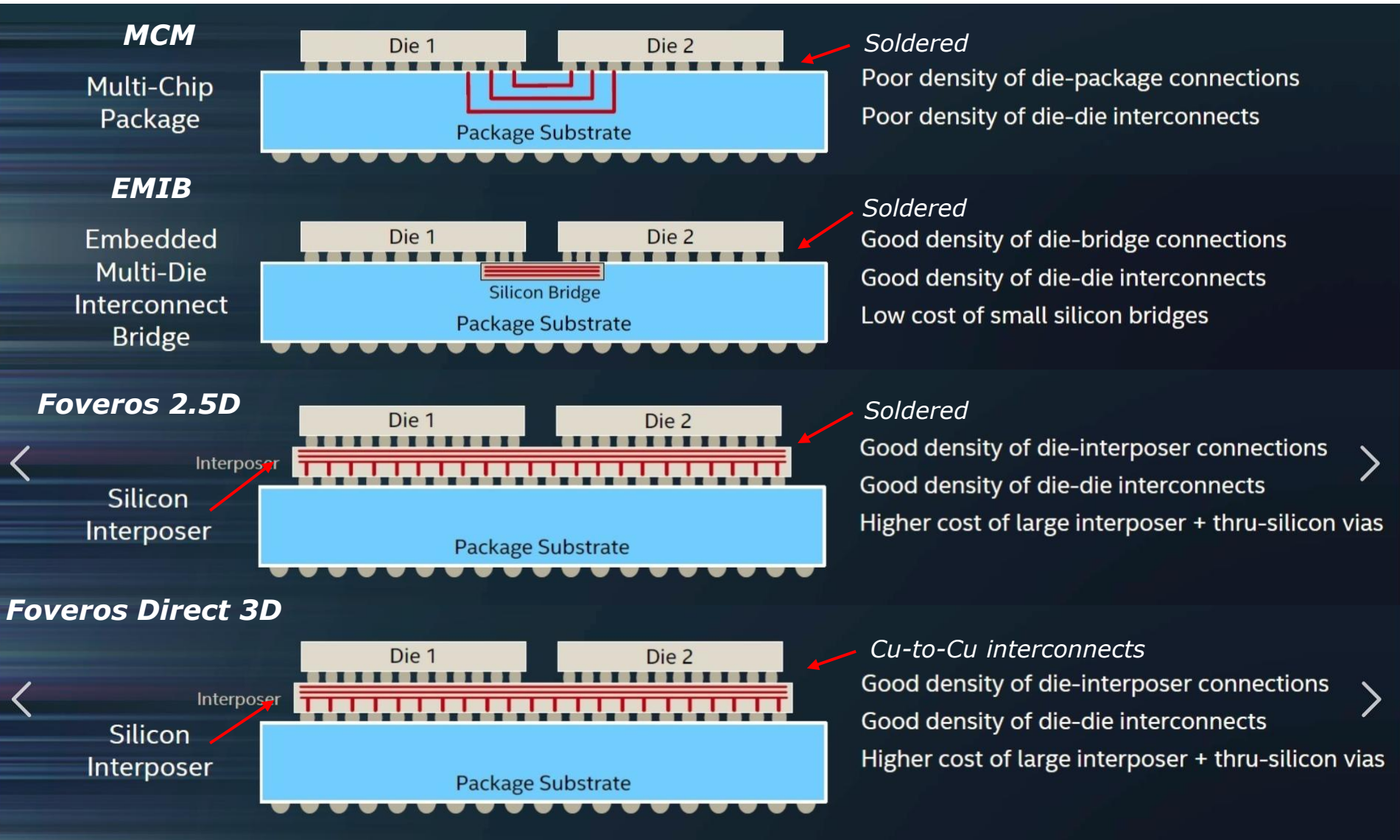
3.4 How are Intel's recent multicore processors packaged? (9)

Összesített eredmények



3.4 How are Intel's recent multicore processors packaged? (8)

The MCM technology and Intel's packaging technologies [215]



3.4 How are Intel's recent multicore processors packaged? (9)

How are Intel's and AMD's processors packaged?

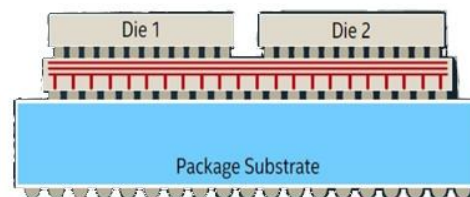
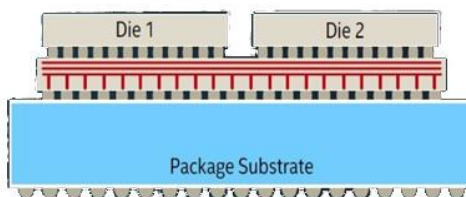
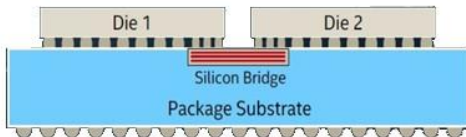
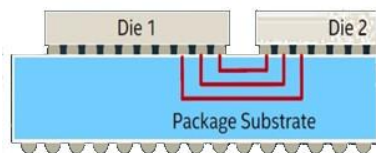
Major packaging technologies used by Intel and AMD

**2D packaging
with soldered dies**

**2.5D packaging
with soldered dies
and using a Si-bridge**

**2.5D packaging
with soldered dies
and a Si interposer**

**3D packaging
with Cu-to-Cu bonded dies
and a Si interposer**



Intel

Early dual-die Packaging
(2005)

EMIB (2.5D)

Stratix 10 FPGA (2017)
Multi-die server lines
(2023 -)

Foveros 2.5D

Multi-die client lines:
Lakefield (pilot) (2019)
Meteor Lake (2023)
and subsequent lines

Foveros Direct 3D

Multi-die server lines
(since 2026)

AMD

MCM

(Multi-chip Module)
(2010)
Zen family

EFB

(Elevated Fanout Bridge)
(for GPUs)
(2021)

3D V-Cache

(L3 stacked over proc. die)
(2022)

EMIB: Embedded Multi-die Interconnect Bridge Package substrate: Polimer-alapú tokozó anyag
Cu routing layers in the Package Substrate: réz vezetősávok a tokozó anyagban

3.5 How has the CPU core width increased?

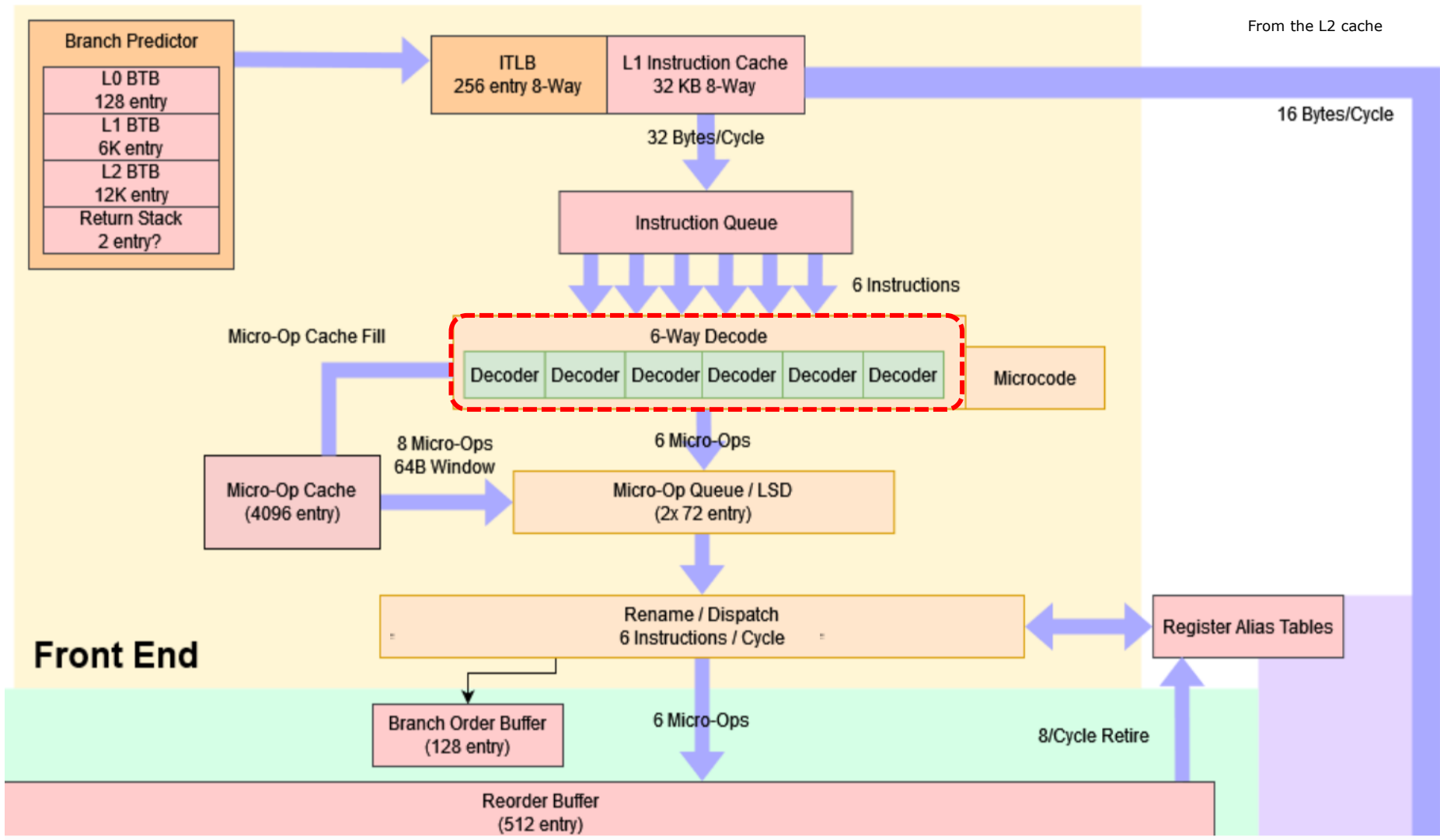
3.5 How has the CPU core width increased? (1)

3.5 How has the CPU core width increased?

- Before discussing this point, let's clarify the concept of the "CPU core width".
- The "CPU core width" usually refers to the number of instruction decoders implemented in the "front-end" of the CPU cores.
- Remarks on the notion of "CPU core width"
It is important to note that this notion is also referred to as "front-end width" or simply "core width".
- As an example, the following Figure shows that the P-core of Meteor Lake is 6-wide, as it contains 6 decoders.

3.5 How has the CPU core width increased? (2)

Example: The 6-wide P-core of client processors in the Meteor Lake line [54]



LSD: Loop Stream Detector (Detects loops and inserts decoded micro-op sequences for them.)

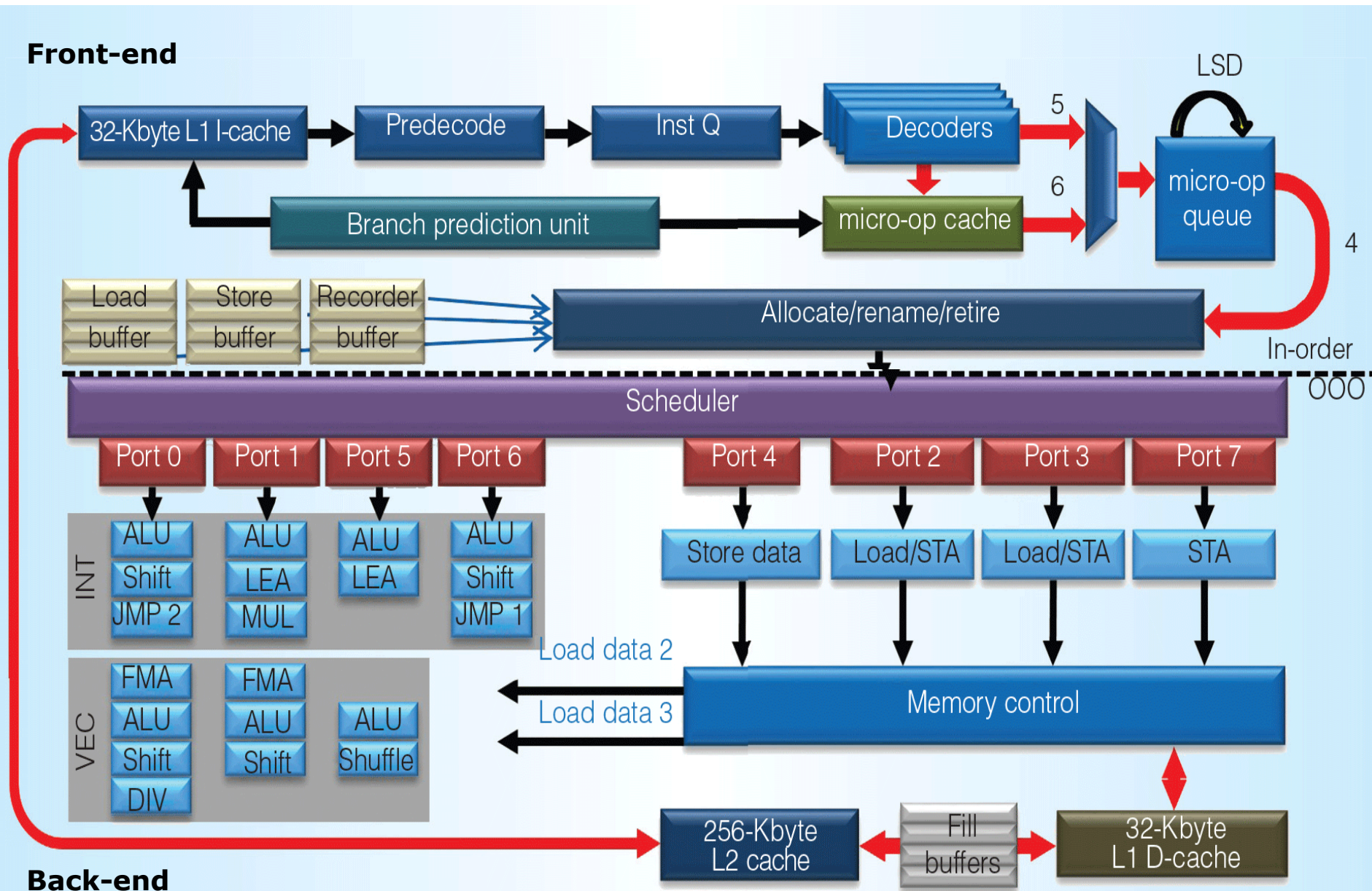
3.5 How has the CPU core width increased? (3)

Remark on the notions “front-end” and “back-end”

CPU cores can typically be divided into **two main parts**: the **front-end** and the **back-end**, as illustrated in the following Figure.

3.5 How has the CPU core width increased? (4)

Block diagram of Intel's 6th gen. Sky Lake core (2015) [145] -1



3.5 How has the CPU core width increased? (5)

The front-end and back-end of CPU cores exemplified by Intel's 6th gen. Sky Lake CPU core (2015) [145] -2

- The **front-end** of a processor is responsible for **preparing instructions for execution**.
- It primarily handles **instruction fetching**, **decoding them into microoperations**, and **performs register renaming**.

The **rename process** eliminates false data dependencies, such as write-after-write and write-after-load dependencies, enabling more efficient out-of-order execution.

- The **micro-op cache** **stores decoded instructions along with their corresponding micro-op sequences**.

During instruction decoding, the processor checks the micro-op cache to determine whether the current instruction has already been decoded and stored.

If a match is found, the associated micro-op sequence is directly inserted into the micro-op stream, bypassing the decoding stage.

This approach significantly **reduces decode latency** and lowers **energy consumption**.

- The **back-end** of the processor core is responsible for **executing micro-operations** and **updating the program state** (or program states in the case of multithreading) with the instruction results.
- Instruction results are **committed** in program order by using a **ROB (ReOrder Buffer)**.
(In the current example, the ROB's role in updating program state is not illustrated.)

3.5 How has the CPU core width increased? (6)

Why is the number of decoders decisive for the performance potential of CPU cores?

- The **number of instruction decoders** in a core determines the **maximum number of microoperations** that can be delivered to the core's back-end **per cycle**.
This directly **limits the peak single-core (or single-thread) performance**.
- While adding **a micro-op cache can boost decode bandwidth by about 5 to 15 %**, the **overall decode bandwidth remains fundamentally constrained by the number of decoders** implemented in the core [163].

3.5 How has the CPU core width increased? (7)

Increasing the width of the CPU cores in Intel's and AMD's client processors

Wider cores improve performance, e.g. in terms of **IPC** (Instructions Per Cycle), therefore **both Intel and AMD have strived to widen their cores** over time, as shown in the Figure below.

Width of Intel's and AMD's (x86-based) CPU cores

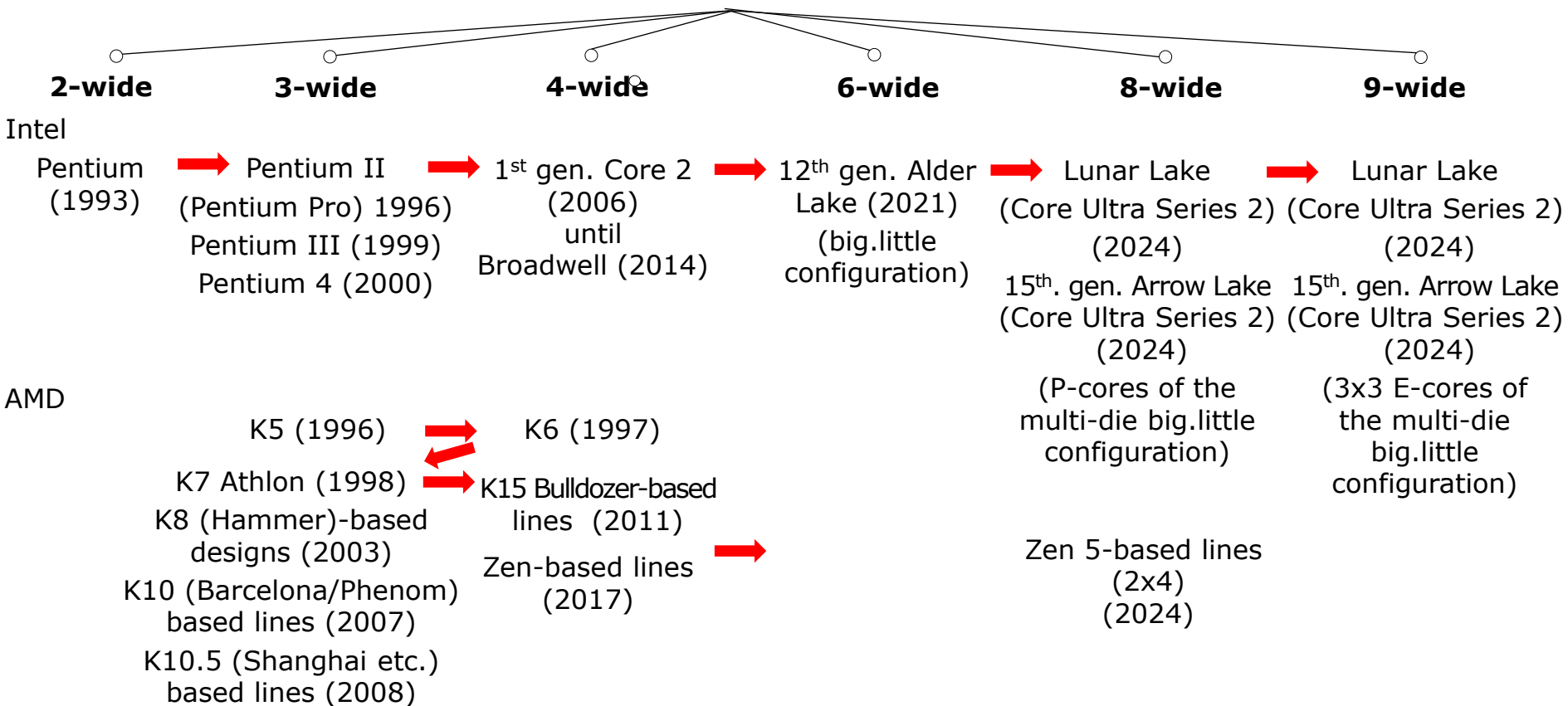


Figure: Increasing the CPU core widths in Intel's and AMD's client processors

3.6 How has the core count in Intel's client CPUs increased?

3.6 How has the core count in Intel's client CPUs increased? (1)

3.6 How has the core count in Intel's client CPUs increased?

Intel has increased the max. **core counts** of its client processors in **four consecutive phases**.

These phases are discussed below for your reference.

3.6 How has the core count in Intel's client CPUs increased? (2)

a) Phase 1: Increasing the maximum core count up to four -1

- Soon after introducing the dual-core Core family in 2006, Intel increased the core count of its client processors to four.

The following Table illustrates this.

3.6 How has the core count in Intel's client CPUs increased? (3)

a) Phase 1: Increasing the max. core counts up to 4 -2

Gen.	Processor	Technology	Date of intro.	Segment	Max core count	Connecting memory	Mem. speed (up to)	Mem. channels	Socket	MCH/PCH
1.	Core 2	65 nm	6/2006	M D	2C	Via the NB	DDR2-800	2	LGA 775	945-975
	Core 2 Quad	65 nm	6/2007	D	2x2C		DDR3-1067		LGA 775	3 Series
	Penryn	45 nm	6/2008	D	2x2C		DDR3-1067		LGA 775	4 Series
	2. G. Nehalem (Lynnfield)	45 nm	9/2009	D	4C	Via the processor	DDR3-1333		LGA 1156	5 Series
	Westmere (M: Arrandale) D: Clarkdale/	32 nm	1/2010	M D	2C+G				M: BGA 1288 D: LGA 1156	5 Series
2.	Sandy Bridge	32 nm	1/2011	M D	2C/4C+G		DDR3-1333		M: PGA 988 D: LGA 1155	6 Series
3.	Ivy Bridge	22 nm	4/2012	M D	4C+G		DDR3-1600			7 Series
4.	Haswell	22 nm	6/2013	M D	4C+G		DDR3-1600			D: 8 Ser.
4.	Haswell Refr.	22 nm	5/2014	D	4C+G		DDR3-1666		M: BGA 1168 D: LGA 1150	D: 9 Ser.
5.	Broadwell	14 nm	9/2014	M D	4C+G		DDR3-1866			D: 9 Ser.
6.	Skylake	14 nm	10/2015	M D	4C+G		DDR4-2133		M: BGA 1356 D: LGA 1151	D: 100 Series
7.	Kaby Lake	14 nm	8/2016	M D	4C+G		DDR4-2400			D: 200 Series
8.	Coffee Lake	14 nm	10/2017	M D	6C+G		DDR4-2666		M: BGA 1440 D: LGA 1151	D: 300 Series
8.	Amber Lake	14 nm	8/2018	M	2C+G		LPDDR3-2133		BGA 1515	--
8.	Whiskey Lake	14 nm	8/2018	M	4C+G		DDR4-2400		BGA 1528	--
9.	Coffee Lake R.	14 nm	10/2018	D	8C+G		DDR4-2666		LGA 1151	300 Series
10.	Comet Lake	14 nm	8/2019	M D	10C+G		DDR4-2666		M: BGA 1528 D: LGA 1200	D: 400 Series
11.	Rocket Lake	14 nm	3/2021	D	8C+G		DDR4-3200		LGA 1200	

3.6 How has the core count in Intel's client CPUs increased? (4)

a) Phase 1: Increasing the maximum core count up to four -3

- The first quad-core models, released in 2007, were implemented using dual-dies configurations, followed by monolithic (single-die) designs in 2009.
- However, for nearly a decade, Intel maintained this four-core standard, as most software applications at the time offered limited parallelism and did not benefit significantly from additional cores.

3.6 How has the core count in Intel's client CPUs increased? (5)

b) Phase 2: Increasing the maximum core count to ten -1

- In 2017, AMD launched its Zen-based client processors without integrated GPUs, initially with 8 cores, and later expanding its maximum core count to 16.
- This aggressive scaling challenged Intel's dominance in the client processor market.
- In response, beginning in 2017, Intel gradually increased the core count of its client processors first to 6, then to 8, and subsequently to 10, as indicated in the following two Tables.

3.6 How has the core count in Intel's client CPUs increased? (6)

Phase 2: Increasing the maximum core count to 10 -2

Gen.	Processor	Technology	Date of intro.	Segment	Max core count	Connecting memory	Mem. speed (up to)	Mem. channels	Socket	MCH/PCH
1.	Core 2	65 nm	6/2006	L D	2C	Via the NB	DDR2-800	2	LGA 775	945-975
	Core 2 Quad	65 nm	6/2007	D	2x2C		DDR3-1067		LGA 775	3 Series
	Penryn	45 nm	6/2008	D	2x2C		DDR3-1067		LGA 775	4 Series
	2. G. Nehalem (Lynnfield)	45 nm	9/2009	D	4C	Via the processor	DDR3-1333		LGA 1156	5 Series
	Westmere (L: Arrandale) D: Clarkdale/	32 nm	1/2010	L D	2C+G				L: BGA 1288 D: LGA 1156	5 Series
2.	Sandy Bridge	32 nm	1/2011	L D	2C/4C+G		DDR3-1333		L: PGA 988 D: LGA 1155	6 Series
3.	Ivy Bridge	22 nm	4/2012	L D	4C+G		DDR3-1600			7 Series
4.	Haswell	22 nm	6/2013	L D	4C+G		DDR3-1600			D: 8 Ser.
4.	Haswell Refr.	22 nm	5/2014	D	4C+G		DDR3-1666		L: BGA 1168 D: LGA 1150	D: 9 Ser.
5.	Broadwell	14 nm	9/2014	L D	4C+G		DDR3-1866			D: 9 Ser.
6.	Skylake	14 nm	10/2015	L D	4C+G		DDR4-2133		L: BGA 1356 D: LGA 1151	D: 100 Series
7.	Kaby Lake	14 nm	8/2016	L D	4C+G		DDR4-2400			D: 200 Series
8.	Coffee Lake	14 nm	10/2017	L D	6C+G		DDR4-2666		L: BGA 1440 D: LGA 1151	D: 300 Series
8.	Amber Lake	14 nm	8/2018	L	2C+G		LPDDR3-2133		BGA 1515	--
8.	Whiskey Lake	14 nm	8/2018	L	4C+G		DDR4-2400		BGA 1528	--
9.	Coffee Lake R.	14 nm	10/2018	D	8C+G		DDR4-2666		LGA 1151	300 Series
10.	Comet Lake	14 nm	8/2019	L D	10C+G		DDR4-2666		L: BGA 1528 D: LGA 1200	D: 400 Series
11.	Rocket Lake	14 nm	3/2021	D	8C+G		DDR4-3200		LGA 1200	

3.6 How has the core count in Intel's client CPUs increased? (7)

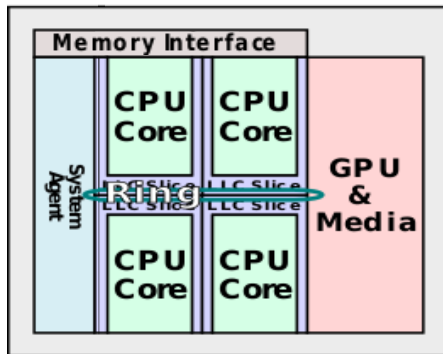
Phase 2: Increasing the maximum core count to 10 -3

	Processor	Technology	Date of intro.	Segment	Max. core count	Mem. speed (up to)	Mem. channels	Socket	PCH
8.	Cannon Lake	10 nm	5/2018	L:U	2C	DDR4-2400	2x64- bit	BGA 1440	--
10.	Ice Lake	10 nm	7/2019	L: G	4C+G	DDR4-3200 LPDDR4- 3733		BGA 1526	--
11.	Tiger Lake	10 nm	09/2020 01/2021 05/2021	L: G L: H L: H	4C+G 4C+G 8C+G	LPDDR4x-4267 DDR4-3200		BGA 1449 BGA 1449 BGA 1787	--
11.	Tiger Lake	10 nm	01/2021	D	8C+G	DDR4-3200		LGA 1700	500 Series
12.	Alder Lake	Intel 7	11/2021	D	8+8C+G	DDR5-4800		D: LGA 1700	600 Series
12.	Alder Lake	Intel 7	02/2022	L:H,P L: U	6+8C+G 2+8+G	DDR5-4800 LPDDR5-5200 LPDDR4x-4267		BGA 1744	--
12.	Alder Lake	Intel 7	05/2022	L: HX	8+8+G	DDR5-4800		BGA: 1781	600 Series
13	Raptor Lake	Intel 7	10/2022	D	8+16C+G	DDR5-5600		LGA 1700	700 Series
13.	Raptor Lake	Intel 7	01/2023	L:HX	8+16C+G	DDR5-5600		BGA 1964	700 Series
13.	Raptor Lake	Intel 7	01/2023	L:H,P L:U	6+8C+G 2+8C8C	DDR5-5200 LPDDR5x-6400 LPDDR4x-4267		BGA 1744	--
14	Raptor Lake Refresh	Intel 7	10/2023	D	8+16C+G	DDR5-5600		LGA 1700	700 Series
14	Raptor Lake Refresh	Intel 7	01/2024	L:HX	8+16C+G	DDR5-5600 DDR4-3200		BGA 1964	700 Series
14.	Raptor Lake Refresh	Intel 7	01/2024	L:U	2+8C+G	DDR5-5200 LPDDR5x-6400 LPDDR4x-4267		BGA-1744	--
14	Meteor Lake	Intel 4	12/2023	L:H L:U-15	6+8+2C+G 2+8+2+G	DDR5-5600 LPDDR5X-7467		BGA 2049	--
14	Meteor Lake	Intel 4	12/2023	L:U-9	2+8+2+G	LPDDR5X-6400		BGA 2551	--

3.6 How has the core count in Intel's client CPUs increased? (8)

Examples: Increasing core counts in Intel's Core-based client processors [24]

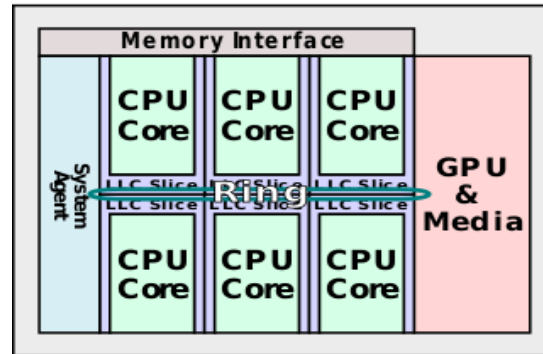
2nd Sandy Bridge



(2011)

Up to 4 cores

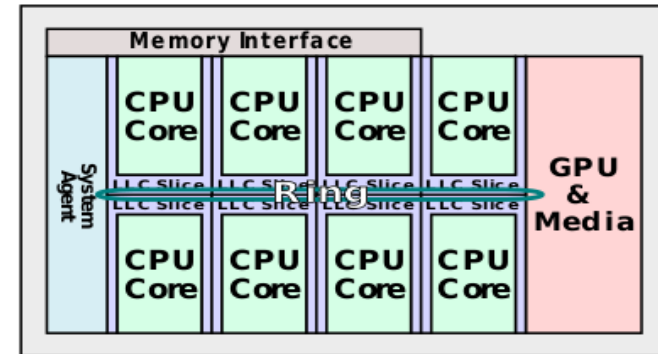
8th Coffee Lake



(2017)

Up to 6 cores

9th Coffee Lake Refresh



(2018)

Up to 8 cores

3.6 How has the core count in Intel's client CPUs increased? (9)

c) Phase 3: Increasing the maximum core count over 10 in Intel's monolithic big.little lines -1

- **AMD's** gaming DT processors traditionally do not include GPUs.
- In 2019, their Zen 2-based generation introduced 16 core models, intensifying competition and prompting Intel to push its client processor core count beyond 10.
- In response, **Intel** launched its 12th gen. Alder Lake line in 2021, featuring a hybrid architecture with up to 8 performance cores (P) and 8 efficiency cores (E), as shown in the following Table.

3.6 How has the core count in Intel's client CPUs increased? (10)

Phase 3: Increasing the maximum core count over 10 in Intel's monolithic big.little lines -2

	Processor	Technology	Date of intro.	Segment	Max. core count	Mem. speed (up to)	Mem. channels	Socket	PCH
8.	Cannon Lake	10 nm	5/2018	L:U	2C	DDR4-2400	2x64- bit	BGA 1440	--
10.	Ice Lake	10 nm	7/2019	L: G	4C+G	DDR4-3200 LPDDR4- 3733		BGA 1526	--
11.	Tiger Lake	10 nm	09/2020 01/2021 05/2021	L: G L: H L: H	4C+G 4C+G 8C+G	LPDDR4x-4267 DDR4-3200		BGA 1449 BGA 1449 BGA 1787	--
11.	Tiger Lake	10 nm	01/2021	D	8C+G	DDR4-3200		LGA 1700	500 Series
12.	Alder Lake	Intel 7	11/2021	D	8+8C+G	DDR5-4800		D: LGA 1700	600 Series
12.	Alder Lake	Intel 7	02/2022	L:H,P L: U	6+8C+G 2+8+G	DDR5-4800 LPDDR5-5200 LPDDR4x-4267		BGA 1744	--
12.	Alder Lake	Intel 7	05/2022	L: HX	8+8+G	DDR5-4800		BGA: 1781	600 Series
13	Raptor Lake	Intel 7	10/2022	D	8+16C+G	DDR5-5600		LGA 1700	700 Series
13.	Raptor Lake	Intel 7	01/2023	L:HX	8+16C+G	DDR5-5600		BGA 1964	700 Series
13.	Raptor Lake	Intel 7	01/2023	L:H,P L:U	6+8C+G 2+8C8C	DDR5-5200 LPDDR5x-6400 LPDDR4x-4267		BGA 1744	--
14	Raptor Lake Refresh	Intel 7	10/2023	D	8+16C+G	DDR5-5600		LGA 1700	700 Series
14	Raptor Lake Refresh	Intel 7	01/2024	L:HX	8+16C+G	DDR5-5600 DDR4-3200		BGA 1964	700 Series
14.	Raptor Lake Refresh	Intel 7	01/2024	L:U	2+8C+G	DDR5-5200 LPDDR5x-6400 LPDDR4x-4267		BGA-1744	--

3.6 How has the core count in Intel's client CPUs increased? (11)

c) Phase 3: Increasing the maximum core count over 10 in Intel's monolithic big.little lines -3

- Intel's 13th gen. Raptor Lake line, expanded the core configuration to a maximum of 8+16 cores, as shown in the following Table.

3.6 How has the core count in Intel's client CPUs increased? (12)

d) Phase 4: Providing additional CPU cores on the SOC die in Intel's heterogeneous big.little lines

Beginning with the 14th gen. Meteor Lake line, introduced in 2023, Intel provides 2 or 4 **additional CPU cores on the SOC die**, to allow switching off the CPU die during low loads.

W	Processor	Technology	Date of intro.	Segment	Max. core count	Mem. speed (up to)	Mem. channels	Socket	PCH
14 CUS1	Meteor Lake	Intel 4 TSMC N5/N6	12/2023	L: H L: U-15W	6+8+2C+G 2+8+2C+G	DDR5-5600 LPDDR5X-7467	2x64-bit	BGA 2049	--
14 CUS1	Meteor Lake	Intel 4 TSMC N5/N6	12/2023	L: U-9W	2+8+2C+G	LPDDR5X-6400		BGA 2551	--
14 CUS1	Meteor Lake	Intel 4 TSMC N5/N6	04/2024	L: UL	2+8C+G	LPDDR5X-7467		BGA 2049	--
CUS2	Lunar	TSMC N3B/N6	09/2024	L: V	4+4C+G	LPDDR5X-8533		BGA 2833	--
CUS2	Arrow Lake	TSMC N3B/N5P/N6	10/2024	D: G	8+16C+G	DDR5-5600 CUDIMM-6400		LGA1851	--
CUS2	Arrow Lake	TSMC N3B/N5P/N6	01/2025	L: U L: H L: HX	2+8+2C+G 6+8+2C+G 8+16C+G	DDR5-5600 CUDIMM-6400		BGA 2114	--
CU3	Panther Lake	Intel 1.8A/3, TSMC N3E/N6	01/2026	L: H	4+8+4+G	DDR5-7200 LPDDR5X-9600		BGA 2540	-

CUS: Core Ultra Series

3.7 How has the memory subsystem been enhanced?

3.7 How has the memory subsystem been enhanced?

The memory subsystem covers several issues, that we will discuss subsequently.

- 3.7.1 Evolution of the memory connection
- 3.7.2 At which rate have memory speeds been increased?
- 3.7.3 Introduction of laptop-oriented new memory types
- 3.7.4 Introduction of high-speed CUDIMM memories
- 3.7.5 Introducing in-package integrated main memory

3.7.1 Evolution of the memory connection

- 3.7.1 Evolution of the memory connection
- 3.7.2 At what rate have memory speeds been increased?
- 3.7.3 Introduction of laptop-oriented new memory types
- 3.7.4 Introducing on-package integrated main memory

3.7.1 Evolution of the memory connection (1)

3.7.1 Evolution of the memory connection

For connecting memory to the processor, there are **two options**, as depicted below.

Options for connecting memory to the processor

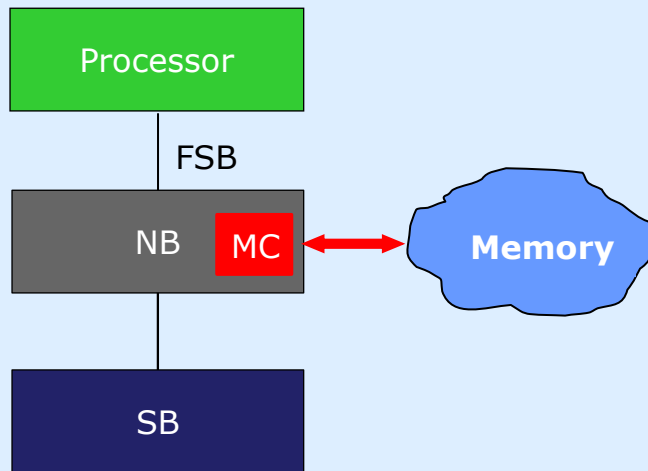
Connecting memory through the NB to the processor



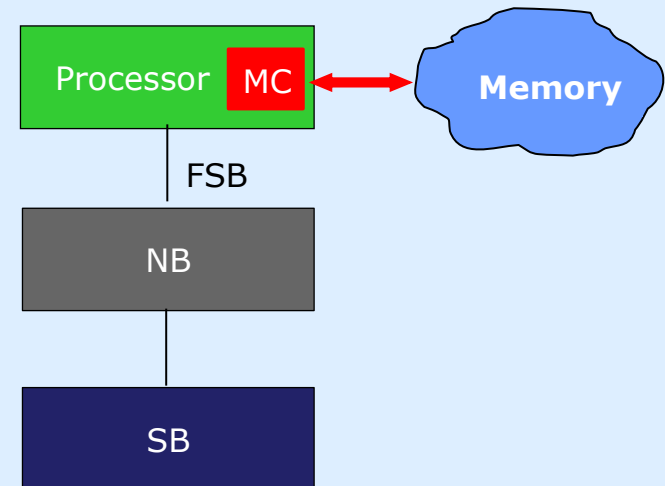
Connecting memory directly to the processor

The MC is integrated into the NB.
Data is transmitted via the FSB between the processor and memory.

The MC is integrated onto the processor die.
Data is transmitted directly between the processor and memory.



Up to the Penryn (2008)



From the Nehalem on (2009)

MC: Memory Controller NB: North Bridge SB: South Bridge

3.7.1 Evolution of the memory connection (2)

Benefits of connecting memory directly to the processor

- a) Connecting memory directly to the processor offers notable advantages, primarily by reducing memory latency.
- b) Another significant advantage - especially in multiprocessor server architectures - is the substantial increase in memory bandwidth.

Namely, when memory is connected via the Northbridge (NB), the FSB must handle all traffic for all processors, this leads to a memory bottleneck, as illustrated in the following Figure.

3.7.1 Evolution of the memory connection (3)

Memory bandwidth bottleneck in a 4S server when the memory is connected through the NB and a single FSB (based on [55])

In this case, the **FSB is shared** among all four processors, the average per-core bandwidth is **one-fourth** of the total FSB bandwidth, as indicated in the Figure below.

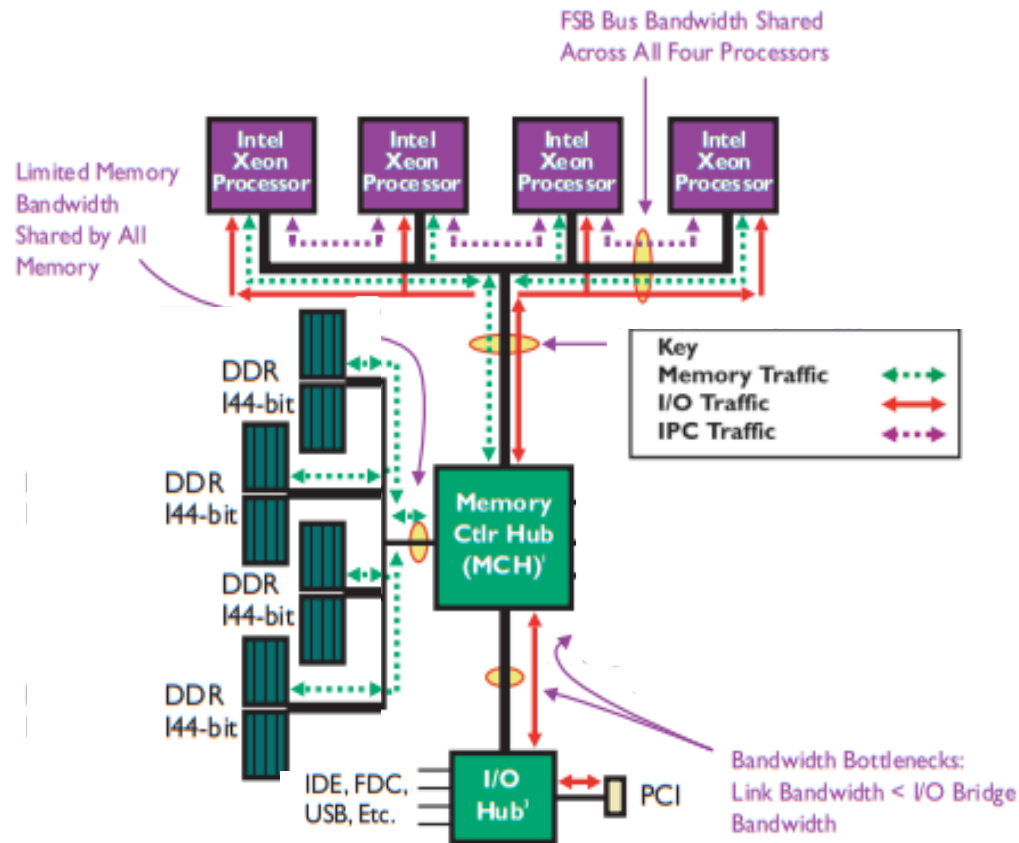


Figure: Connecting memory through the NB and a single FSB in a 4S server

4S: 4-Socket FSB: Front-Side Bus (Rendszerbusz)

3.7.1 Evolution of the memory connection (4)

Eliminating the memory bandwidth bottleneck discussed in 4S servers by directly connecting memory to the processors (based on [55])

When memory is directly connected to the processors, the **FSB does not limit the memory bandwidth anymore** in multi-processor servers, and **each processor gets the full bandwidth of the memory channel connected to it**, as shown in the Figure below.

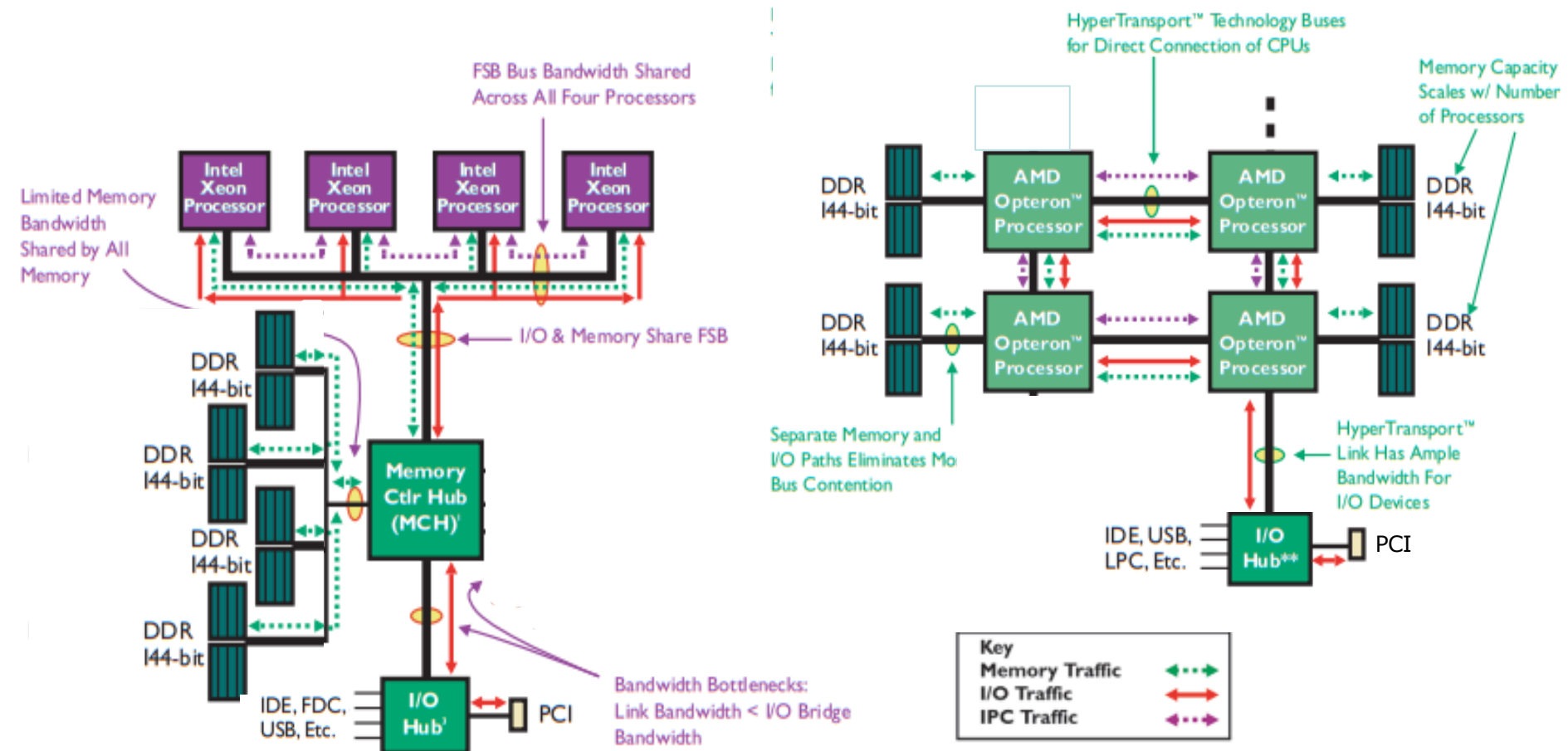


Figure: Connecting memory via the FSB Figure: Connecting memory directly to the processors

3.7.1 Evolution of the memory connection (5)

The transition to direct memory connections in processors of various vendors
Due to its benefits, **processor vendors gradually shifted to direct memory connection**, as depicted in the Table below.

Options for connecting memory to processors		
	Connecting memory through the NB to the processor	Connecting memory directly to the processor
Examples	HP PA-7000 (1991)	HP PA-7100 (1992) HP PA-7150 (1994) and subsequent PA models
	UltraSPARC II (1C) (~1997)	UltraSPARC III (2001) and all subsequent Sun models
	AMD's K7 lines (1C) (1999-2003)	Opteron server lines (2C) (2003) and all subsequent AMD lines
	POWER4 (2C) (2001)	POWER5 (2C) (2005) and subsequent POWER families
	Core 2 Duo line (2C) (2006) and all preceding Intel lines	Nehalem lines (4) (2008) 2. gen. client-oriented Nehalem line and all subsequent Intel lines
	Core 2 Quad line (2x2C) (2006/2007)	
	Penryn line (2x2C) (2008)	
	Montecito (2C) (2006)	Tukwila (4C) (2010)

3.7.1 Evolution of the memory connection (6)

Comparing Intel's decision to connect memory directly to the processor with similar developments from other firms

- As shown in the previous Figure, Intel began connecting memory directly to the processors with its server-oriented Nehalem line in 2008, followed by the client-oriented 2nd gen. Nehalem family in 2009.
- It is worth noting that Intel was relatively late in adopting direct memory connection; other companies transitioned earlier, for example, AMD implemented direct memory connections approximately five years prior.
- The Table below illustrates Intel's shift toward a direct memory connection.

3.7.1 Evolution of the memory connection (7)

Intel's transition to direct memory connection in their Core-line

Gen.	Processor	Technology	Date of intro.	Segment	Max core count	Connecting memory	Mem. speed (up to)	Mem. channels	Socket	MCH/PCH
1.	Core 2	65 nm	6/2006	L D	2C	Via the NB 	DDR2-800	2	LGA 775	945-975
	Core 2 Quad	65 nm	6/2007	D	2x2C		DDR3-1067		LGA 775	3 Series
	Penryn	45 nm	6/2008	D	2x2C		DDR3-1067		LGA 775	4 Series
	2. G. Nehalem (Lynnfield)	45 nm	9/2009	D	4C	Via the processor	DDR3-1333		LGA 1156	5 Series
	Westmere (L: Arrandale) D: Clarkdale/	32 nm	1/2010	L D	2C+G				L: BGA 1288 D: LGA 1156	5 Series
2.	Sandy Bridge	32 nm	1/2011	L D	2C/4C+G		DDR3-1333		L: PGA 988	6 Series
3.	Ivy Bridge	22 nm	4/2012	L D	4C+G		DDR3-1600		D: LGA 1155	7 Series
4.	Haswell	22 nm	6/2013	L D	4C+G		DDR3-1600		L: BGA 1168 D: LGA 1150	D: 8 Ser.
4.	Haswell Refr.	22 nm	5/2014	D	4C+G		DDR3-1666			D: 9 Ser.
5.	Broadwell	14 nm	9/2014	L D	4C+G		DDR3-1866			D: 9 Ser.
6.	Skylake	14 nm	10/2015	L D	4C+G		DDR4-2133		L: BGA 1356 D: LGA 1151	D: 100 Series
7.	Kaby Lake	14 nm	8/2016	L D	4C+G		DDR4-2400			D: 200 Series
8.	Coffee Lake	14 nm	10/2017	L D	6C+G		DDR4-2666		L: BGA 1440 D: LGA 1151	D: 300 Series
8.	Amber Lake	14 nm	8/2018	L	2C+G		LPDDR3-2133		BGA 1515	--
8.	Whiskey Lake	14 nm	8/2018	L	4C+G		DDR4-2400		BGA 1528	--
9.	Coffee Lake R.	14 nm	10/2018	D	8C+G		DDR4-2666		LGA 1151	300 Series
10.	Comet Lake	14 nm	8/2019	L D	10C+G		DDR4-2666		L: BGA 1528 D: LGA 1200	D: 400 Series
11.	Rocket Lake	14 nm	3/2021	D	8C+G		DDR4-3200		LGA 1200	

3.7.2 At what rate have memory speeds been increased?

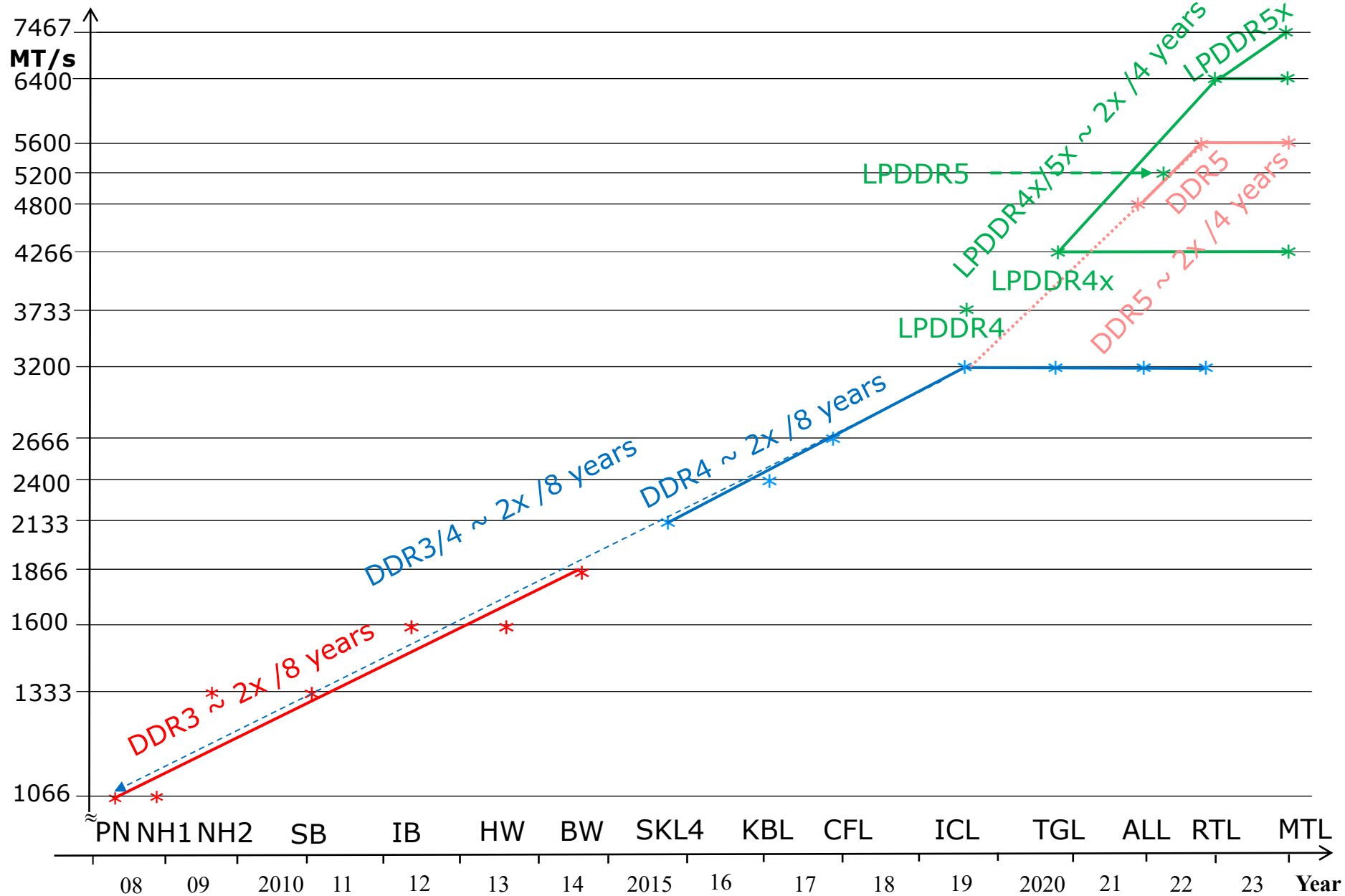
3.7.2 At what rate have memory speeds been increased? (1)

3.7.2 At what rate have memory speeds been increased?

- **Memory technology** has steadily evolved over time, but **at what rate?**
- The following Figure provides insight into this question **for the client processors of Intel's Core family.**
- It shows the progression of **maximum memory transfer rates across memory generations** **using a logarithmic scale** to highlight its exponential growth.

3.7.2 At what rate have memory speeds been increased? (2)

At what rate have memory speeds been raised in Intel's client processors? -1



3.7.2 At what rate have memory speeds been increased? (3)

At what rate have memory speeds been raised in Intel's client processors? -2

- The graph above indicates that the **transfer rate of DDR memories** within the same technology rises **exponentially**.
- As shown,
DDR3 and DDR4 memories typically require around **8 years to double** their transfer rates, while **DDR5 and LPDDRx** memories only need about **4** years for doubling.
- Here, we note that regarding the transfer rates of DDR5 and LPDDR memories, currently, only a limited number of data points are currently available.
Therefore, the related statement should be taken with caution.
- Additionally, it is worth mentioning that **Intel's Alder Lake (ALL) and Raptor Lake (RTL) processors support both the DDR4 and DDR5 memory types**.
Notably, both processor families have the same DDR4 transfer rate of DDR4-3200.

LPDDR: Low Power DDR

3.7.3 Introduction of laptop-oriented new memory types

3.7.3 Introduction of laptop-oriented new memory types

In recent years, Intel has introduced **three new memory types**, primarily **for its laptop processors**, which are:

- a) the LPDDR and LPDDR_x memories and
- b) the LPCAMM2 memory and
- c) the CUDIMM memory.

In this Section, we will describe these new memory types.

LPCAMM: pronounced as $\varepsilon l . p i : ' k \ae m$.

CUDIMM: pronounced as $k j u : d i m$.

3.7.3 Introduction of laptop-oriented new memory types (2)

a) What are LPDDR and LPDDR_x memories, and why are they important?

- As their names suggest, **LPDDR** (Low Power DDR) and its successor, **LPDDR_x** memories are designed to **consume significantly less power** than standard DDR memories, making them ideal **for laptop and embedded applications**.
This energy efficiency is primarily achieved through **reduced operating voltages**, which lower both the dynamic and static dissipation.
- Despite their lower power requirements, **LPDDR memories deliver the same data transfer rates to conventional DDR counterparts**, ensuring high performance without compromising battery life.
- **LPDDR_x** memories have an **even more advanced memory design** than their LPDDR predecessors, offering even **higher data rates** while consuming **less power compared to standard LPDDR memories**.
- Note that both LPDDR and LPDDR_x memories are typically packaged using **BGA sockets** that are **soldered directly to the mainboard**.

This design choice **enhances compactness but eliminates upgrading and replacing**.

- The Table below provides a summary of **the main features** of LPDDR and LPDDR_x memories.

- **Remark**

Additionally, there were **more advanced LPDDR memories** designed by several vendors referred to as **LPDDRE** memories.

They provide **higher transfer rates** than traditional LPDDR memories (see next Table).

3.7.3 Introduction of laptop-oriented new memory types (3)

Types and main features of subsequent LPDDR/LPDDR_x generations [56] -1

LPDDR	1	2	3	4	4X	5	5X
Memory array clock (MHz)	200	200	200	200	266.7	400	533
Data width	x16/x32	x16/x32	x16/x32	x16			
I/O bus clock frequency (MHz)	200	400	800	1600	2133	3200	4267
Peak Data transfer rate for LPDDR (MT/s)	400	800	1600	3200	4267	6400	8533
Peak Data transfer rate for LPDDR _E (MT/s)	533	1066	2133	4267			
Supply voltage(s)	1.8 V	1.2 V	1.2 V	1.1 V	0.6 V, 1.1 V	0.5 V, 1.05 V	0.5 V, 1.05 V
Introduction	2009	2009	2012	2014	2017	2019	2021

- **LPDDR3/LPDDR4** memory modules operate at significantly **lower voltages** than traditional **DDR** memories (**DDR**: 2.5 V, **DDR2**: 1.8 V, **DDR3**: 1.5 V, **DDR4**: 1,2 V).
- **Note** that despite their lower power consumption, **LPDDR** and **standard DDR** memory generations offer the **same maximum data rates**.
- **LPDDR_x** memories introduce further advancements, including the use of **dual voltages**, which enable **even higher transfer rates** than earlier LPDDR designs.
- Notably, **LPDDR5** memories switched to **dual voltage designs** as well.

3.7.3 Introduction of laptop-oriented new memory types (4)

Principle of the LPDDR4X memory (released by Samsung in 01/2017) [56]

- The **LPDDR4X** design introduces a **key innovation** by **separating the output driver voltage (V_{DDQV}) from the supply voltage (V_{DDQ})**, **reducing V_{DDQV} from 1.1 V to just 0.6 V**, as illustrated in the Figure below.

This results in both **higher transfer rates** and **greater power savings**, as the next Figure shows.

- However, the new design comes with a trade-off, it needs an **additional power rail**.

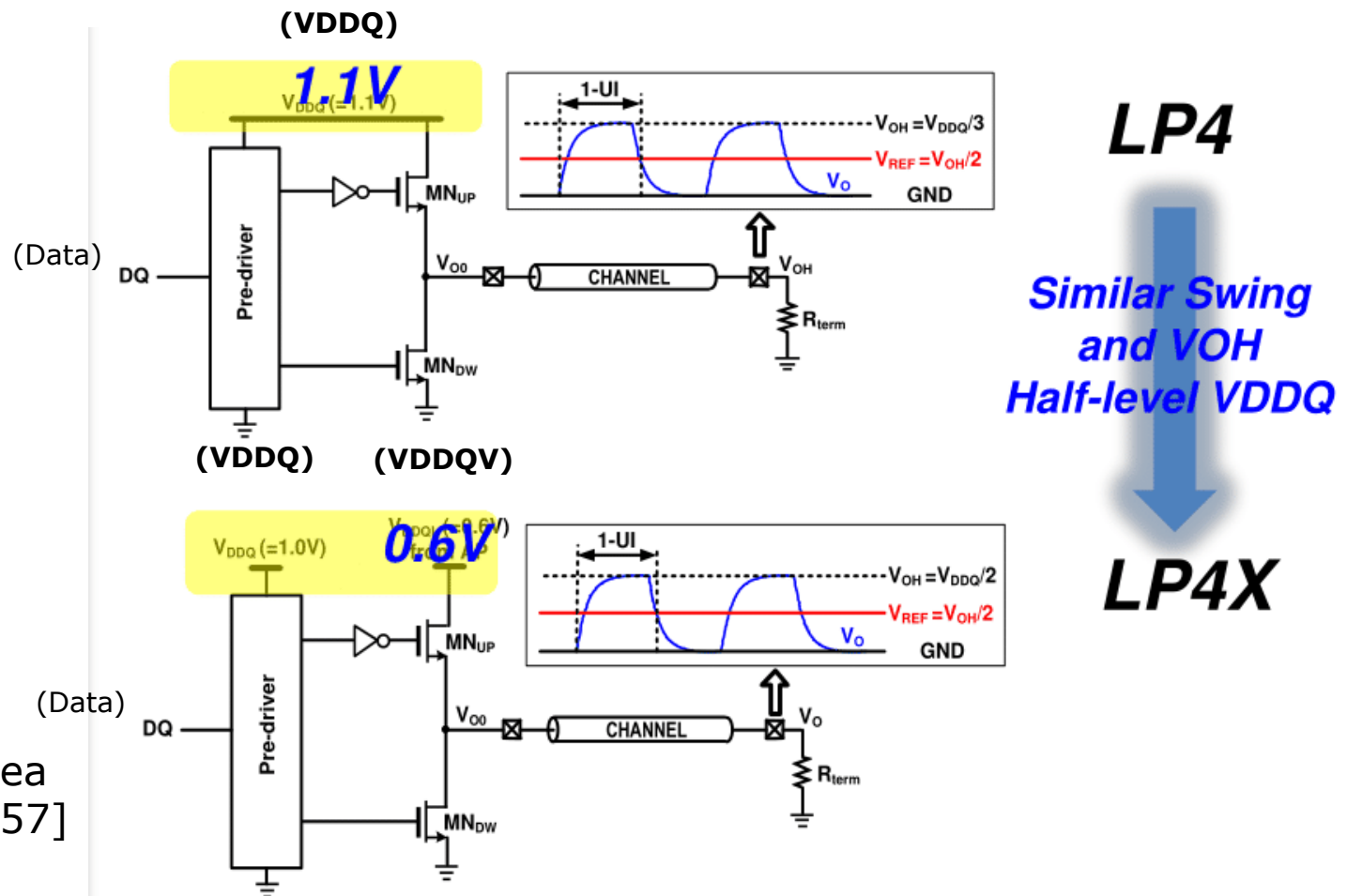


Figure: The idea of LPDDR4X [57]

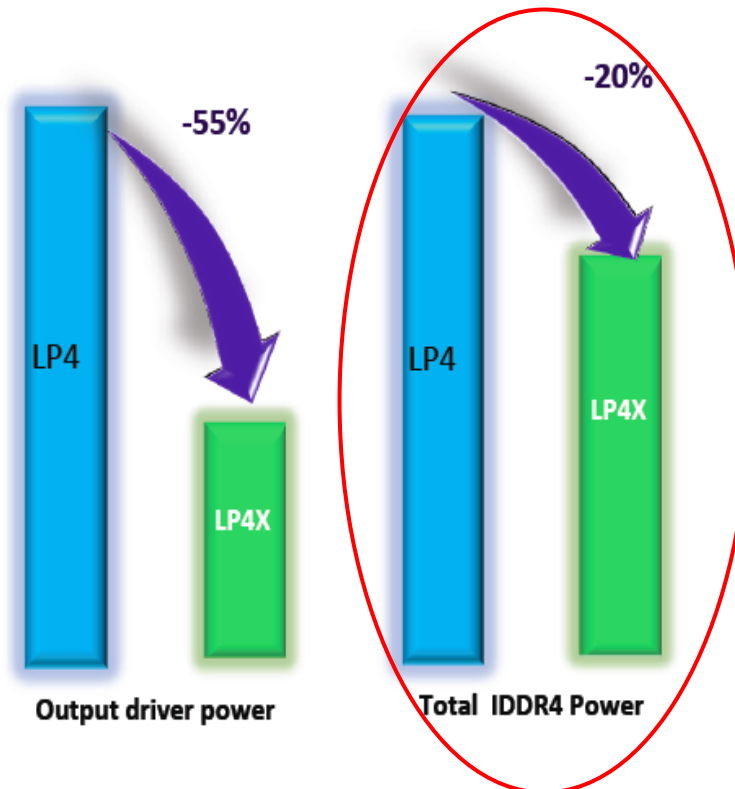
3.7.3 Introduction of laptop-oriented new memory types (5)

Power consumption of LPDDR4x memories compared to LPDDR memories [58]

LP4/LP4x Spec Comparison

Spec	LPDDR4	LPDDR4X
Function	Same	
Speed*	3200/3733/4266	3200/3733/ 4266
AC Timing	Same	
VDD1/VDD2	Same(VDD1=1.8V/VDD2=1.1V)	
VDDQ	1.1V	0.6V
Power Eff. (mW/GB/s)	1X	0.8X

Power reduction by LPDDR4x



As shown in the Figure, LPDDR4x memories consume approximately 20 % less power than LPDDR4 memories.

3.7.3 Introduction of laptop-oriented new memory types (6)

Standardization of the LPDDR and LPDDR4X memories [56]

- Initially, [Micron](#) and [Samsung](#) introduced [low-power memory solutions](#) for laptop processors, branding them as [Mobile DDR memories](#).

By 2009, this naming was changed to [LPDDR1](#), marking the beginning of a new memory technology.

Following the success of the early implementations of this memory type, [JEDEC](#) released formal specifications starting with [LPDDR2](#) in 2009.

In the following years, [JEDEC](#) released the subsequent LPDDR standards, such as the [LPDDR3 to LPDDR5](#) specifications.

Recently, in 07/2015 JEDEC released the LPDDR6 standard as well.

The specified peak data transfer rate is 14.4 Gb/s and the sub-channel width is x12.

- In 2016, [Samsung](#) unveiled its [LPDDR4x](#) design, which introduced [dual operating voltages](#), to achieve both [lower power consumption](#) and [higher speed](#).
- JEDEC later standardized this innovation [with the LPDDR4x specifications](#) released in 2017, followed by the [LPDDR5x](#) standard in 2021.

JEDEC (dʒɛdɛk) : It is the leading organization for setting open standards for the microelectronics industry.

Use of LPDDR memories in client processors

- LPDDR4 memories were initially developed for smartphones and tablets, where low power consumption is critical.

Over time, its efficiency led to broader adoption, including into client processors.

- More recently, both LPDDR and LPDDR_x memories have been widely utilized in Intel's client processors, as indicated in Section 3.7.2 before.

This shift marks the growing demand for energy-efficient, high-performance memory solutions across computing platforms.

3.7.3 Introduction of laptop-oriented new memory types (8)

b) The LPCAMM memories [59]

- **LPCAMM** stands for **Low-Power Compression Attached Memory Module**, a new **memory technology** standardized by JEDEC for energy efficient computing.

Its distinctive feature lies in its **electrical interface**, **LPCAMM modules establish electrical contacts through compression** rather than insertion into a socket or soldering.

- This approach enables a compact, **replaceable design in contrast to the BGA socketed memories**.
- So far, **two generations** of LPCAMM memories have been released:
 - The **1st gen.** LPCAMM memory module is a **single-channel LPDDR5 design**, standardized by JEDEC in **03/2023**.
 - The more advanced **2nd gen.** LPCAMM2 memory module supports **dual-channel LPDDR5 or LPDDR5x memory**, it was standardized in **12/2023**.
- Next, we will give a **brief overview of the 2nd gen. LPDDR2** memory module.

JEDEC: It is the leading body for establishing microelectronic standards.

3.7.3 Introduction of laptop-oriented new memory types (9)

The LPCAMM2 memory module [59]

- **LPCAMM2 memory modules** provide dual 64-bit LPDDR5/x or DDR5 memory channels. LPCAMM places the memory chips on thin, replaceable circuit boards (as seen in the Figure below).



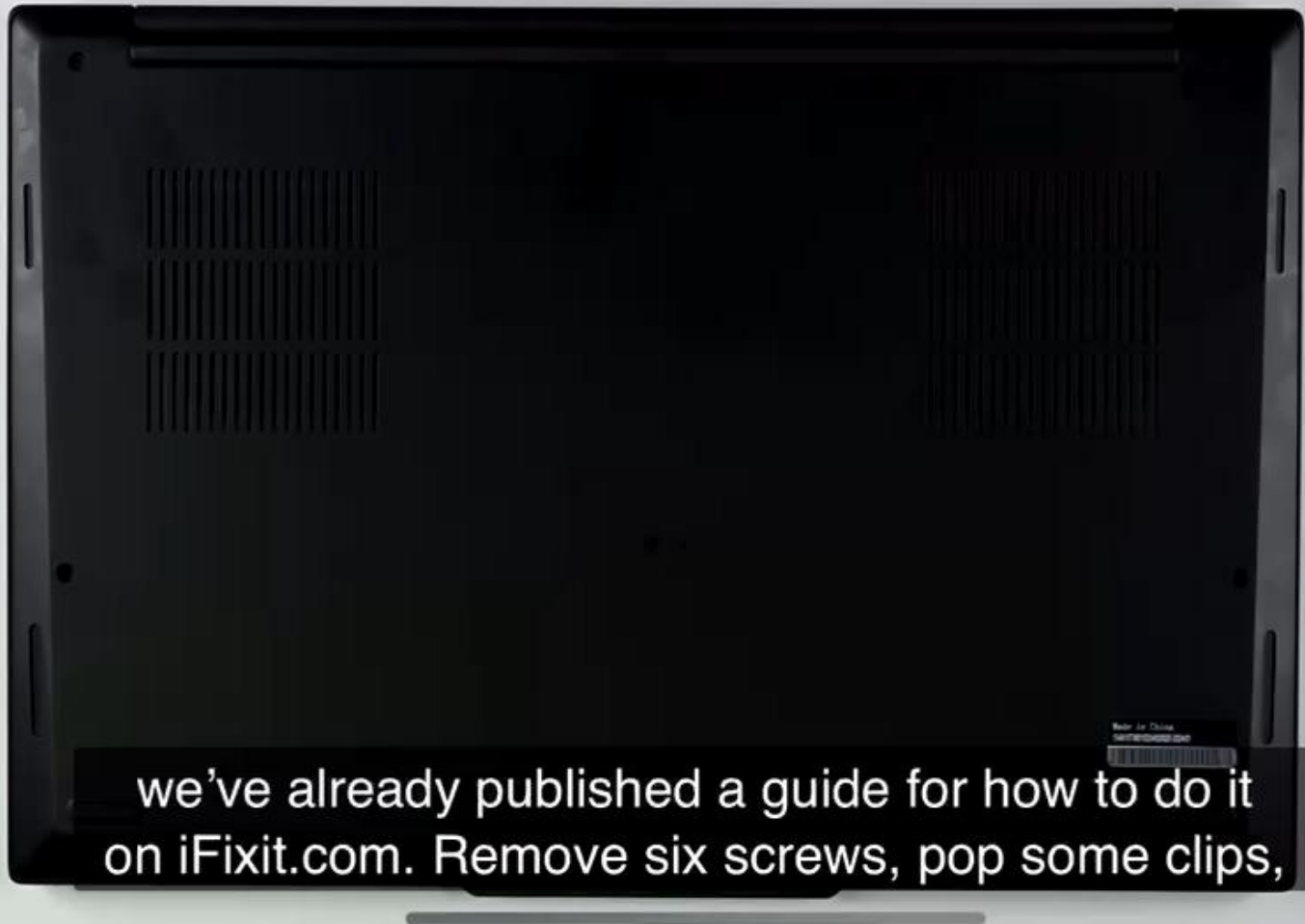
(Összenyomással
csatlakoztatott)

Figure: LPCAMM2 memory module [59]

- Unlike traditional memory modules that are either inserted into sockets or soldered directly onto the mainboard, **LPCAMM memory technology** establishes **electrical contacts through compression**, as shown in the following video.

3.7.3 Introduction of laptop-oriented new memory types (10)

Demonstration of removing and replacing an LPCAMM2 memory module [124]



we've already published a guide for how to do it on iFixit.com. Remove six screws, pop some clips,

Use of the LPCAMM memory technology

- The LPCAMM2 memory made its debut in 2024 with the Lenovo (ThinkPad P1 Gen 7) laptop which provides 64 GB of LPDDR5x memory.
- As of Q4 2024, several companies, including Micron, Samsung, and Hynix, are offering or preparing LPCAMM2 memories with capacities of 32 GB or 64 GB.

3.7.3 Introduction of laptop-oriented new memory types (12)

c) CUDIMM memories

- **CUDIMM** means **C**locked, **U**nbuffered **DDR5** DIMM memory modules.
- They provide **higher speeds** than the conventional DDR5 memory by integrating a **clock driver (CKD)** directly onto the memory module, as shown below.
- This component **reshapes both the timing and voltage of the incoming clock signal**, significantly **improving signal integrity**.

The new layout allows a **doubling of the clock frequency** delivered by the memory controller (QCK is twice as high as DCK).

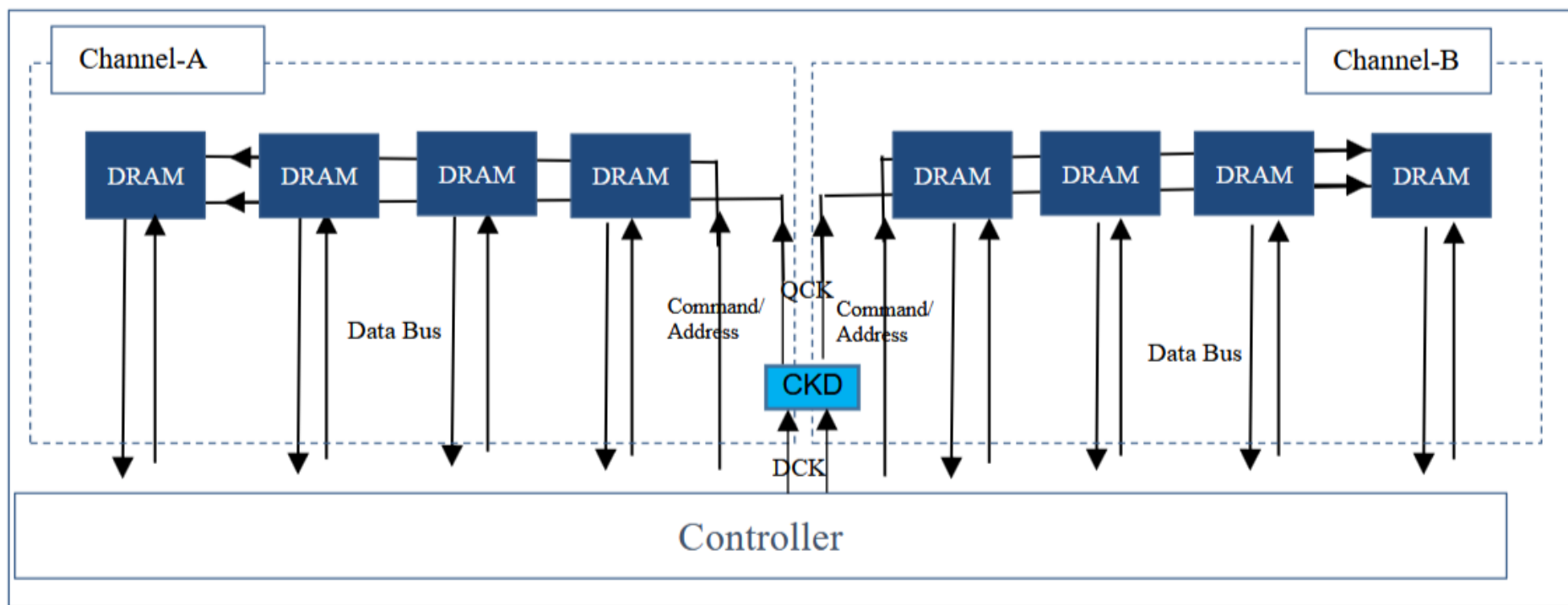


Figure: Example topology of a CUDIMM memory [146] JEDEC Standard No. 323

3.7.3 Introduction of laptop-oriented new memory types (13)

Example CUDIMM module [147]

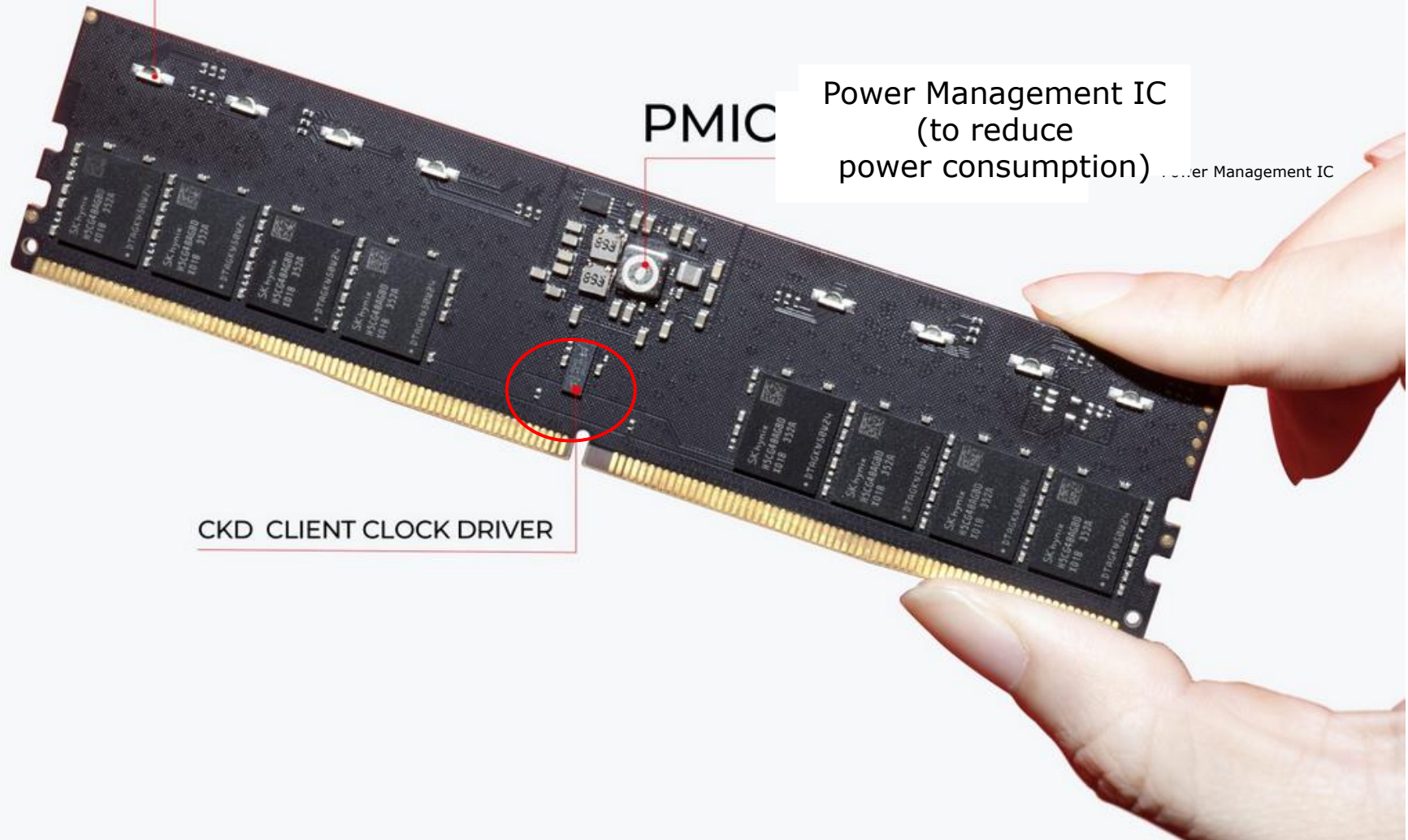
16 LED LIGHT FOR BETTER
RGB EXPERIENCE

PMIC

Power Management IC
(to reduce
power consumption)

Power Management IC

CKD CLIENT CLOCK DRIVER



Standardization of CUDIMM memories

- In early 2024, JEDEC standardized DDR5 CUDIMM memories.

These modules are drop-in compatible with existing DDR5 UDIMMs, using the same 288-pin socket and operating at 1.1.V supply voltage.

- The standard specifies memory capacities of 16/24/32 or 64 GB.
- CUDIMM memory modules feature two operation modes.
 - In one mode, they double (and potentially quadruple in future designs) the clock frequency received from the memory controller.
 - In the other, they simply bypass the clock frequently received.
- This dual-mode capability allows CUDIMMs to deliver improved performance while maintaining compatibility with existing DDR5 platforms.

DIMM: Dual-In-Line Memory Module

UDIMM: Unbuffered DIMM

JEDEC: (Joint Electron Device Engineering Council) is a semiconductor standardization body.

Emergence and use of CUDIMM memories

- **First CUDIMM memories** appeared on the market from various vendors, such as Micron (Crucial), Kingston, and others with speeds up to DDR5-9600 **by the end of 2024**.
- **As of 01/2025**, several OEMs offer **mainboards** with **2x16/2x24 or 2x 32GB CUDIMM** memories operating at speed grades of **up to DDR5-9600** for
 - Intel's 14th gen. **Arrow Lake (2024)** and
 - AMD's Zen 4-based **Ryzen 8000G DT APU series (2024)** as well as Zen 5-based **Ryzen 9000X (2024)** gaming DT processors.

3.7.3 Introduction of laptop-oriented new memory types (16)

Remarks on memory types targeting server processors -1

- **Server processors** must support **large memory capacities**, which requires connecting multiple memory modules per channel.

However, **parallel connected memory modules** introduces **high electrical load**, especially **on the clock, address and control signal lines** of memory controllers.

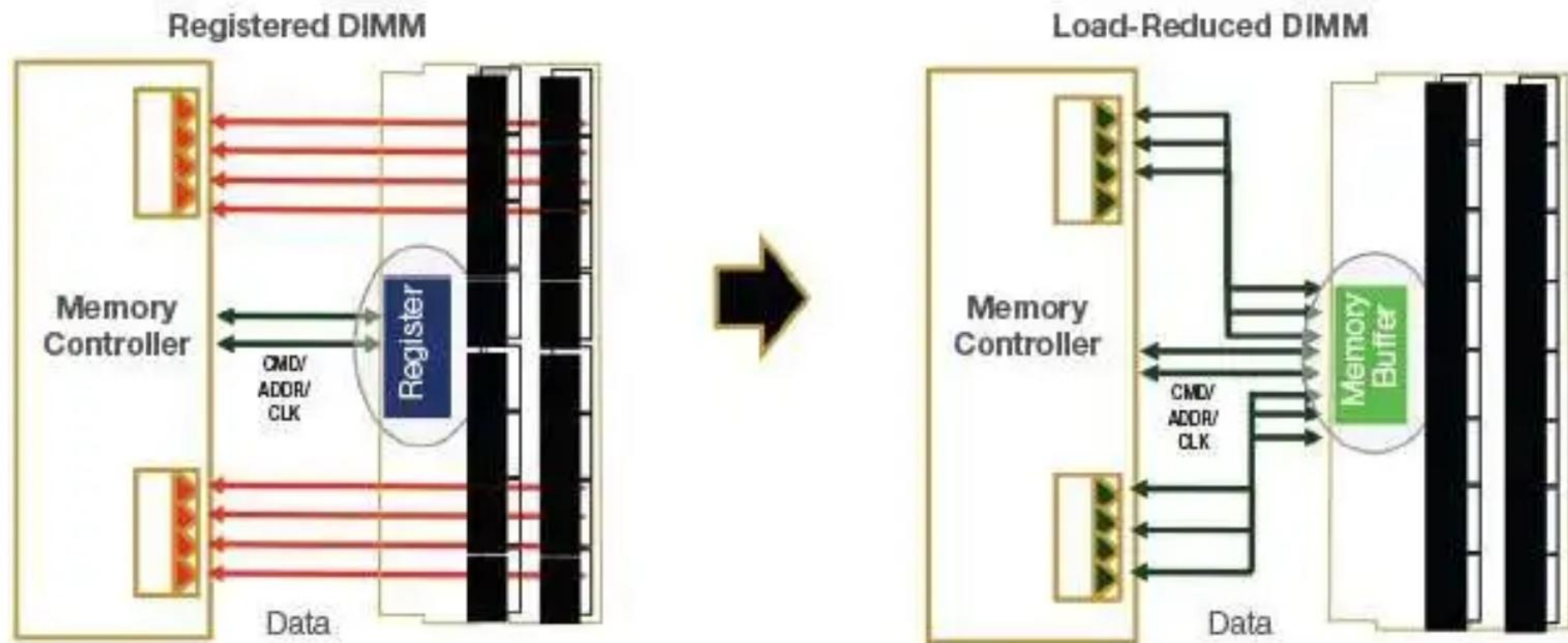
Excessive load can lead to **signal degradation and timing errors**, which cause memory errors.

- **RDIMM and LRDIMM** are memory types especially developed to mitigate these issues **by incorporating buffering mechanisms** to preserve signal integrity.

RDIMM Registered DIMM (Buffering of clock, command, and address signals)

3.7.3 Introduction of laptop-oriented new memory types (17)

Contrasting the layouts of RDIMM and LRDIMM memories [173]



- **RDIMM** memory modules include a dedicated component on the module (called Register in the Figure), which buffers the command, address, and clock signals arriving from the memory controller. to the DRAM chips.

However, RDIMMs do not buffer data signals, data flows directly between the MC and the DRAM chips.

- In contrast, **LRDIMM** memory modules extend buffering to the data signals as well, as indicated by the Memory Buffer component in the Figure.

3.7.3 Introduction of laptop-oriented new memory types (18)

Comparing the use of various buffering techniques implemented in memories of client and server processors

DIMM Type	Clock driver	Buffering of clock, command and address signals	Buffering of data signals	Typical use
UDIMM	✗ No	✗ No	✗ No	Clients, workstations
CUDIMM	✓ Yes	✗ No	✗ No	Clients
RDIMM	✗ No	✓ Yes	✗ No	Standard servers
LRDIMM	✗ No	✓ Yes	✓ Yes	High-capacity servers

UDIMM: Unbuffered DIMM

3.7.4 Introducing on-package integrated main memory

3.7.4 Introducing on-package integrated main memory (1)

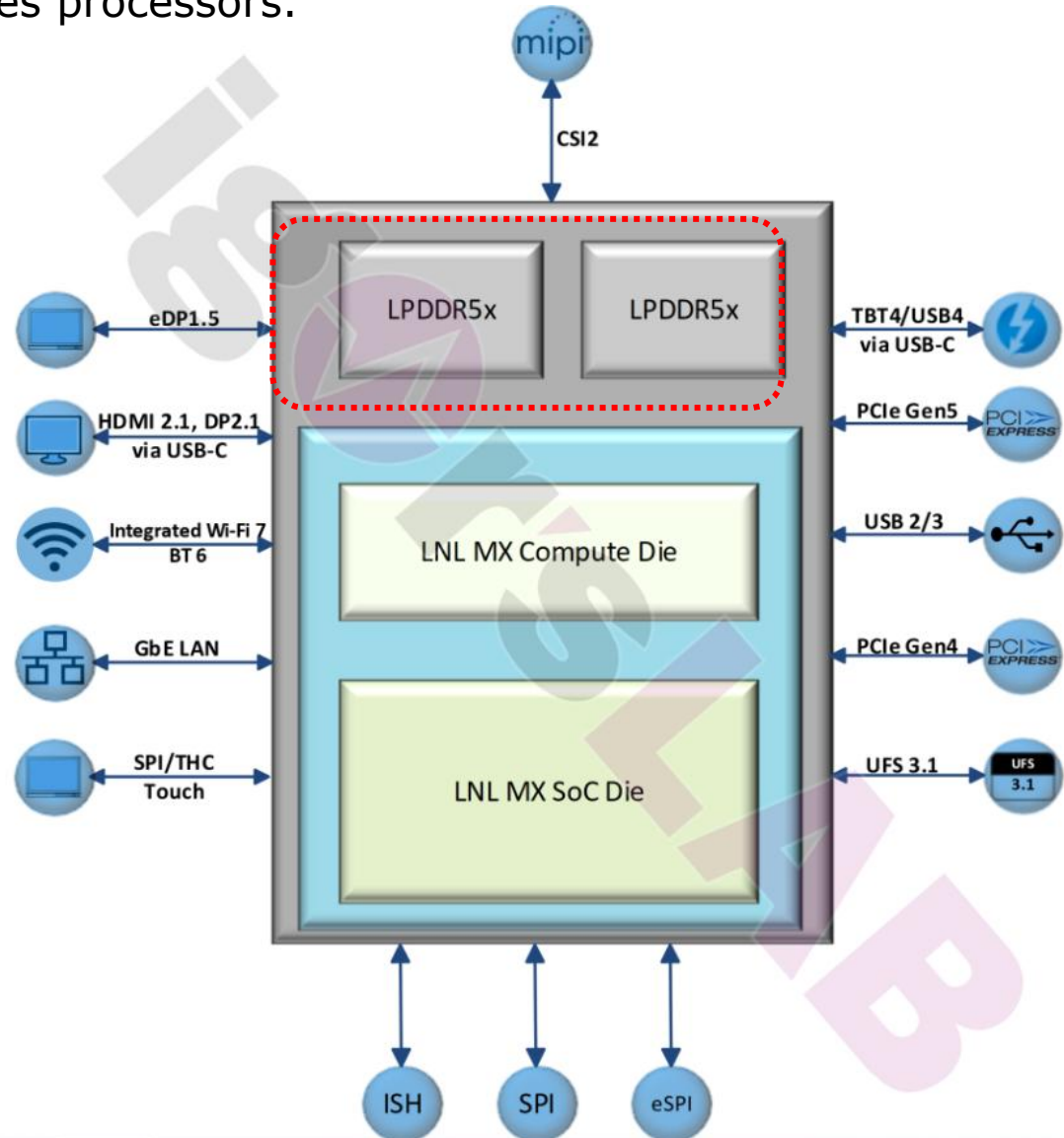
3.7.4 Introducing on-package integrated main memory

- In 2024 Intel released its **Lunar Lake** laptop processor, designed to compete with ARM-based **Snapdragon X** series processors.

A key innovation of the **Lunar Lake** line of processors is the **integration of 2x8 GB or 2x16 GB LPDDR5x-8533 main memory into the package**, as, illustrated on the right.

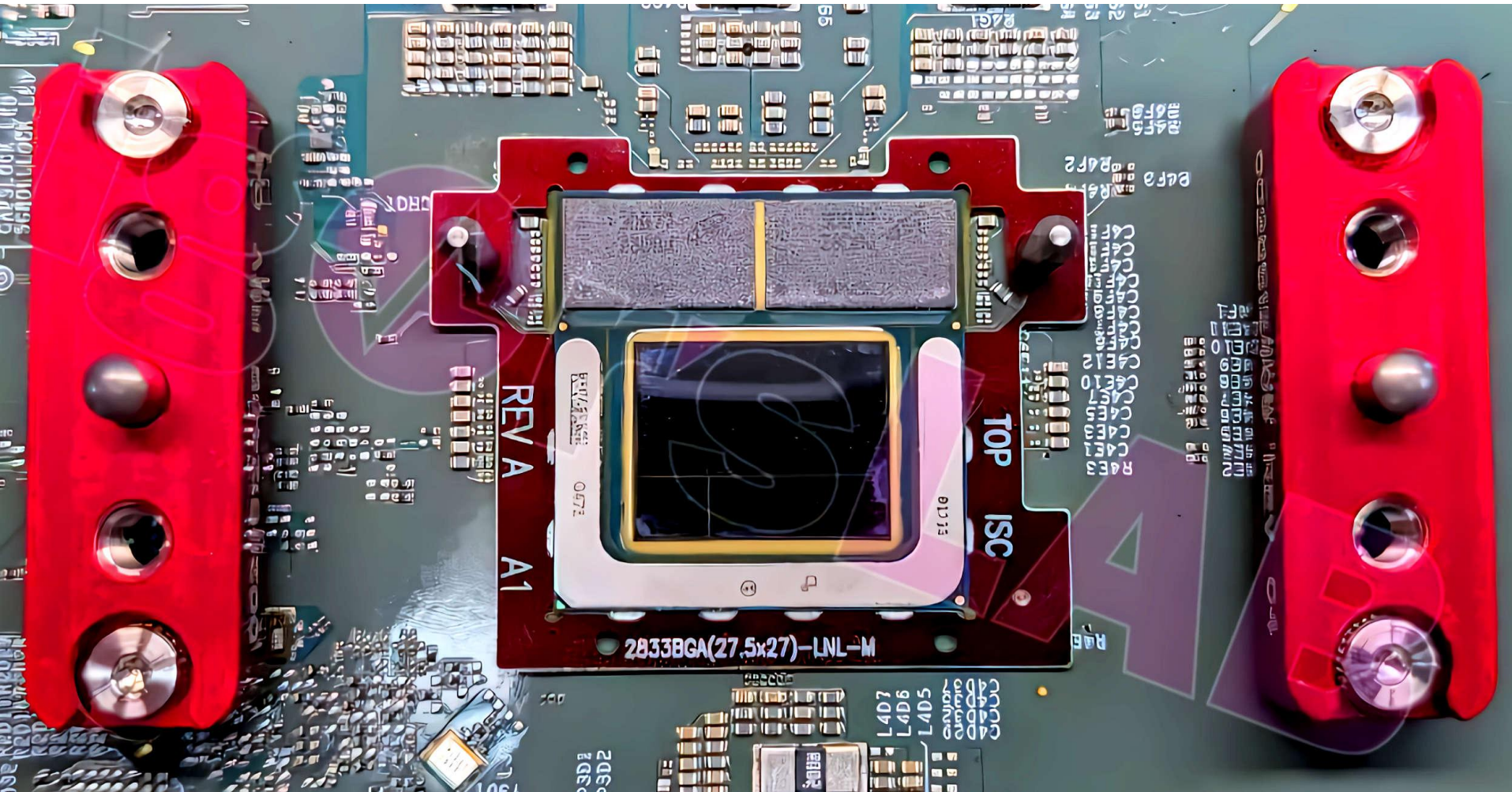
On-Package integrated memories offer **lower memory access latencies**, but they prevent the ability to replace or expand the main memory.

[157].



3.7.4 Introducing on-package integrated main memory (2)

Intel's Lunar Lake processor with dual on-package LPDDR5x memory dies [164]



3.7.5 Introducing on-package integrated main memory (3)

Remark 1

Apple's M-series processors also came with **on-package-integrated main memories**.

Apple's M1 processor, launched in 2020, comprises three on-package integrated dies: the CPU die, and two low-power LPDDR4x-4266 memory dies, each of 8 GB size, as depicted in the Figure below.

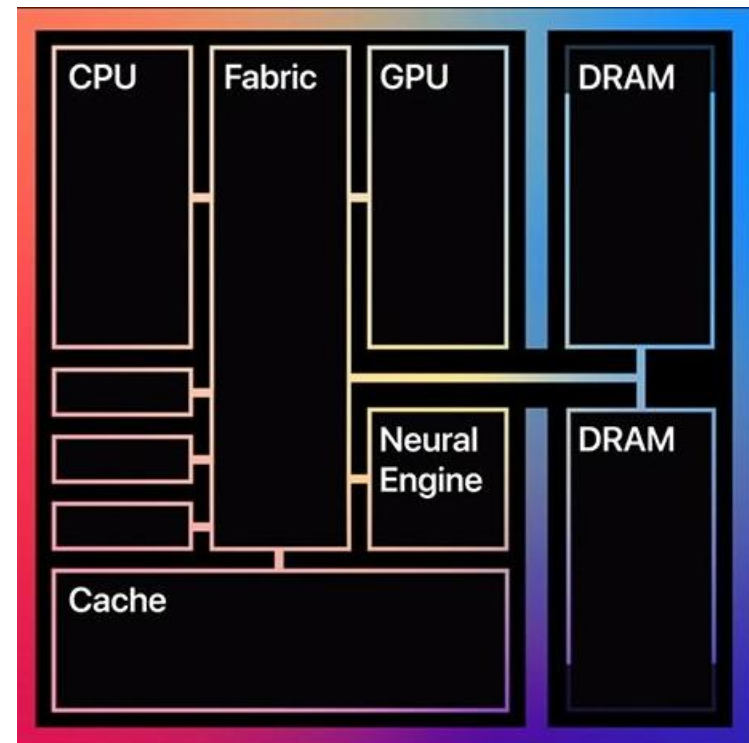
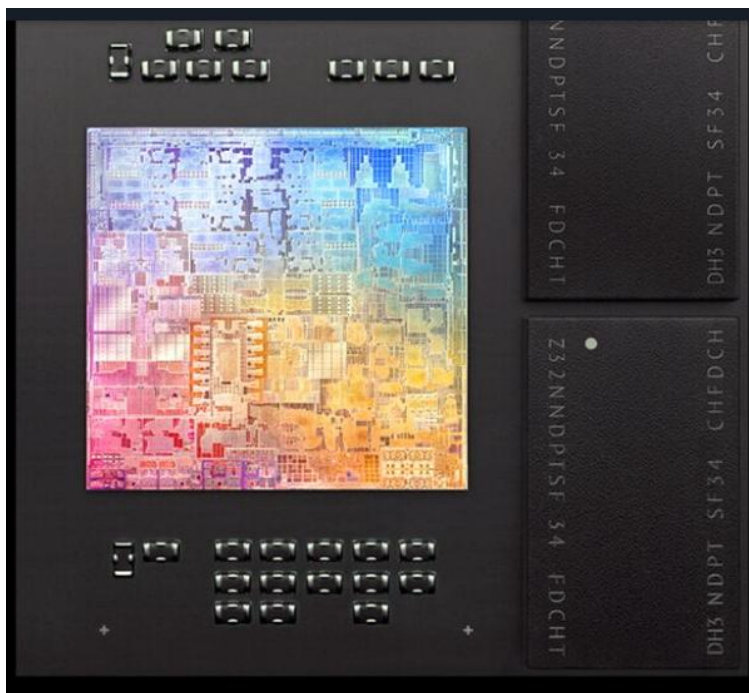


Figure: Layout of Apple's M1 chip (2020) [132]

3.7.5 Introducing on-package integrated main memory (4)

Remark 2

- Smartphone processors utilize another type of on-package integrated main memories, known as **PoP (Package-on-Package) memories**.
- This technology involves **stacking dies vertically and properly interconnecting their solder balls**, as illustrated in the Figure below.

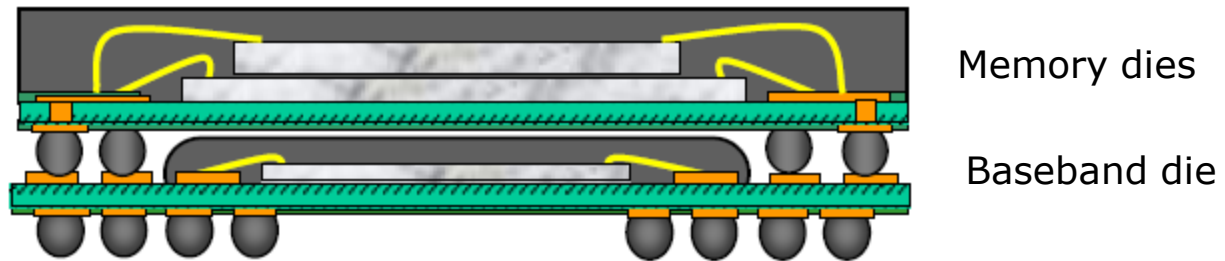


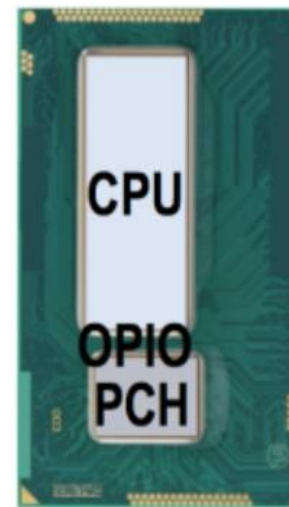
Figure: Possible PoP structure for implementing a baseband or application processor and two memory dies [133]

- The first PoPs emerged in cell phones around 2004 [133].
- Apple's original iPhone was an early example of using the PoP technology.

3.7.5 Introducing on-package integrated main memory (5)

Remark: Introduction of on-package integrated PCHs in Intel's Core-based "laptop" lines

- Intel introduced **on-package-integrated PCHs** in their **4th gen. "laptop" Haswell series in 2013**, as shown in the Figure below.
- Subsequently, **all of Intel's laptop-oriented processors** (characterized by low TDP values of less than 50 W), implement **in-package integrated PCHs**, with a few exceptions.



CPU-PCH:
2 GT/s

4 GB/s
32 mW
(1 pJ/bit)

Figure: On-package integrated PCH

- In this solution, the CPU die, and the PCH die are interconnected by an **IO interface** called **OPIO (On-Package I/O)**.
- The OPIO interface allows for high-bandwidth (4 GB/s) low-power (1 pJ/bit) communication between the two dies [148].
- The two dies are **enclosed by a Multi-Chip Packaging (MCP)**.

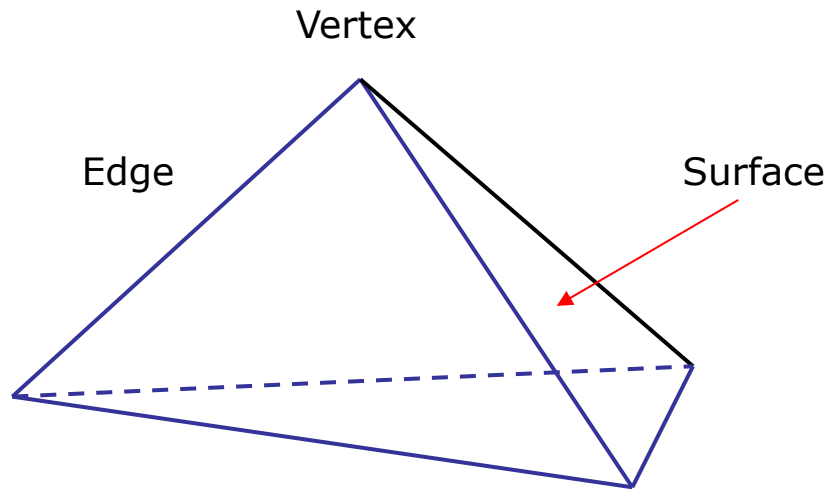
PCH: Platform Control Hub MCP: Multi-Chip Package

3.8 Basics of 3D graphics processing

3.8 Basics of 3D graphics processing (1)

3.8 Basics of 3D graphics processing

In 3D graphics processing, objects are represented by a set of triangles, as indicated in the Figure below.



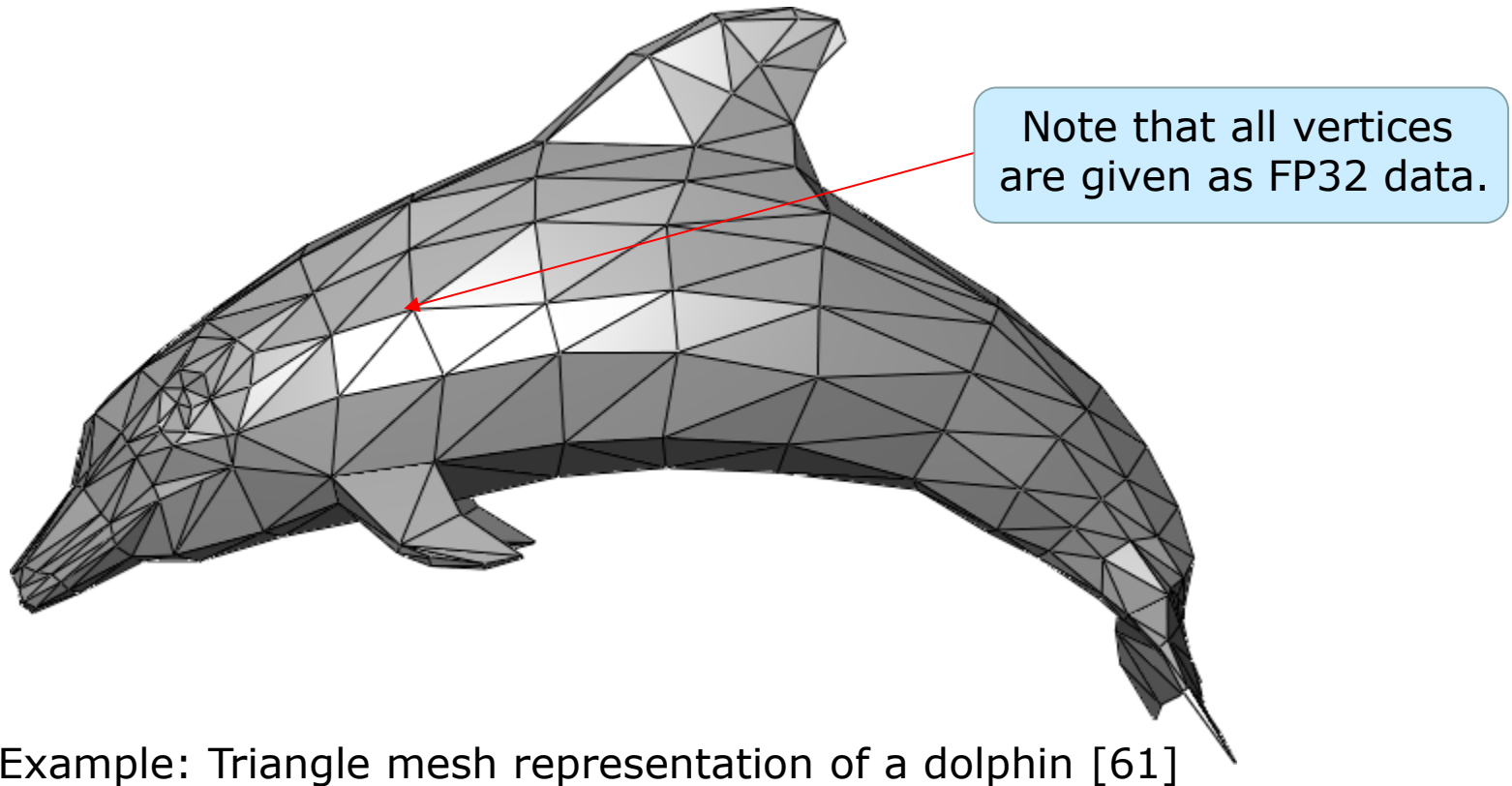
Vertices in graphics

- have three spatial coordinates (represented by FP32 data), and
- supplementary information necessary to render the object, such as
 - color (RGB)
 - texture
 - reflectance properties
 - etc.

3.8 Basics of 3D graphics processing (2)

Triangle mesh representation of objects in graphics processing

Graphics software and hardware operate **more efficiently on a group of triangles**, called **triangle meshes** rather than on individual triangles, so **graphics objects are typically represented by triangle meshes**.



Example: Triangle mesh representation of a dolphin [61]

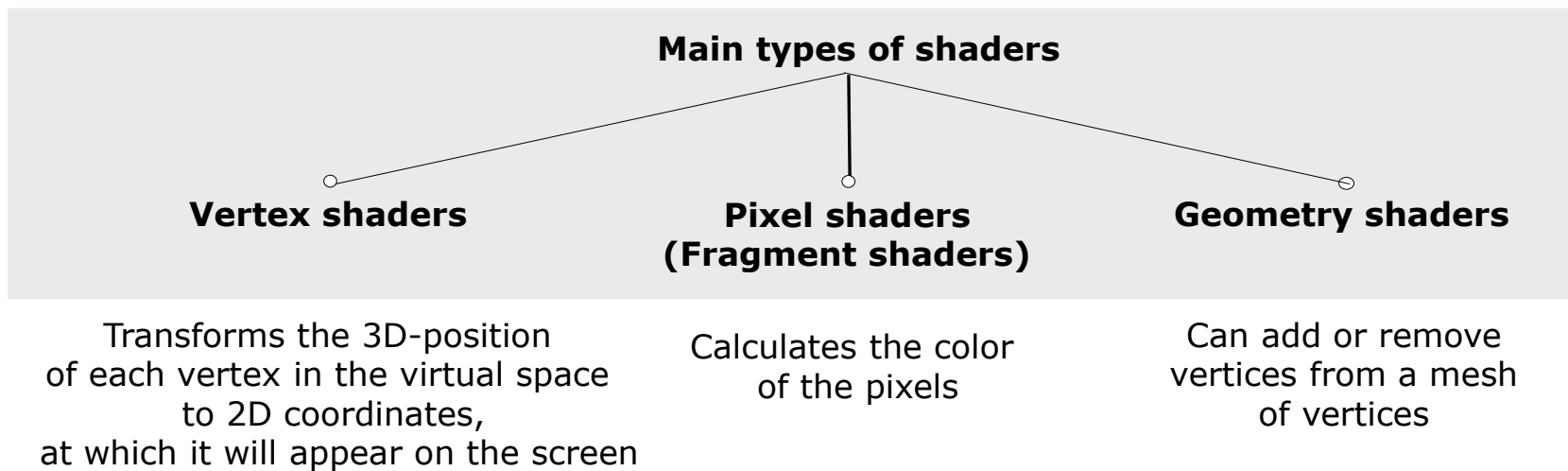
Shaders (in graphics)

- The term “**shader**” refers to graphics routines that run on the GPU’s graphics cores and perform tasks such as processing vertices or pixels of 2D/3D objects.

3.8 Basics of 3D graphics processing (4)

The main types of shaders and their basic tasks

There are **three main types of shaders**, as shown below.



- We **note** that there are further shader types as well, like compute shaders, ray tracing shaders, but they are beyond the scope of our Lecture.
- **Graphics APIs** (like Microsoft's **DirectX** or **OpenGL**) assume the **triangle representation of graphics objects**.

Shaders: Graphics programs running on the GPU hardware

Shader languages

- Shaders are written using **shader languages** and will be **run on the GPU**.
- **Examples** of the most common shader languages are:
 - **HLSL (High-Level Shader Language):**
Used primarily with **DirectX**.
Developed by **Microsoft**.
Its syntax is similar to C/C++.
 - **GLSL (OpenGL Shading Language):**
Used with **OpenGL**.
Developed by the **Khronos Group**.
Also has a C-like syntax.
 - **Cg (C for Graphics):**
Developed by **NVIDIA**.
Similar to HLSL and GLSL.

3.8 Basics of 3D graphics processing (6)

Historical remark

- In early graphics processing, in the beginning of the 2000s' different shaders run on different parts of the graphics hardware, as seen below on the left.
- However, since about 2006, all shaders have been running on the same hardware, which is the GPU, instead of using dedicated HW units for each shader, as shown below on the right.

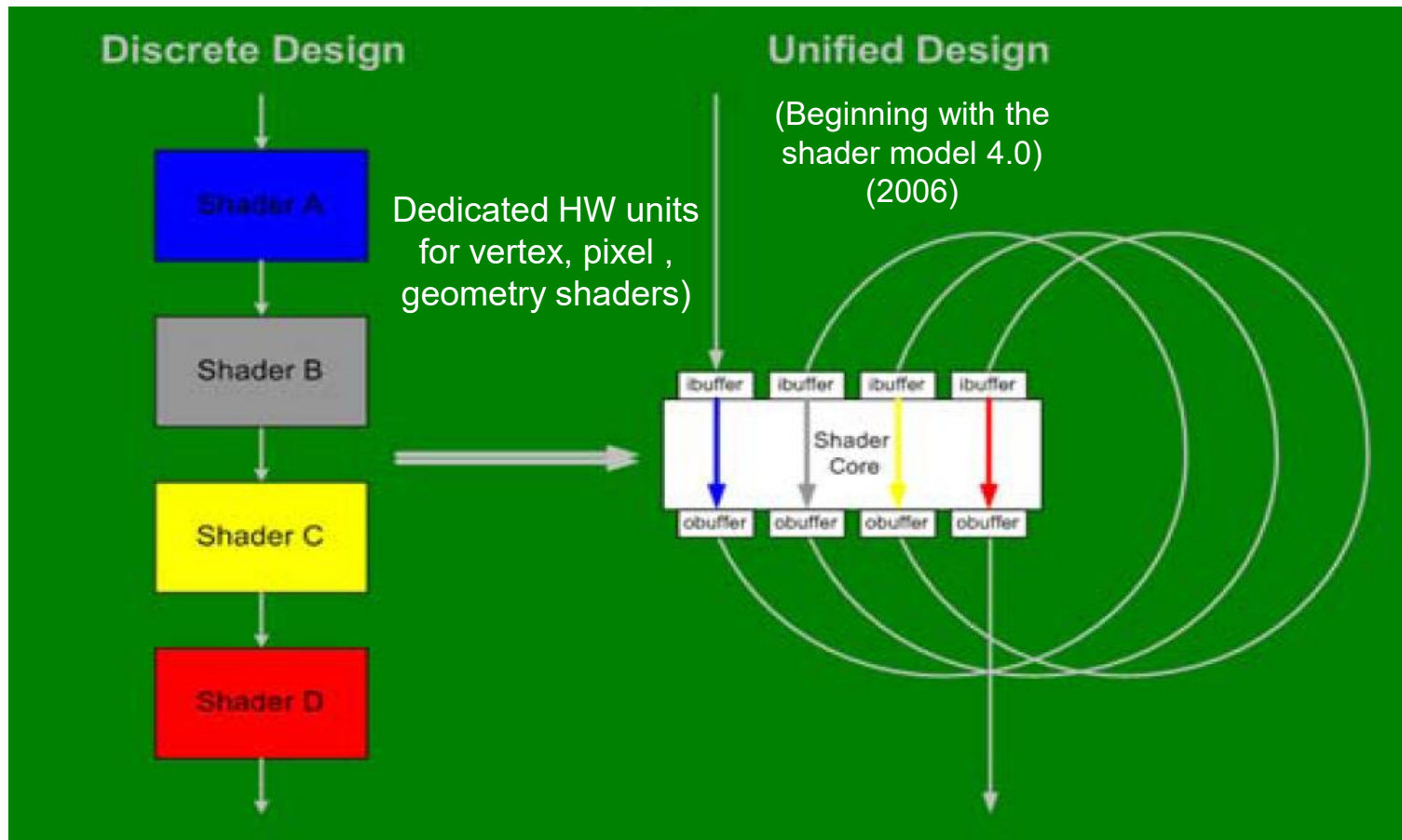


Figure:
Basic
shader
designs
[62]

Remark on graphics processing

- **Note** that due to the huge number of vertices to be processed for objects, **graphics computations** need a **vast number of FP32 execution resources**.
- This is why integrated GPUs can execute a few thousand FP32 operations per cycle, while recent graphic cards have tens of thousands of FP32 operations per cycle.
- E.g. NVIDIA's GeForce RTX 5090 can perform 21760 FP32 MAC operations per cycle, that is 43520 FP32 operations per cycle.

MAC: Multiply and Accumulate ($axb + c$) operation.

4. Case examples for recent client processors

- 4.1 Case example 1: Intel's Core Ultra Serie 2 (Lunar Lake) laptop family (2024)
- 4.2 Case example 2: Intel's Core Ultra (Panther Lake) laptop family (2026)

4.1 Case example 1: Intel's Core Ultra Serie 2 (Lunar Lake) laptop family (2024)

(Optional reading, not part of the exam.)

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (1)

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (2024)

- The Lunar Lake processor is designed to compete with Qualcomm's ARM-based low-power, high-performance Snapdragon X processor optimized for Windows on ARM PCs.
 - Its key features are as follows:
 - heterogeneous, dual-die big.little design,
the tiles are: the Compute tile and the Platform controller tile.
The Compute tile is manufactured using TSMC's N3B process, while the Platform controller tile is fabricated using TSMC's N6 technology.
 - 4 P-cores and 4 LPE-cores (Low-Power E-cores),
8-wide P-cores and 3x3-wide E-cores.
- no multithreading,
 - On-package 16 or 32 GB DDR5x-8522 DRAM providing low latency, low-power memory,
 - NPU 4 with up to 48 TOPS,
 - Xe2 GPU (8 cores, 8 MB L2)
 - TDP: 17-30 W

Remark

In the Lunar Lake processor, Intel designates its E-cores as LPE (Low-Power E-cores) to highlight their improved power efficiency compared to previous E-cores.

In future processors (such as the Panther Lake, expected in Q1/2026), Intel plans to implement both LPE-cores and traditional E-cores alongside P-cores.

4.1 Case example 1: Intel’s Core Ultra Series 2 (Lunar Lake) laptop family (2)

Key features of Intel’s Core Ultra 200V laptop processors (2024) [158]

Branding	SKU	Cores (threads)		Clock rate (GHz)			Arc Graphics		NPU (TOPS)	Smart cache ¹ (MB)	RAM	TDP			Release date
		P	LPE	Base	P	LPE	Xe cores (XVEs)	Max. freq. (GHz)				Base	Turbo	cTDP	
Core Ultra 9	<u>288V</u>			3.3	5.1	3.7	8 (64)	2.05	48	12 MB	32 GB	30 W		17-37 W	
Core Ultra 7	<u>268V</u>				5.0			2.0	48		32 GB				
	<u>266V</u>			2.2		3.7	8 (64)			12 MB	16 GB	17 W		8-37 W	
	<u>258V</u>	4 (4)	4 (4)		4.8			1.95	47		32 GB		37 W		
	<u>256V</u>										16 GB				
Core Ultra 5	<u>238V</u>				4.7						32 GB				
	<u>236V</u>			2.1		3.5	7 (56)	1.85	40	8 MB	16 GB	17 W		8-37 W	
	<u>228V</u>				4.5						32 GB				
	<u>226V</u>										16 GB				

Note the lack of multithreading.

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (3)

The physical implementation of the Lunar Lake processors [135]

Lunar Lake Construction

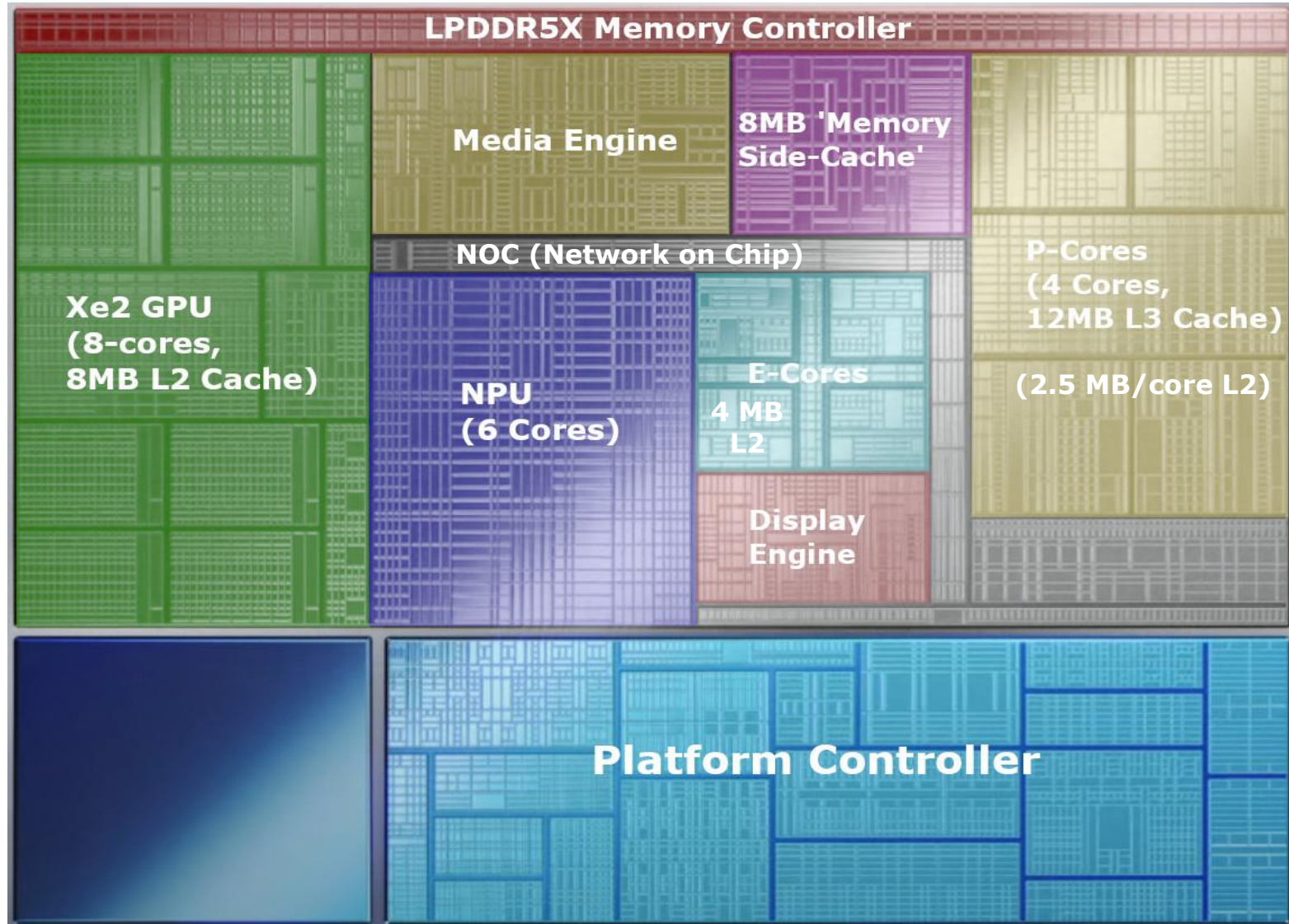
Built with advanced packaging



As shown, Intel's Lunar Lake has a **multi-die big.little architecture**, like Meteor Lake.

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (4)

The annotated die plot of the Lunar Lake processor without the dual memory dies [149]



4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (5)

Assumed block diagram of the Core Ultra 200V Lunar Lake laptop processors -1

This architecture resembles the 14th gen. Meteor Lake's processor, but it is implemented using only two tiles and has a few differences, as shown below.

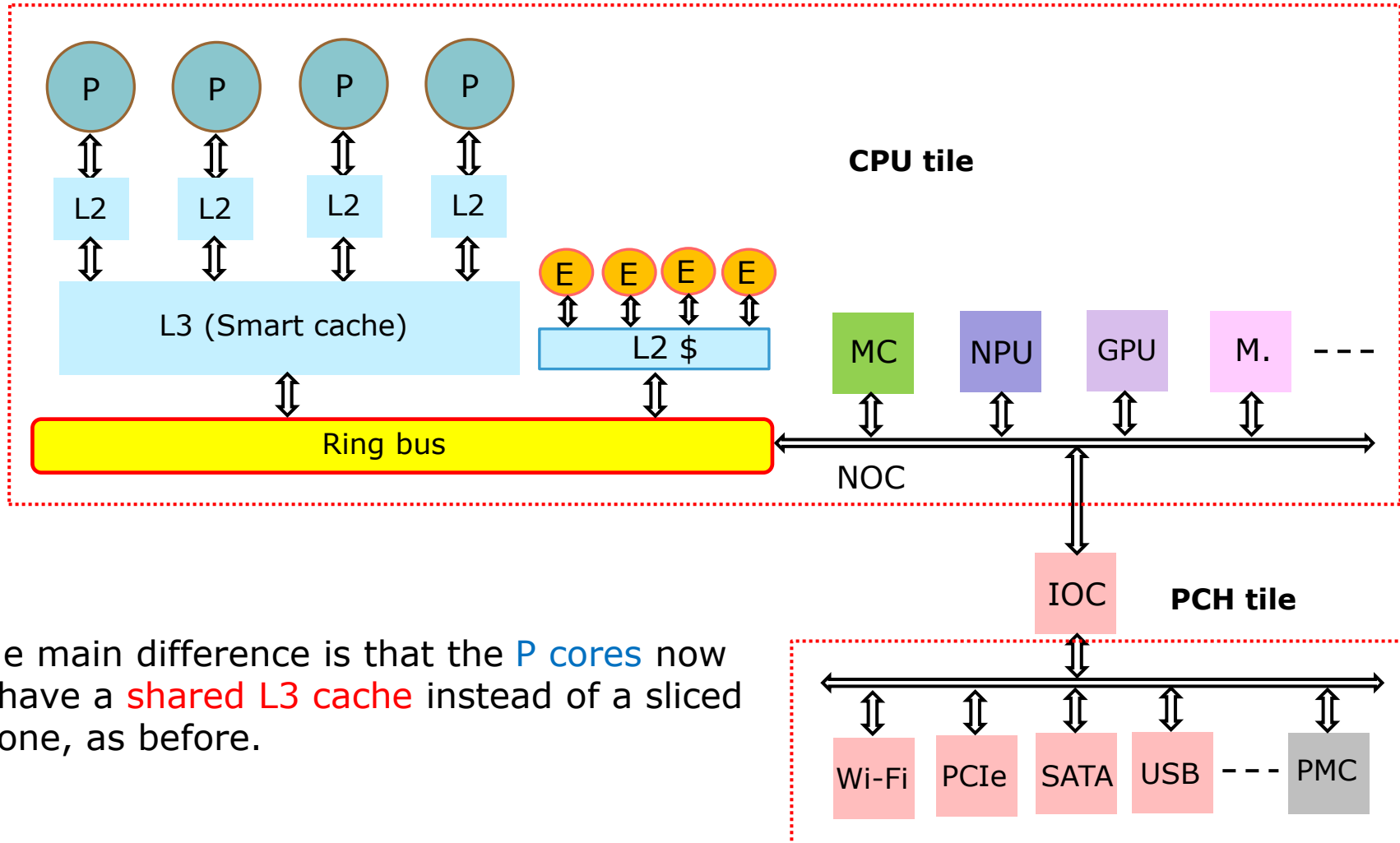


Figure: Assumed block diagram of the Core Ultra 200V Lunar Lake laptop processors

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (6)

Remark

- The CPU tile of the Lunar Lake processor includes a memory-side cache (system level cache) that is 8 MB in size.
- This cache is accessible to all agents of the processor, including the CPU, GPU, and NPU. However, there is no reference to how this cache is integrated into the block diagram, so it is not indicated there.

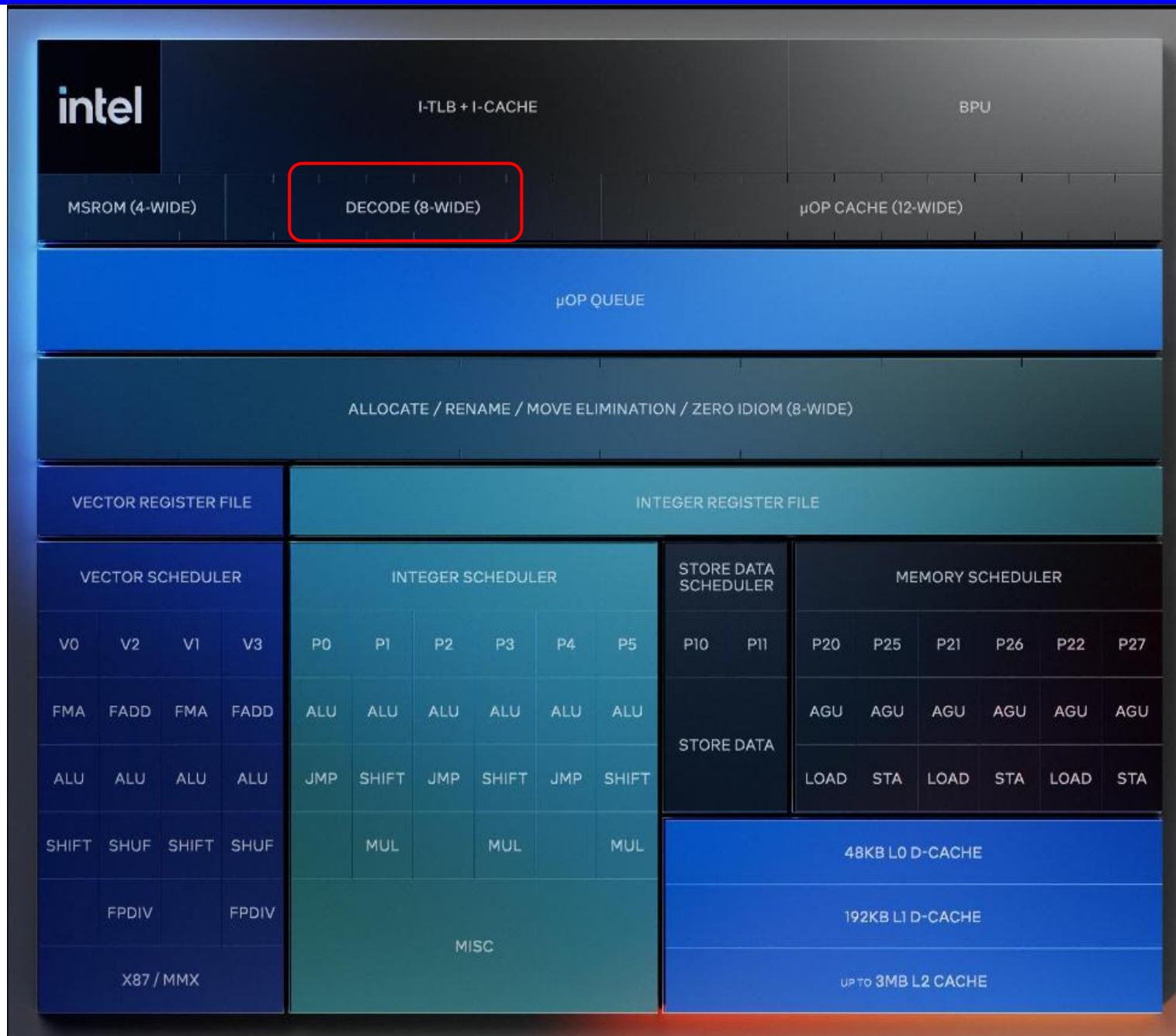
4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (7)

The P-core
(Lyon core)
[150]

8-wide decode
+14 % perf.
over
Redwood core
(Meteor Lake)

No multithreading
12 MB shared L3
cache

16.67 MHz
clock intervals
in the CPPC
for more
efficiency



4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (8)

The LPE core (Skymont core)
[150]

It is **the same core** as found
in the Arrow Lake S DT
processor

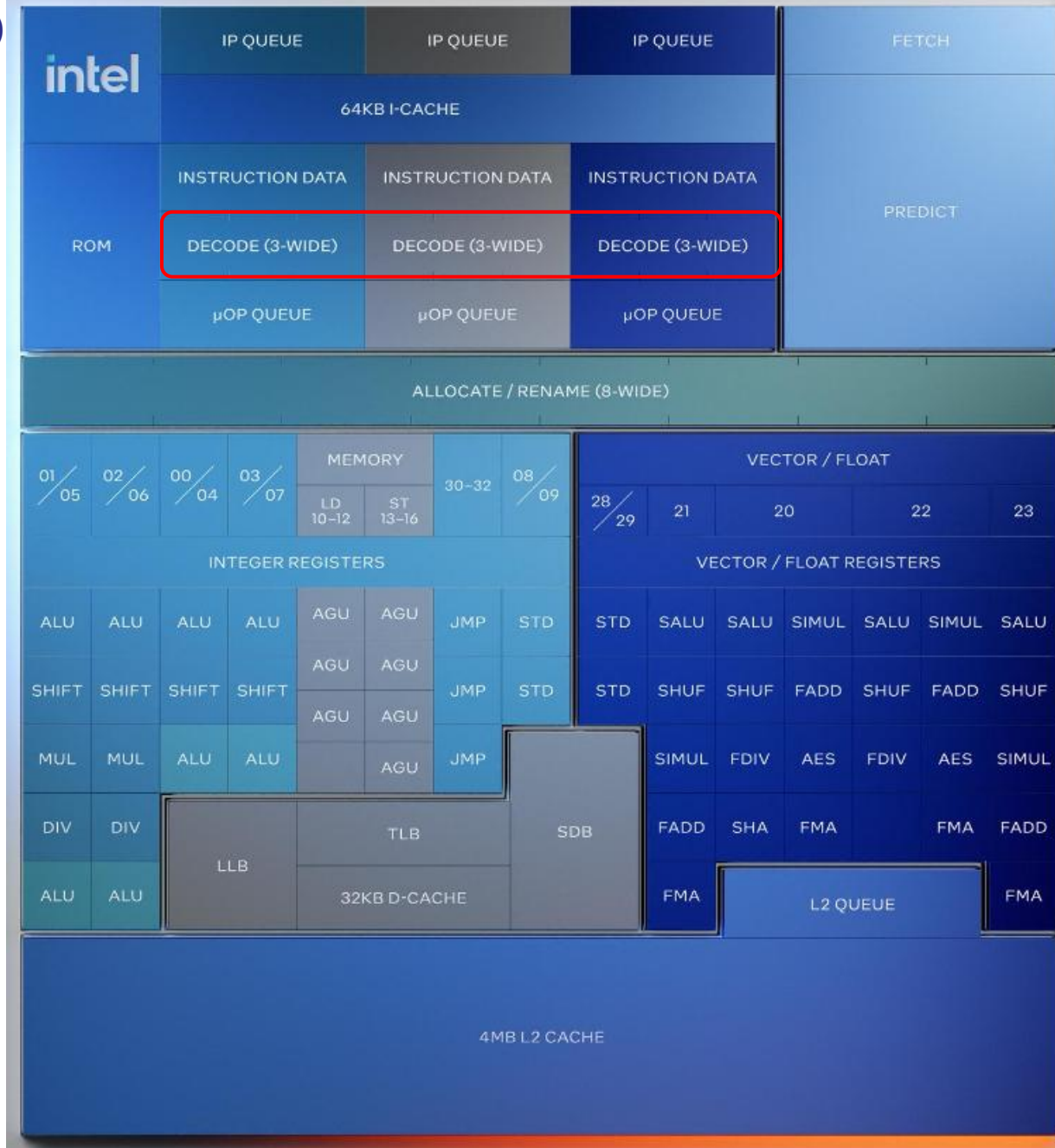
3x3 wide decode

1.38x Meteor Lake LP-E

SPECrate2017_int_base est /

1.68x Meteor Lake LP-E

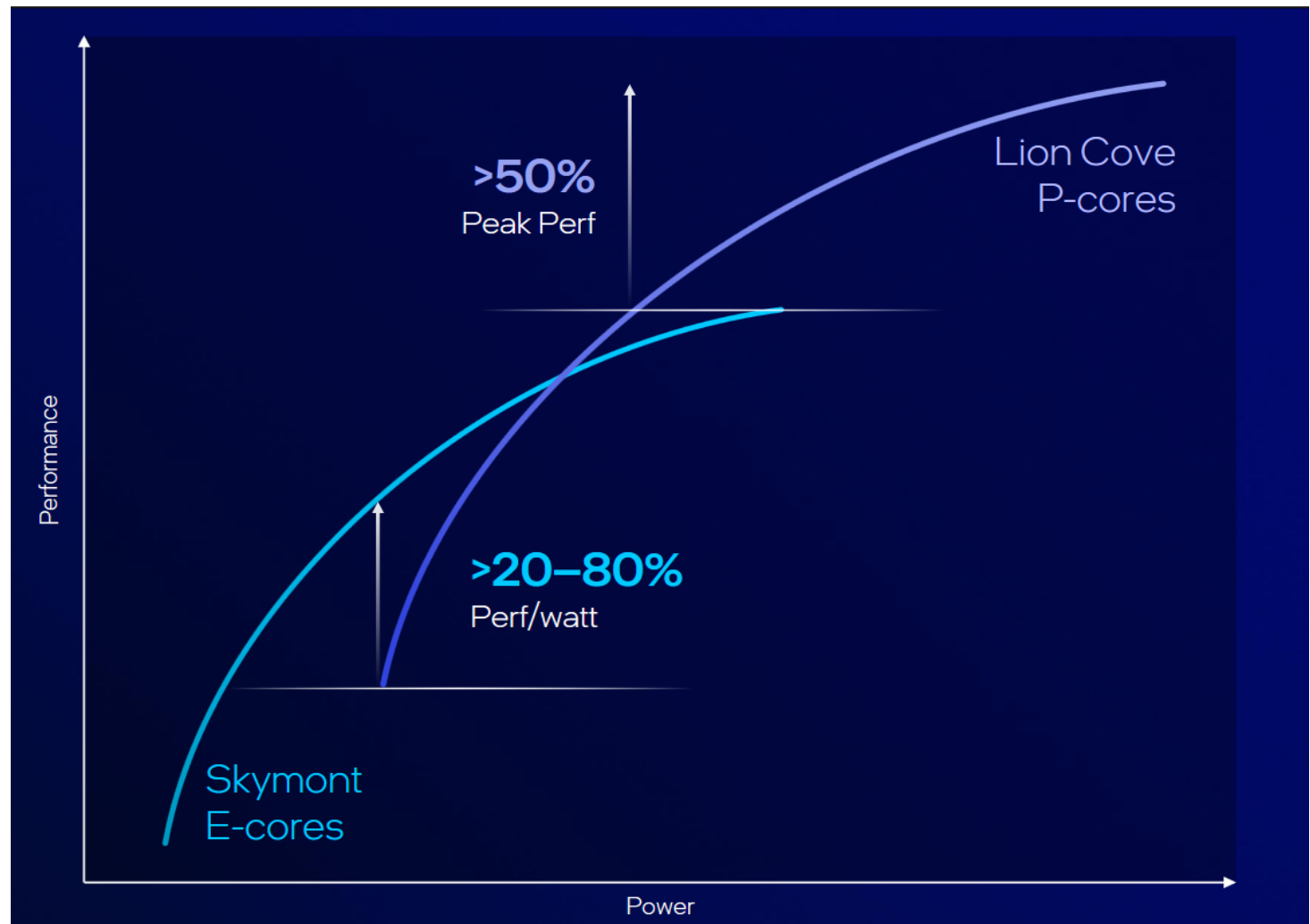
SPECrate2017_fp_base est /



Note that Lunar Lakes E cores
offer **significantly higher**
performance than the E cores
in the Meteor Lake processor

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (9)

The efficiency graphs of Lunar Lake's P- and E-cores [150]



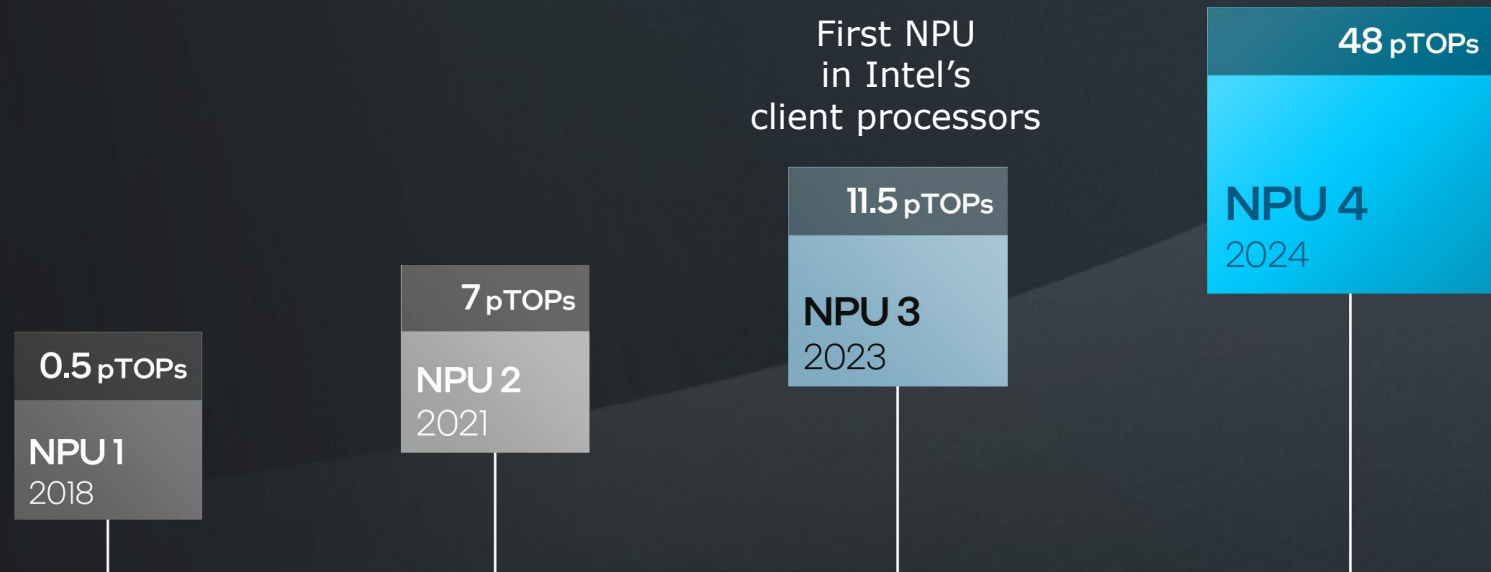
For lighter workloads, the E-cores, while for more intensive ones the P-cores are used.

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (10)

Evolution of the NPUs in Intel's Core family over time [151]

Continuous NPU Improvements

Across 4 generations of IP



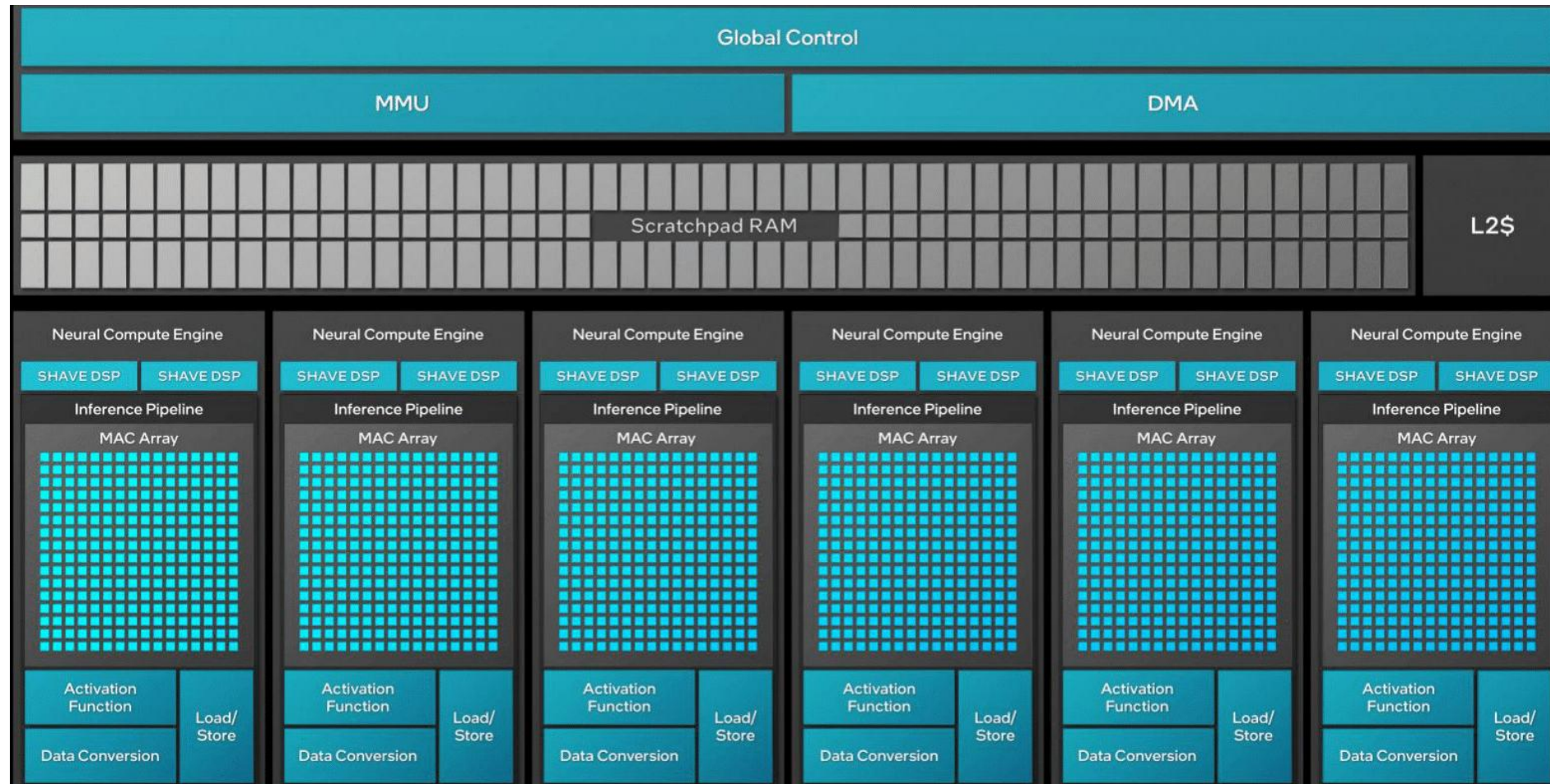
14th gen.
Meteor Lake

Core Ultra 200V
For Copilot+ PCs

TOPS: Tera Operations Per Second (10^{12})

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (11)

The central part of the NPU unit of the Lunar Lake processor -1 [151]



- It includes 8x512 MAC Processing Engines (MPEs), but above only 6 are indicated.
- Each MPE is capable of performing four INT8 or two FP16 operations per cycle [152].
- It follows that the NPU can execute $8 \times 512 \times 4 = 16 \text{ K}$ INT8 operations per cycle.

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (12)

The central part of the NPU unit of the Lunar Lake processor -2 [151]

NPU 4 MAC array

Matrix multiplication
& convolution

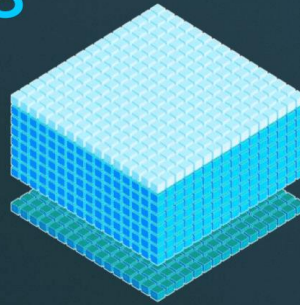
2048 MAC/cycle int8

1024 MAC/cycle FP16

Up to 2x¹ efficiency
driving better perf/watt

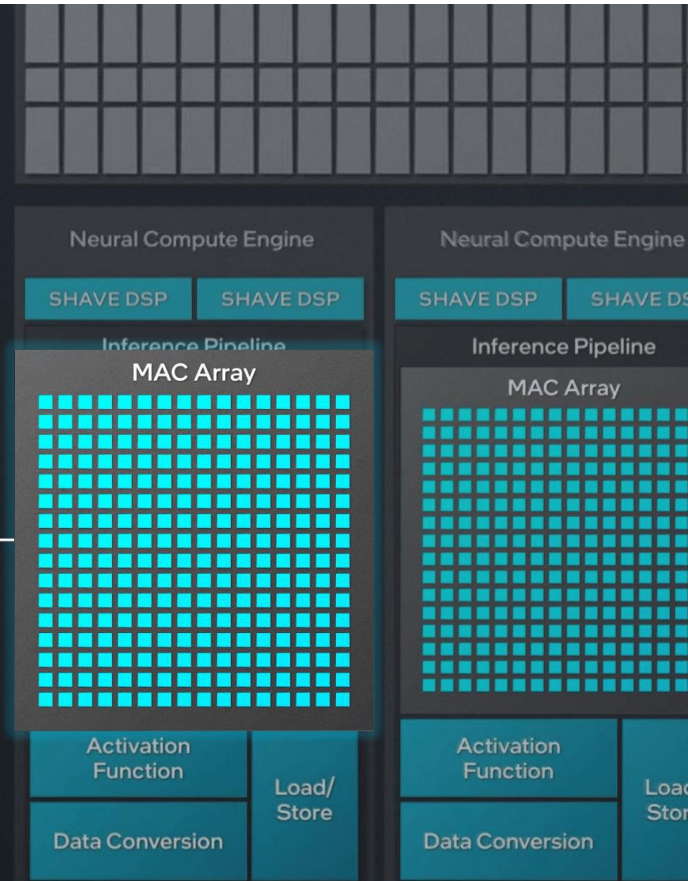
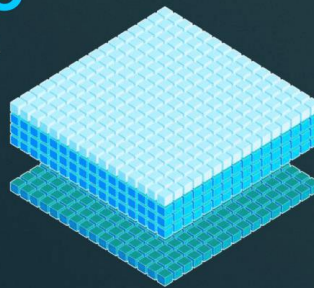
INT8

16x16x8



FP16

16x16x4



Assuming an NPU 4 clock cycle of 300 MHz, the NPU performance would be calculated as 16 K operations x 300 MHz, resulting in 48 TOPS (relating to FP8 operations).

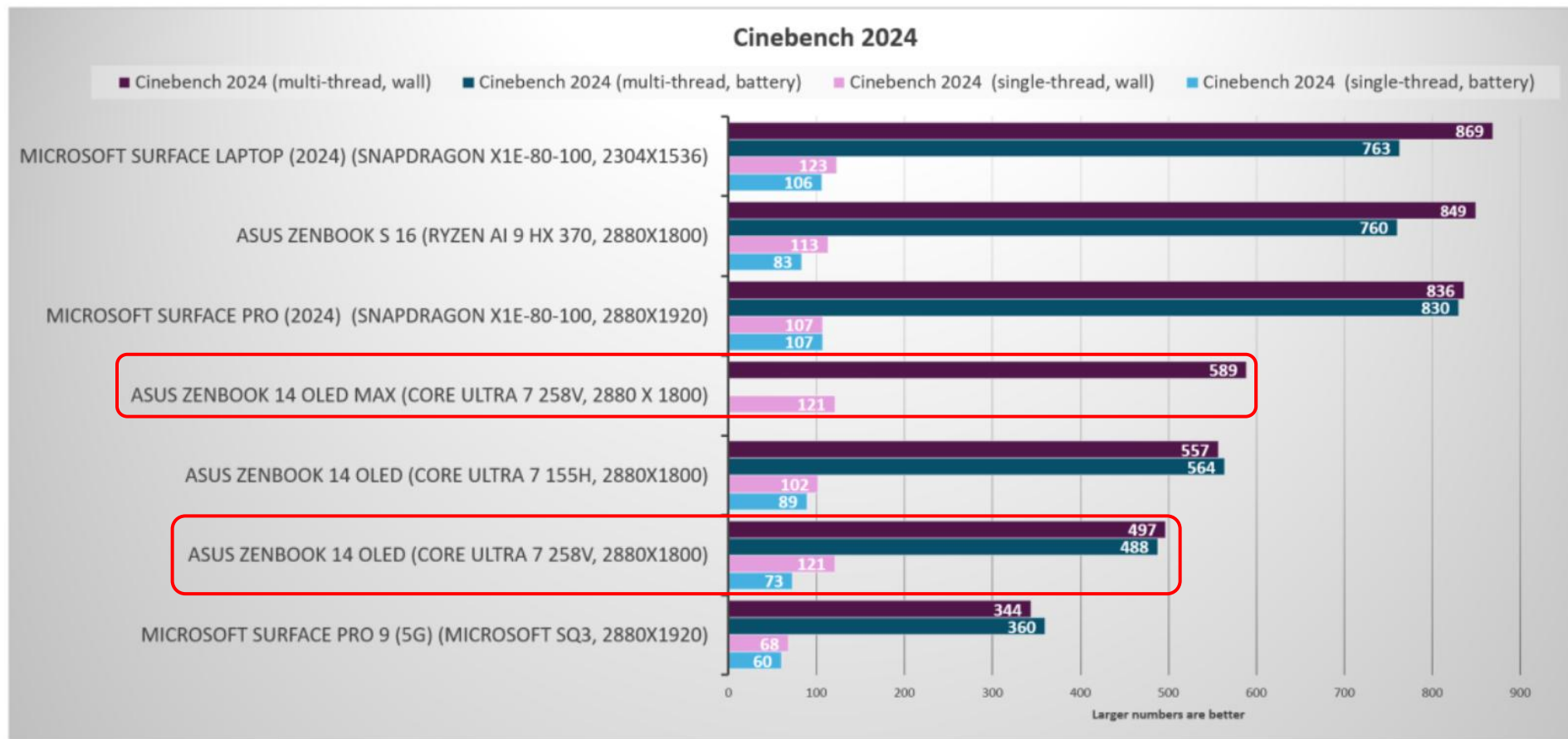
4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (13)

Ahmed Y., Introducing Copilot+ PCs



4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (14)

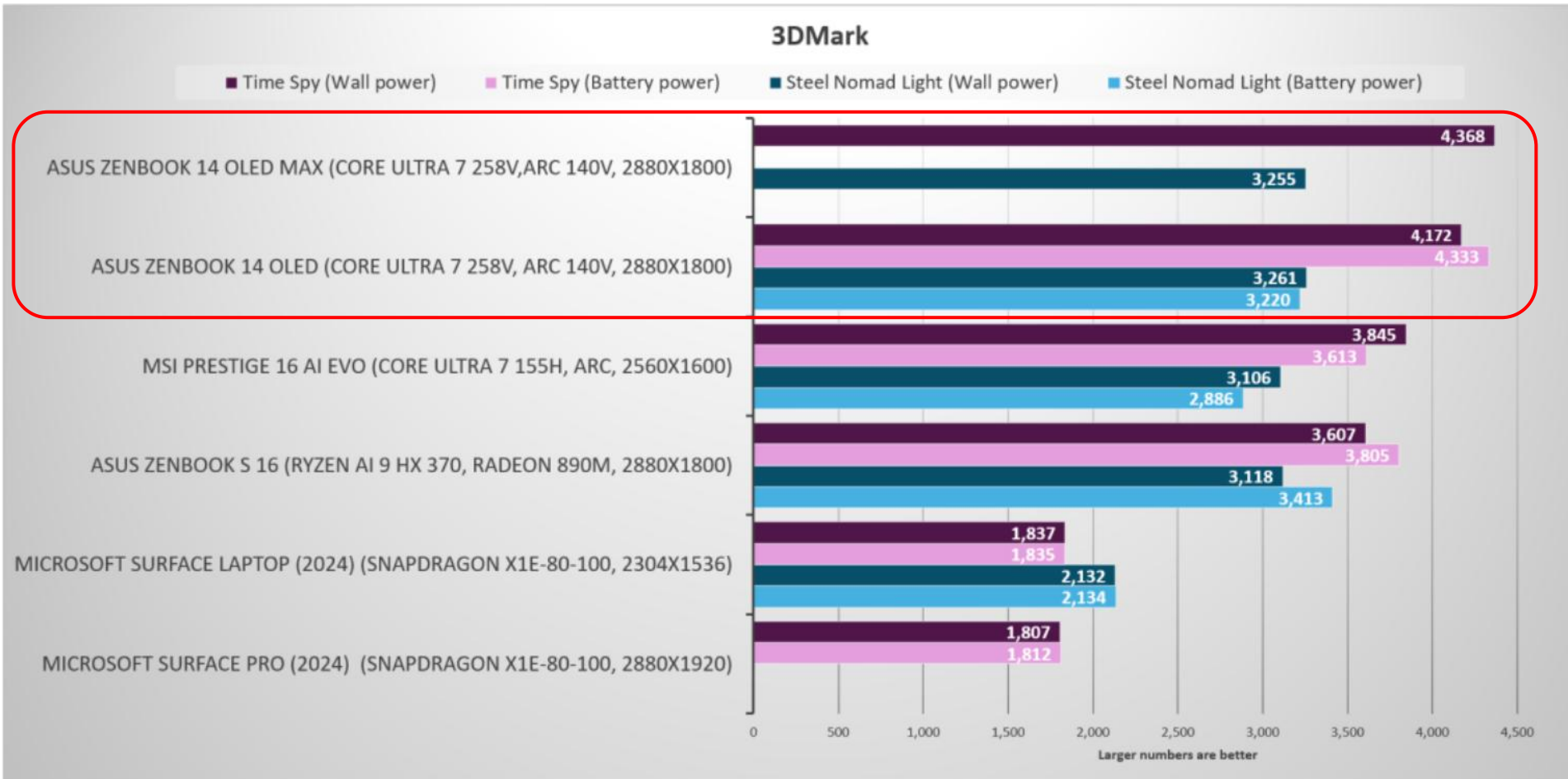
Comparing the CPU performance of advanced laptop processors [165]



As shown in the 4+4 core Figure, the Lunar Lake processor (Ultra 7 258V) has [similar ST CPU](#) performance compared to the competing Qualcomm Snapdragon X1E 80-100 and AMD Ryzen AI9 HX370 processors, but it provides [lower MT](#) performance.

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (15)

Comparing the GPU performance of advanced laptop processors [165]



As shown in the Figure, the Lunar Lake processor (Ultra 7 258V) demonstrates **higher GPU performance** compared to the AMD Ryzen AI 9 HX370 processor as well as **significantly higher** performance compared to Qualcomm's Snapdragon X1E 80-100 processor.

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (16)

Comparing battery-life performance of advanced laptop processors [165]

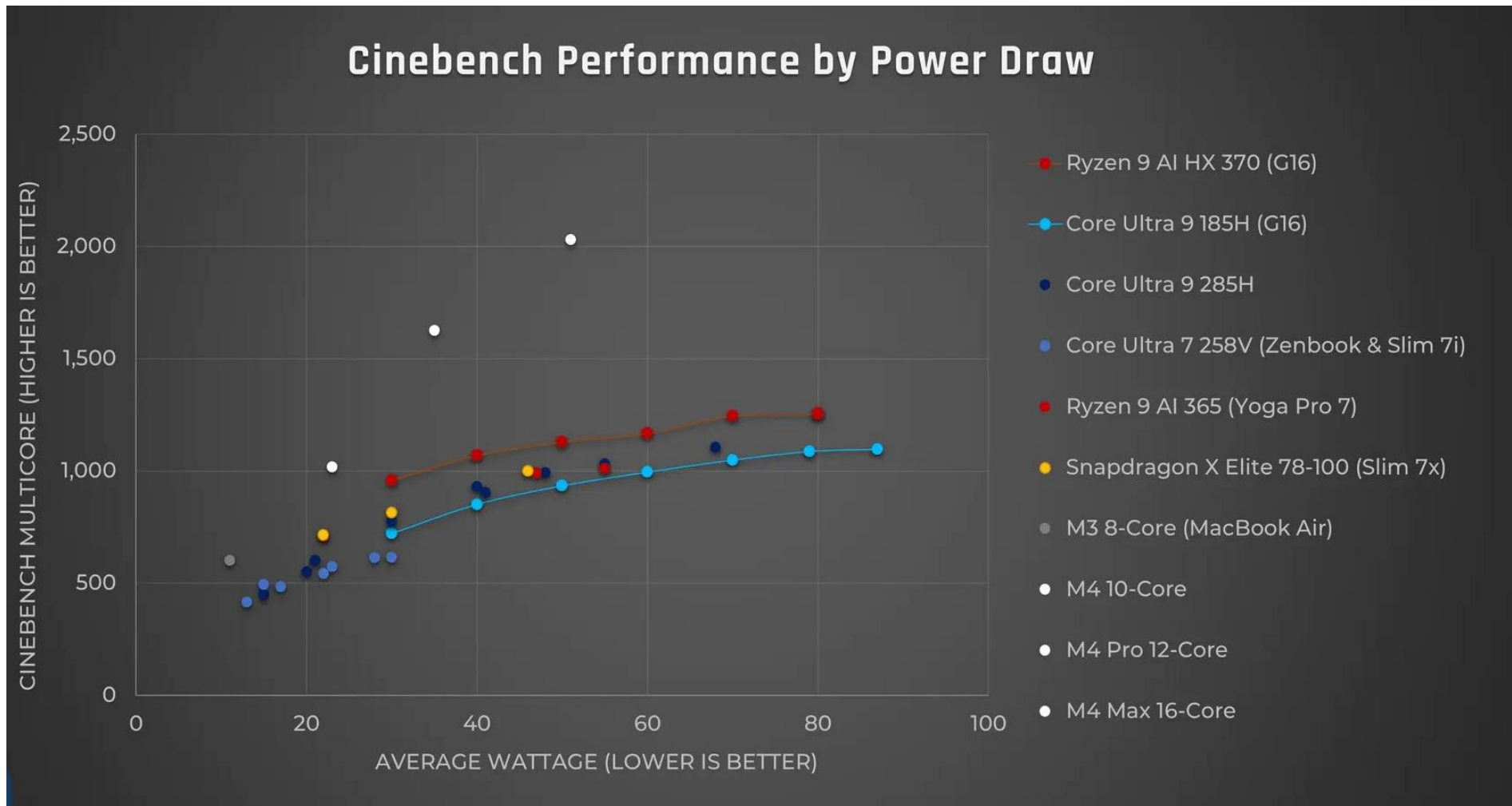
- This test was conducted using [UL's Procyon battery-life test](#), which [simulates working on the laptop by cycling through Office applications until the battery is down](#).
- The **results** of the tests are as follows:
 - **Intel Lunar Lake:** 17 hours, 7 minutes
 - **Qualcomm Snapdragon X Elite:** 16 hours, 20 minutes
 - **AMD Ryzen AI 300:** 10 hours, 42 minutes
 - **Intel Meteor Lake:** 10 hours, 35 minutes
- The above results demonstrate that Intel's [Lunar Lake processor offers the longest battery life, slightly outpacing Qualcomm's Snapdragon X Elite processors](#). Both of these processors have [significantly longer battery lives compared to AMD's Ryzen AI 300 and Intel's Meteor Lake processors](#).

The test configurations were:

the Asus ZenBook S 16 (with a Ryzen AI 300 chip inside and a 78 Wh battery),
the 13.8-inch Surface Laptop 7th Edition (with a Snapdragon X Elite chip and a 54 Wh battery),
the Asus ZenBook 14 OLED (with a first-gen Core Ultra Core 155H and a 75 Wh battery) versus the
current Asus ZenBook OLED (with its Lunar Lake chip and a 73 Wh battery).

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (17)

Efficiency of laptop processors in terms of Cinebench 24 Multi-Core/W [171]



As shown in the Figure, Apple's M4 processors are the most efficient, followed by AMD's Ryzen 9 AI, Qualcomm's Snapdragon X Elite and finally by Intel's Core Ultra lines.

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (18)

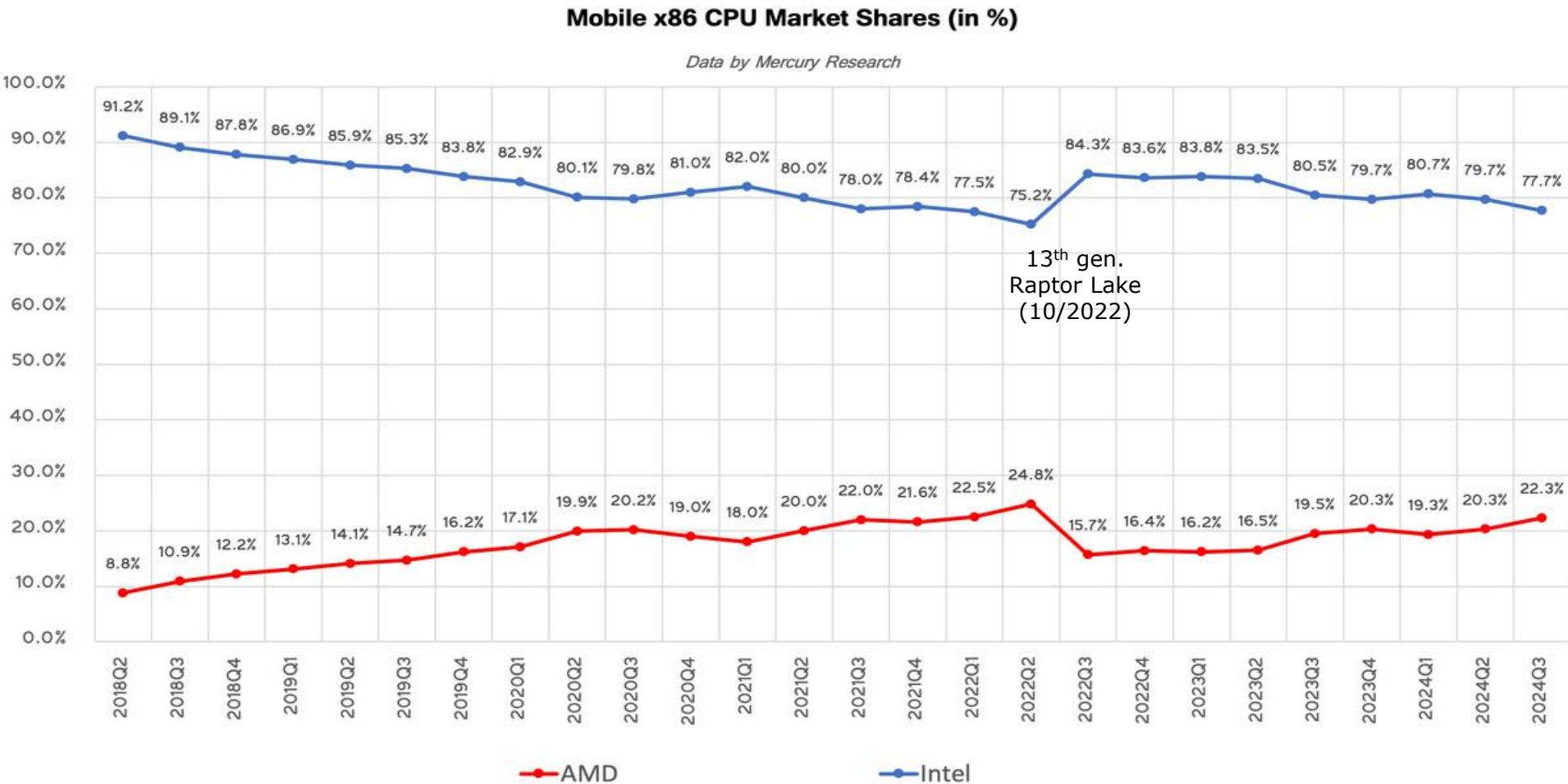
Revealed vulnerability of recent Intel processor lines [131].

- In 11/2023, Google released a bulletin reporting a **serious vulnerability** in several processor lines of Intel, including its 10th gen. Ice Lake, 12th gen. Alder Lake and 13th gen. Raptor Lake lines.
- Potentially, this vulnerability can lead to denial of service or disclosure of sensitive data.
- Intel acknowledged it and fixed it with microcode patches.
- As a result of this vulnerability, Intel experienced a significant loss of market share.

vulnerability: sérülékenység

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (19)

Global laptop processor revenue market share: Intel vs. AMD [153]

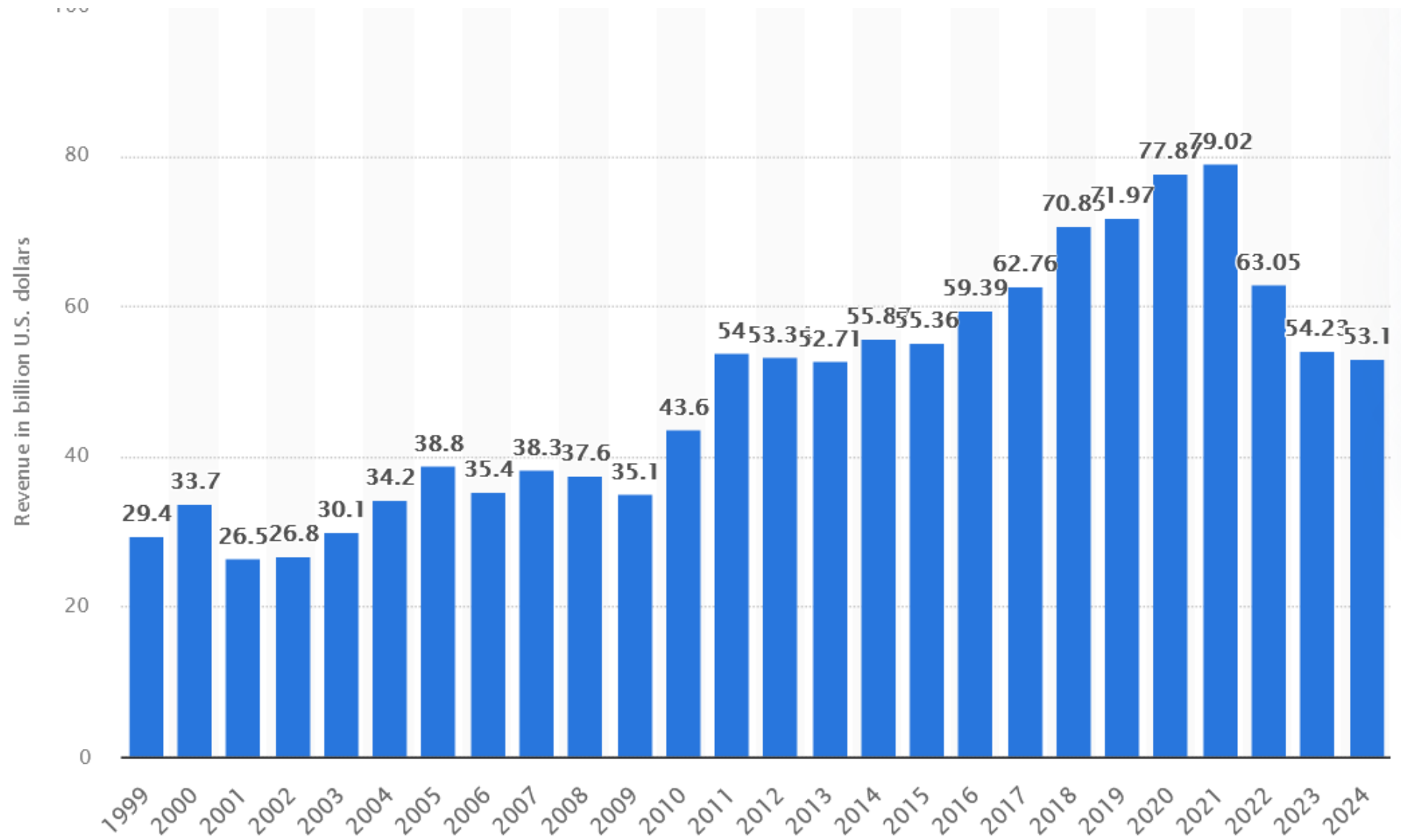


The Figure above indicates **Intel's dominance** in the global market revenue share for client processors, although to a gradually lesser extent.

Please note in the Figure how **significantly** the appearance of Intel's **Raptor Lake** line **modified the revenue share** between the two competing vendors.

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (20)

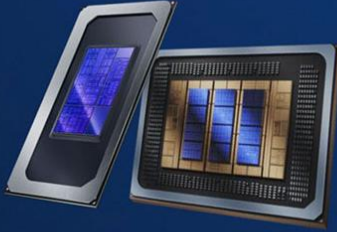
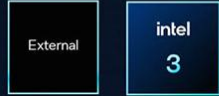
Intel's revenues from 1999 to 2024 (in billion USD) [154]



4.2 Case example 2: Intel's Core Ultra (Panther Lake) laptop family (2026)

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (2)

The evolution of Intel's advanced multi die processor families [225]

				
	Meteor Lake, Sapphire Rapids, Emerald Rapids	Lunar & Arrow Lake	Granite Rapids & Sierra Forest	Panther Lake & Clearwater Forest
Packaging	Foveros 2.5D EMIB 2.D	Foveros 2.5D	EMIB 2.5D	Foveros 2.5D, Foveros 3D Direct, EMIB
Process				
Year	2023	2024	2024	2025-26

All of the **client processor** series mentioned above are **heterogeneous multi-die designs**. The 4th gen. Sapphire Rapids and 5th gen. Emerald Rapids server processors are **homogeneous multi-die designs**; in contrast, the 6th gen. Granite Rapids and Sierra Forest server series are **heterogeneous** layouts.

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (3)

Intel's Core Ultra Series 3 (Panther Lake) laptop family (2026) -2 [225]

- Compared to the Lunar Lake line, the **Panther Lake line** offers **up to 16 cores**, composed of **three die types**, while the Lunar Lake line has 8 cores made up of two core types.

These tiles are:

- the Compute tile (Intel 1.8A)
 - the Graphics tile (Intel3/TSMS N3E) and
 - the SOC tile (TSMC N6)
- Additionally, the Panther Lake line has a **higher TDP** of 25 W rather than 17 W for the Lunar Lake line (except for the highest performance model, which has a TDP value of 30 W).
 - The Panther Lake line is Intel's **first 18A product**, along with its server-oriented 6+ gen. **Clearwater Forest server processor line**, which is scheduled for launch in **H1 2026**.

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (4)

Intel's Ocatillo Campus Arizona (where the Panther Lake and Clearwater Forest chips are fabricated)



4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (5)

Intel's Core Ultra Series 3 (Panther Lake) laptop family (2026) [226] -2

All the **P, E, and LPE Panther Lake cores** are **new designs**, as seen below.

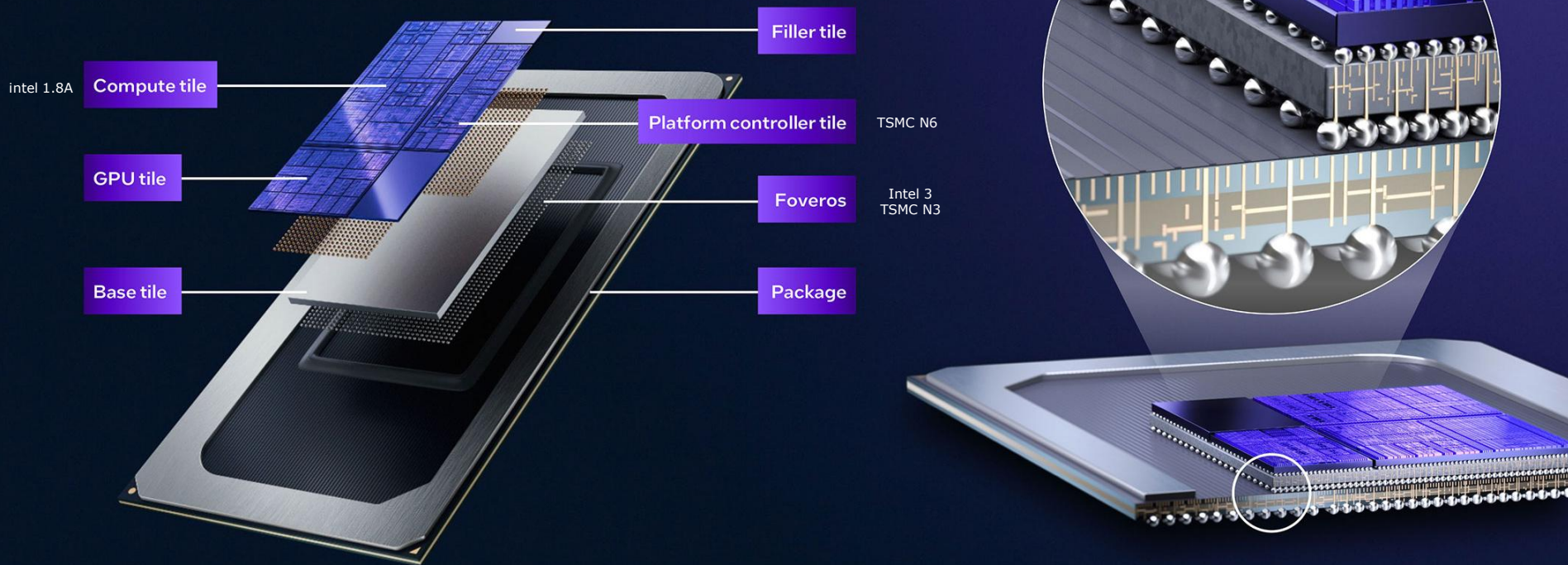


Figure: Intel's heterogeneous big.little client processors [226]

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (6)

Basic layout of the Panther Lake processors [226]

Panther Lake Construction



- As the Figure shows, Panther Lake processors include **three dies**, referred to as: the **Compute tile**, the **GPU tile**, and the **Platform controller tile**.
- These tiles are packaged using the **Foveros 2.5D** technology.

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (7)

Three main configurations of the Panther Lake processors [226]

Panther Lake Package Configurations			
	8 core	16 core	16 core 12Xe
CPU	Up to 8 cores		Up to 16 cores
GPU	Up to 4Xe-cores Xe3		Up to 12Xe-cores Xe3
NPU	NPU 5		
IPU	IPU 7.5		
Media & display	Xe media & display engine		
Memory	LPDDR5X - Up to 6800MT/s DDR5 SO-DIMM - Up to 6400MT/s	LPDDR5X - Up to 8533MT/s DDR5 SO-DIMM - Up to 7200MT/s	LPDDR5X - Up to 9600MT/s
I/O	12x PCIe lanes 8x PCIe Gen 4, 4x PCIe Gen 5	20x PCIe lanes 8x PCIe Gen 4, 12 x PCIe Gen 5	12x PCIe lanes 8x PCIe Gen 4, 4x PCIe Gen 5
Connectivity	Intel® Wi-Fi 7 (R2) & Intel® Bluetooth® Core 6, Thunderbolt™ 5 and 4		

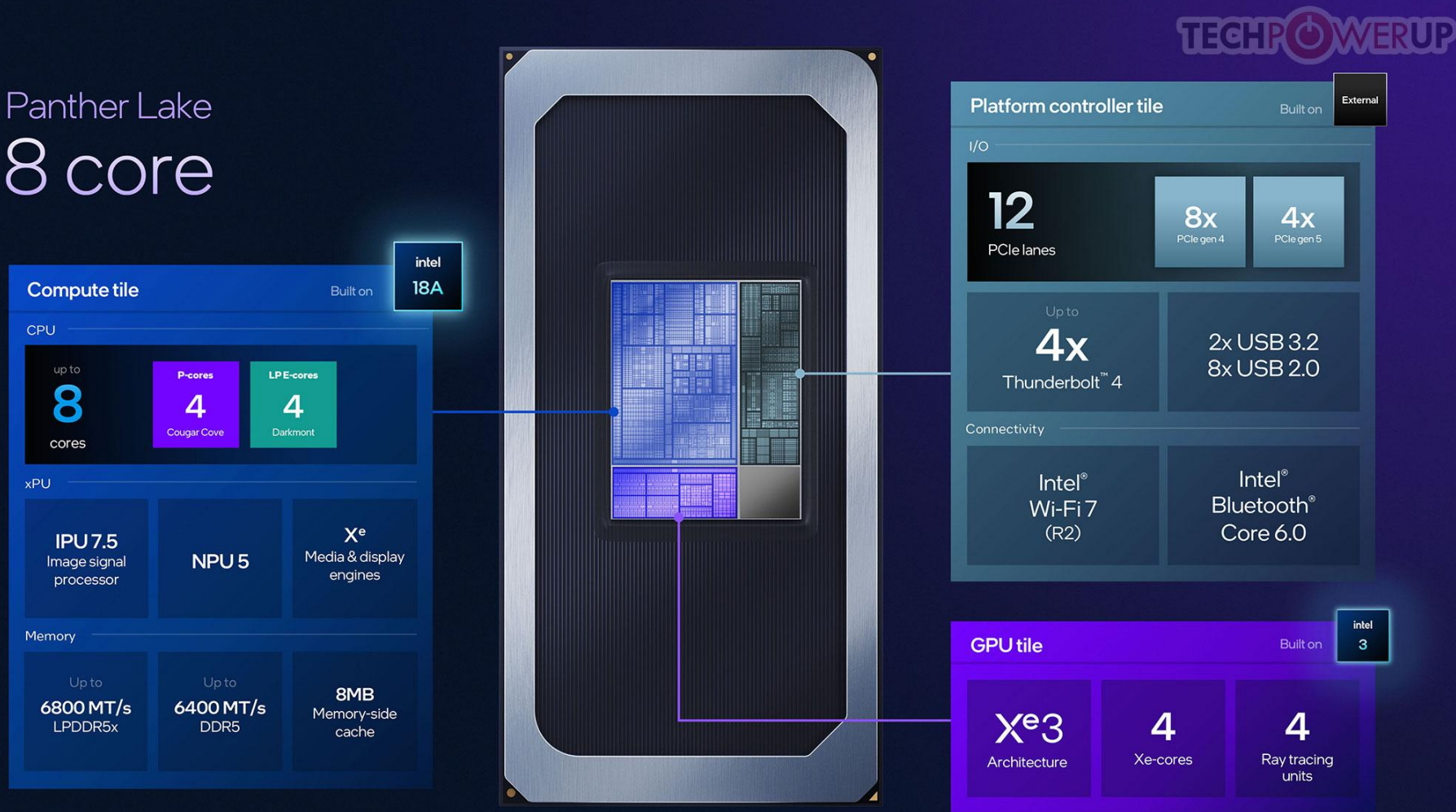
Next, we will outline these basic configurations.

IPU: Image Processing Unit for handling camera, sensor, and multimedia imaging tasks.

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (8)

Main components of the 8-core layout [226]

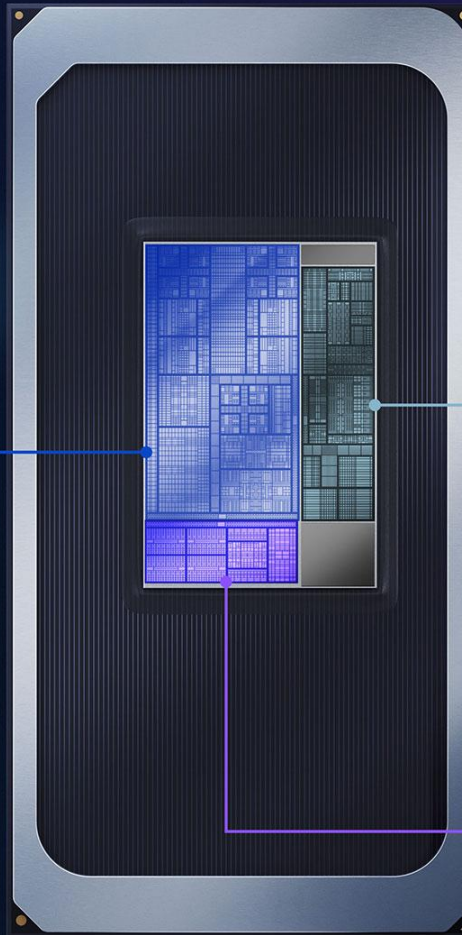
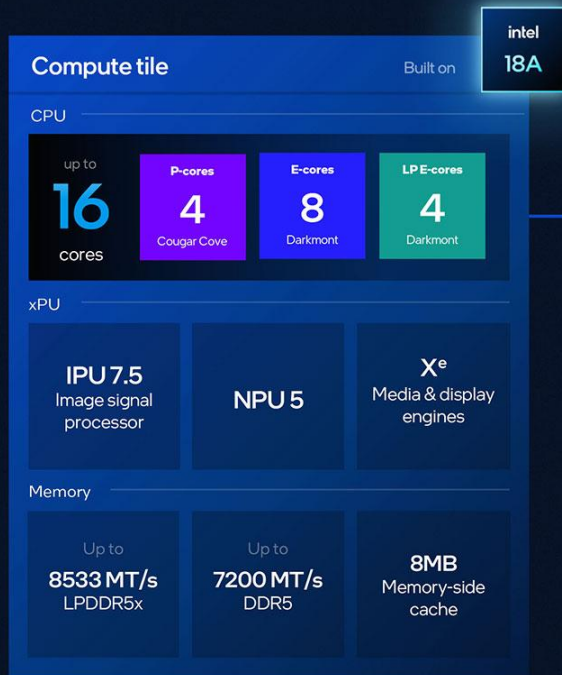
Panther Lake 8 core



4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (8)

Main components of the 16-core layout [226]

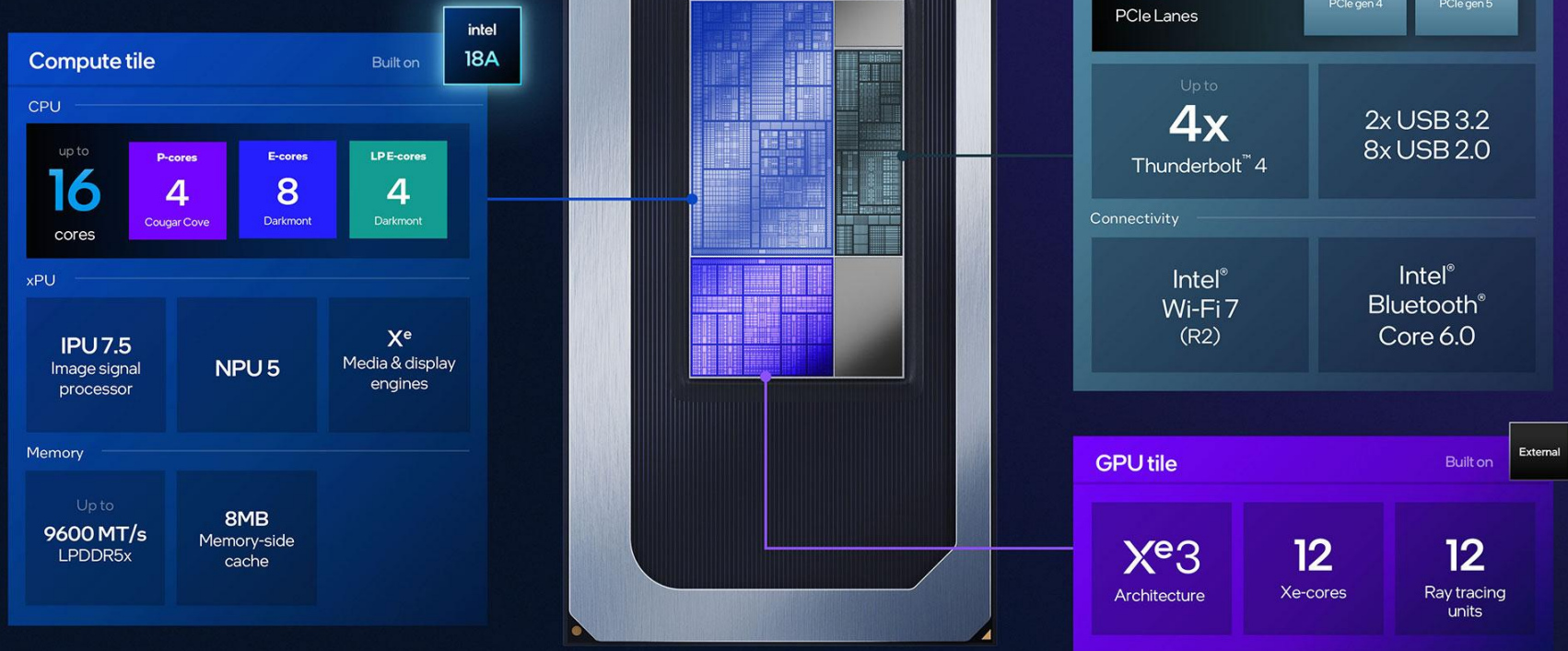
Panther Lake 16 core



4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (9)

Main components of the 16-core 12X layout [226]

Panther Lake 16 core 12X^e



This configuration has 12 Xe cores and 12 ray tracing cores, compared to 4 and 4 cores of the previously discussed versions.

4.2 Case example 2: Intel’s Core Ultra Series 3 (Panther Lake) laptop family (10)

Main features of Intel’s Core Ultra Series 3 (Panther Lake) processor models [226]

Processor Number	CPU			NPU	Memory		Power	
	Total Cores & Threads	P-core Max Turbo Freq (GHz)	Intel® Smart Cache LLC (MB)		Maximum Memory Speed (MT/s)	Maximum Memory Capacity (GB)	Processor Base Power	Maximum Turbo Power (W)
Intel® Core™ Ultra X9 388H	16	5.1	18	50	LP5/X 9600	96 (LP5/x)	25W	65 ⁷ ,80 ⁷
Intel® Core™ Ultra 9 386H	16	4.9	18	50	LP5/X 8533 DDR5 7200	96 (LP5/x) 128 (DDR5)		65 ⁷ ,80 ⁷
Intel® Core™ Ultra X7 368H	16	5.0	18	50	LP5/X 9600	96 (LP5/x)		65 ⁷ ,80 ⁷
Intel® Core™ Ultra 7 366H	16	4.8	18	50	LP5/X 8533 DDR5 7200	96 (LP5/x) 128 (DDR5)		65 ⁷ ,80 ⁷
Intel® Core™ Ultra 7 365	8	4.8	12	49	LP5/X 6800 DDR5 6400			55
Intel® Core™ Ultra X7 358H	16	4.8	18	50	LP5/X 9600	96 (LP5/x)		65 ⁷ ,80 ⁷
Intel® Core™ Ultra 7 356H	16	4.7	18	50	LP5/X 8533 DDR5 7200	96 (LP5/x) 128 (DDR5)		65 ⁷ ,80 ⁷
Intel® Core™ Ultra 7 355	8	4.7	12	49	LP5/X 6800 DDR5 6400			55
Intel® Core™ Ultra 5 338H	12	4.7	18	47	LP5/X 8533	96 (LP5/x)	25W	65 ⁷ ,80 ⁷
Intel® Core™ Ultra 5 336H	12	4.6	18	47	LP5/X 8533 DDR5 7200	LP5/X 6800 DDR5 6400		65 ⁷ ,80 ⁷
Intel® Core™ Ultra 5 335	8	4.6	12	47	96 (LP5/x) 128 (DDR5)			55
Intel® Core™ Ultra 5 325	8	4.5	12	47				55
Intel® Core™ Ultra 5 332	6	4.4	12	46				55
Intel® Core™ Ultra 5 322	6	4.4	12	46				55

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (11)

The 8-wide P-core (Cougar Cove core) of Panther Lake laptop processors [226]

Cougar Cove P-core

Optimizations

Memory disambiguation

More reliable performance

TLB enhancements

1.5x capacity for modern workloads

Branch prediction

Improving performance and energy efficiency

The diagram illustrates the Cougar Cove P-core architecture. At the top, the Intel logo is on the left, followed by the I-TLB + I-CACHE and BPU blocks. Below these are the MSROM (4-WIDE), DECODE (8-WIDE) (highlighted with a red box), and μ OP CACHE (12-WIDE) blocks. A μ OP QUEUE block spans the width below. The main execution stage is labeled ALLOCATE / RENAME / MOVE ELIMINATION / ZERO IDIOM (8-WIDE). Below this are the VECTOR REGISTER FILE and INTEGER REGISTER FILE blocks. The execution units are organized into four main sections: VECTOR SCHEDULER, INTEGER SCHEDULER, STORE DATA SCHEDULER, and MEMORY SCHEDULER. The VECTOR SCHEDULER includes units V0, V2, V1, V3, FMA, FADD, FMA, FADD, ALU, ALU, ALU, ALU, SHIFT, SHUF, SHIFT, SHUF, FPDIV, and FPDIV, with an X87 / MMX block at the bottom. The INTEGER SCHEDULER includes units P0, P1, P2, P3, P4, P5, ALU, ALU, ALU, ALU, ALU, ALU, JMP, SHIFT, JMP, SHIFT, JMP, SHIFT, MUL, MUL, and MUL. The STORE DATA SCHEDULER includes units P10, P11, STORE DATA, and STORE DATA. The MEMORY SCHEDULER includes units P20, P25, P21, P26, P22, P27, AGU, AGU, AGU, AGU, AGU, AGU, LOAD, STA, LOAD, STA, LOAD, and STA. At the bottom, there are three cache blocks: 48KB L0 D-CACHE, 192KB L1 D-CACHE, and UP TO 3MB L2 CACHE.

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (12)

The 3x3-wide E-core (Darkmont core) of Panther Lake laptop processors [226]

Darkmont E-core

Optimizations

Branch prediction
Capacity increases and accuracy refinements

Dynamic prefetcher controls
Responsiveness in workload variation

Nanocode performance
More instruction coverage

Memory disambiguation
More reliable performance



TECHPOWERUP

Note that both core types above [share the same core width](#) as the corresponding cores of the Lunar Lake laptop and Raptor Lake desktop lines.

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (13)

Assumed block diagram of the Core Ultra Series 3 (Panther Lake) processors -1

- Intel has not published the architecture of the Panther Lake processor.
However, available details of this processor suggest that its microarchitecture is similar to that of Meteor Lake processors.
- The corresponding block diagram is shown in the preceding Figure.

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (14)

Assumed block diagram of the Core Ultra Series 3 (Panther Lake) processors -2

This architecture has been implemented across three tiles, as shown below.

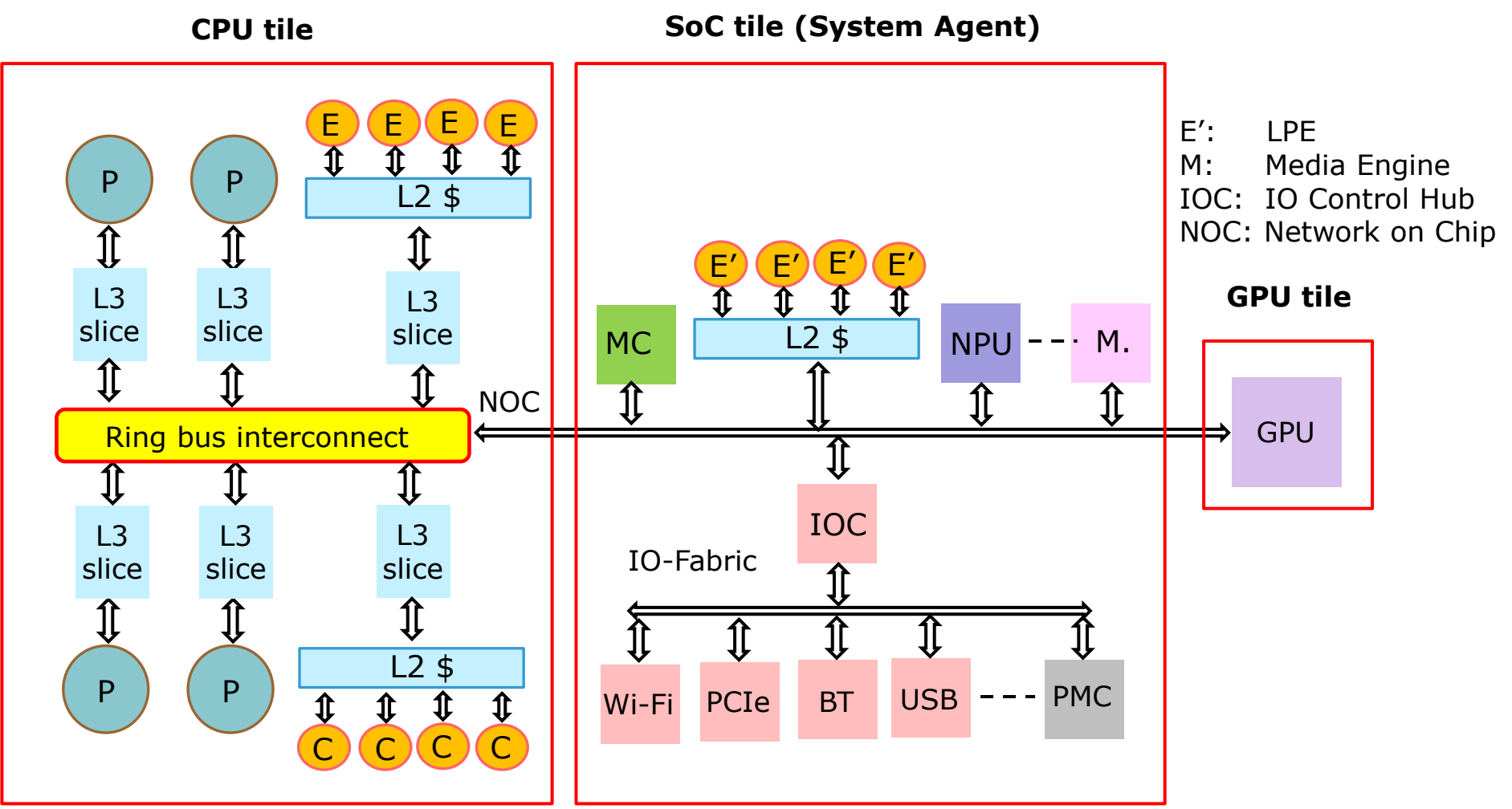


Figure: Assumed block diagram of the Core Ultra Series 3 (Panther Lake) processors

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (15)

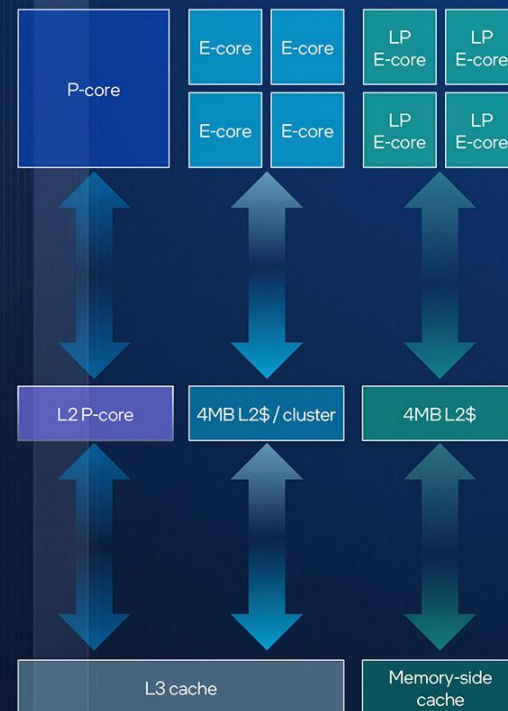
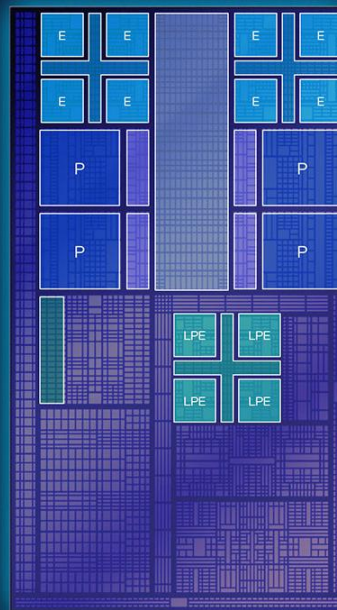
The cache architecture of the Panther Lake processors [226]

Cache & Memory Subsystem

8 E-cores on L3 cache ring
compared to none on Lunar Lake

Double the L2 cache for LP E-cores
compared to Meteor & Arrow Lake

Memory-side cache & controller on compute tile
vs. no memory-side cache and controller on SOC tile on Arrow Lake



- As shown above, the **P cores** have separate L2 caches and L3 cache segments.
- **Every 4 E-cores** are supported by a shared L2 cache and an L3 cache segment, whereas the 4 LPE cores only have a shared L2 cache to the Memory Side Cache.

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (16)

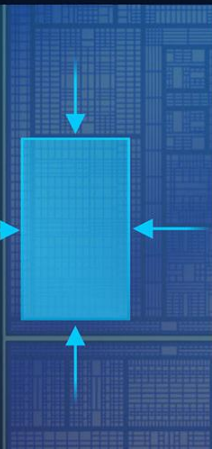
The AI performance of the Panther Lake line [226]

Next-Gen NPU5

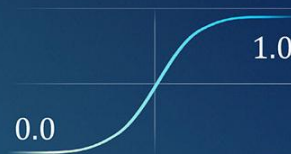
Major leap in efficiency and performance, powering next-gen AI accelerators

>40%

improvement in
TOPS per area



New activation
function support



MAC array
doubled in size



Up to

50

TOPS

Enhanced
datatype
conversion



Native FP8

datatype support



Copilot+PC

3

Neural compute
engines



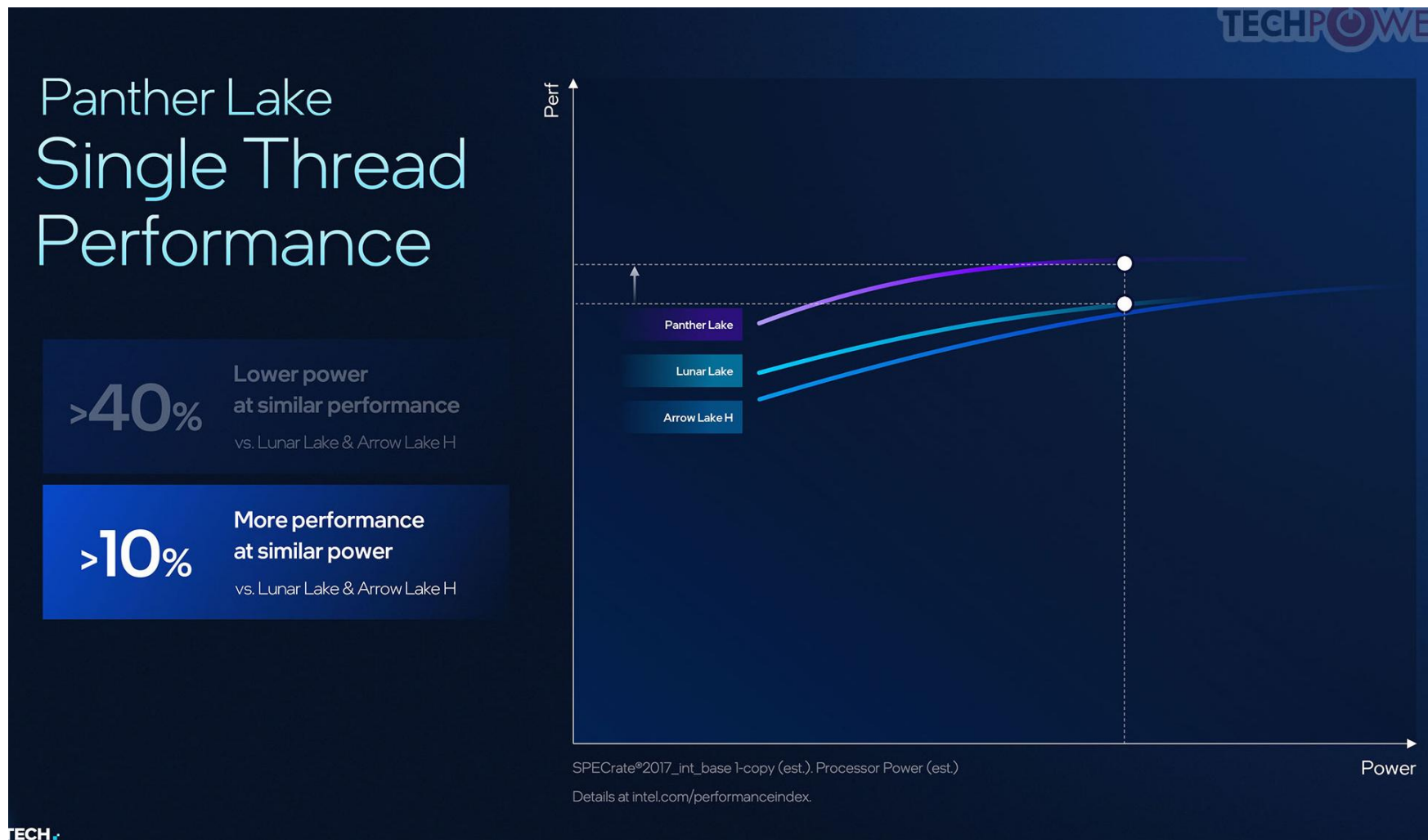
intel

Details at intel.com/performanceindex

As shown above, the Panther Lake Line offers **50 TOPS** AI performance **10 TOPS** more than the Lunar Lake line.

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (17)

Comparison of single-thread SPECInt2007 performance of Intel's 2nd and 3rd Series Core Ultra processors [2026)]



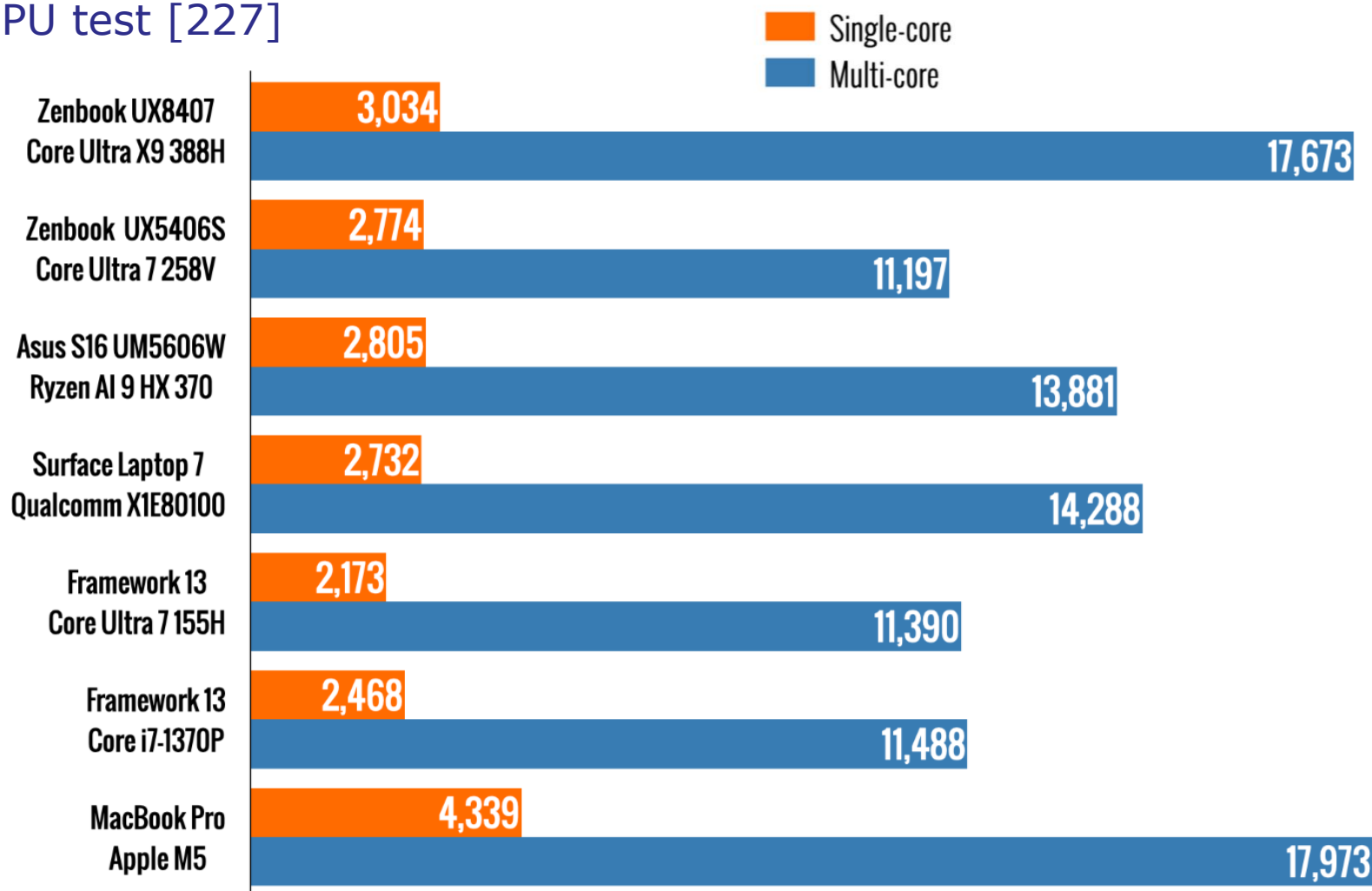
4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (18)

Panther Lakes' performance and power consumption compared to competing processors

Next, we will compare the performance and power consumption of Panther Lake processors against the competition using a few relevant benchmarks.

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (19)

SC and MC performance of the Panther Lake processors using the Geekbench 6 CPU test [227]

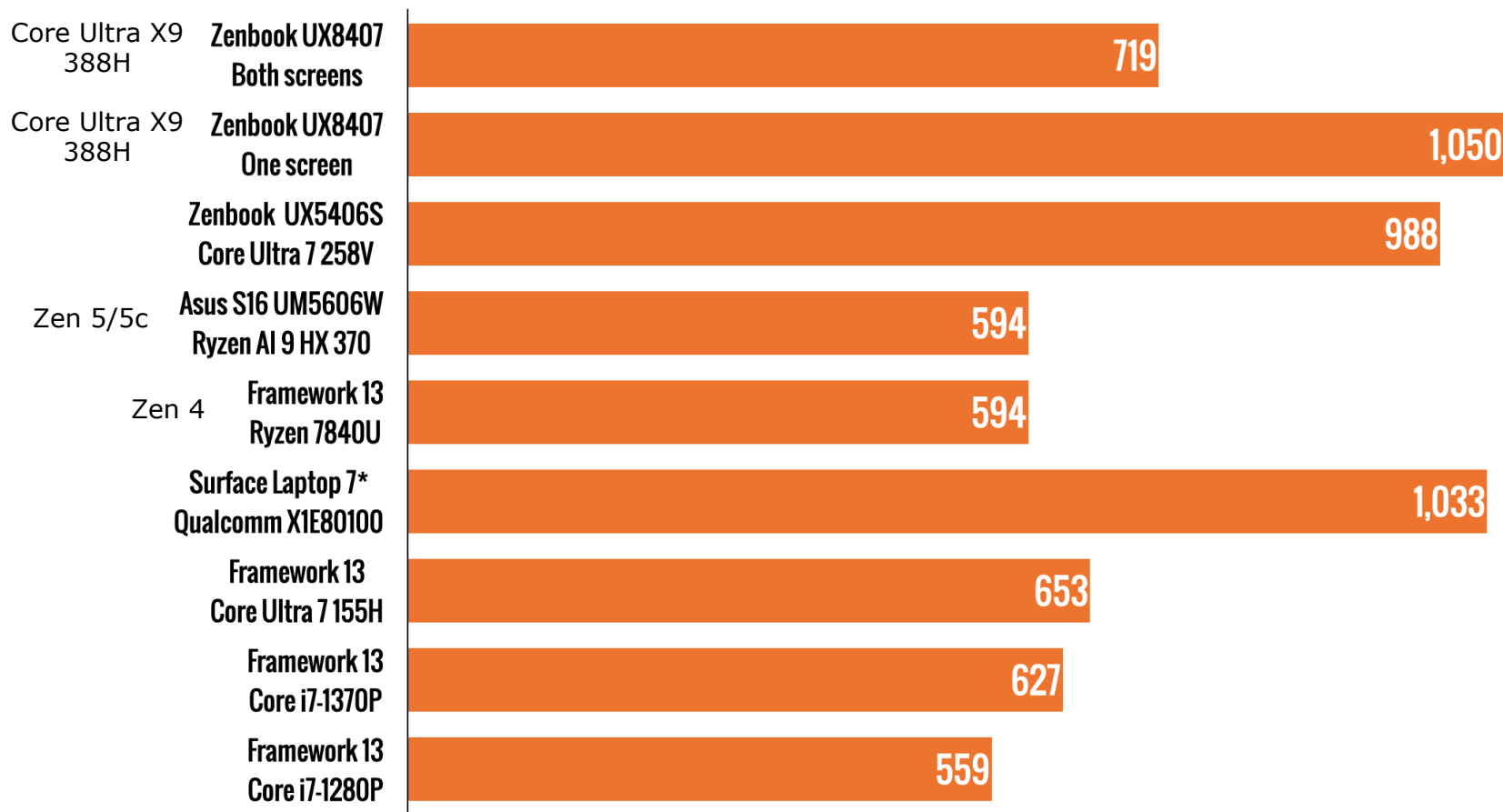


Remark

Geekbench 6 is cross-platform benchmark that evaluates how a CPU handles everyday tasks such as email, photo operations, music playback, augmented reality, and machine-learning-style computations.

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (20)

Battery life comparison using the PCMark Modern Office benchmark [227]
(Results in minutes, higher is better) [227]

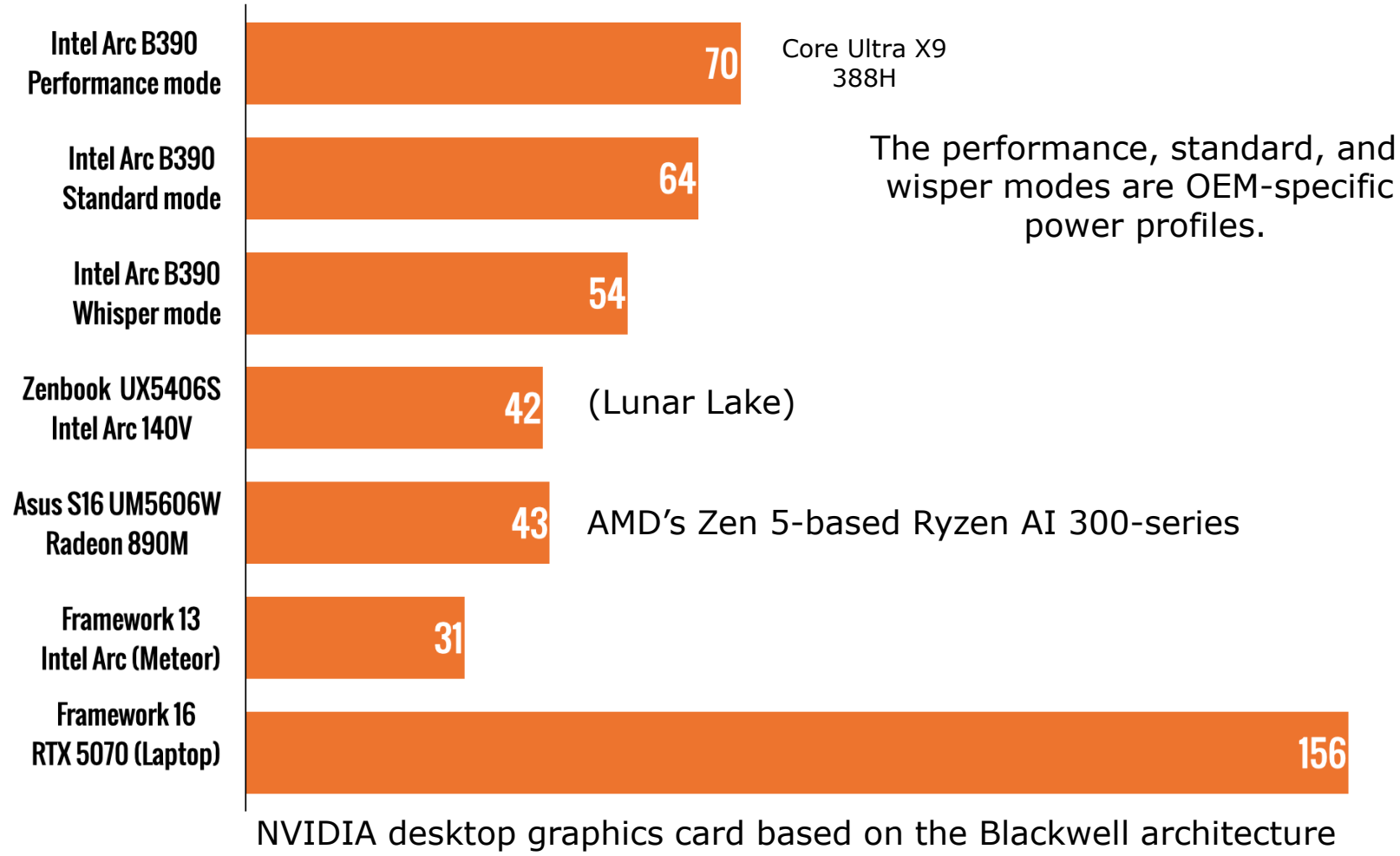


Remark
PCMark measures how well a laptop handles light-to-medium office tasks, such as writing, web browsing, and video conferencing. It is part of the PCMark 10 benchmark (UL Benchmarks).



4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (21)

Graphics performance comparison while running the Shadow of the tomb raider game (Results in FPS, higher is better) [227]



It is a [single-player action-adventure game](#), introduced in 2018.

In the story, Lara Croft races to stop a Maya apocalypse while exploring jungles, tombs, and underwater environments.

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (22)

Assessment of the Core Ultra 3. series Panther Lake processors

The previous Figures demonstrate that Intel's **Panther Lake** processors deliver **outstanding performance and power efficiency**, **surpassing all competitors** in both aspects.

As a result, **Intel's stock price** has seen a **significant increase**, as illustrated in the following Figure.

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (23)

How Intel's stock price responded to market events from Aug. 2025 to Febr. 2026)



The above Figure shows the significant increase in Intel's stock price following NVIDIA's stock buy and Intel's introduction of its Panther Lake family.